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Late Cretaceous–Early Palaeogene tectonic development of SE Asia

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ABSTRACT

The Late Cretaceous–Early Palaeogene history of the continental core of SE Asia (Sundaland) marks the time prior to collision of India with Asia when SE Asia, from the Tethys in the west to the Palaeo-Pacific in the east, lay in the upper plate of subduction zones. In Myanmar and Sumatra, subduction was interrupted in the Aptian-Albian by a phase of arc accretion (Woyla and Mawgyi arcs) and in Java, eastern Borneo and Western Sulawesi by collision of continental fragments rifted from northern Australia. Subsequent resumption of subduction in the Myanmar–Thailand sector explains: 1) early creation of oceanic crust in the Andaman Sea in a supra-subduction zone setting ~95 Ma, 2) the belt of granite plutons of Late Cretaceous–Early Palaeogene age (starting ~88 Ma) in western Thailand and central Myanmar, and 3) amphibolite grade metamorphism between 70 and 80 Ma seen in gneissic outcrops in western and central Thailand, and 4) accretionary prism development in the Western Belt of Myanmar, until glancing collision with the NE corner of Greater India promoted ophiolite obduction, deformation and exhumation of marine sediments in the early Palaeogene. The Ranong strike–slip fault and other less well documented faults, were episodically active during the Late Cretaceous–Palaeogene time. N to NW directed subduction of the Palaeo-Pacific ocean below Southern China, Vietnam and Borneo created a major magmatic arc, associated with rift basins, metamorphic core complexes and strike–slip deformation which continued into the Late Cretaceous. The origin and timing of termination of subduction has recently been explained by collision of a large Luconia continental fragment either during the Late Cretaceous or Palaeogene. Evidence for such a collision is absent from the South China Sea well and seismic reflection record and here collision is discounted. Instead relocation of the subducting margin further west, possibly in response of back-arc extension (which created the Proto-South China Sea) is preferred. Lying between the two subduction related arcs, the Khorat Basin is of predominantly Late Jurassic–Early Cretaceous age but stratigraphic and apatite fission track data also indicates deposition of 1–2 km of Late Cretaceous sediments. The synformal basin geometry probably arose due to the dynamic topography created by converging Tethyan and Palaeo-Pacific subduction zones. The Aptian-Albian slowing of basin subsidence and onset of evaporite deposition coincides with collision of the Mawgyi and Woyla island arcs. Extensive Palaeogene deformation and exhumation (3+ km in places) affected all margins of the Khorat Plateau. Deformation includes folds of the Phu Phan uplift, and strike–slip faults, thrusts and folds on the southern and eastern margins. South of the Khorat Plateau outcrop, and seismic reflection data from the Ton Le Sap Basin (Cambodia), and the Gulf of Thailand, indicate syn-depositional fault-controlled subsidence was important during Cretaceous deposition. The hot, thickened crust developed during the Late Cretaceous–Palaeogene events follows the weak (Indosinian), crustal-scale Inthanon and Sukhothai zones, which persistently guided the location of later structures including Cenozoic extensional, and post-rift basins, and influenced the widespread occurrence of low-angle normal faults, metamorphic core complexes, and eastern Gulf of Thailand super-deep post-rift basins.

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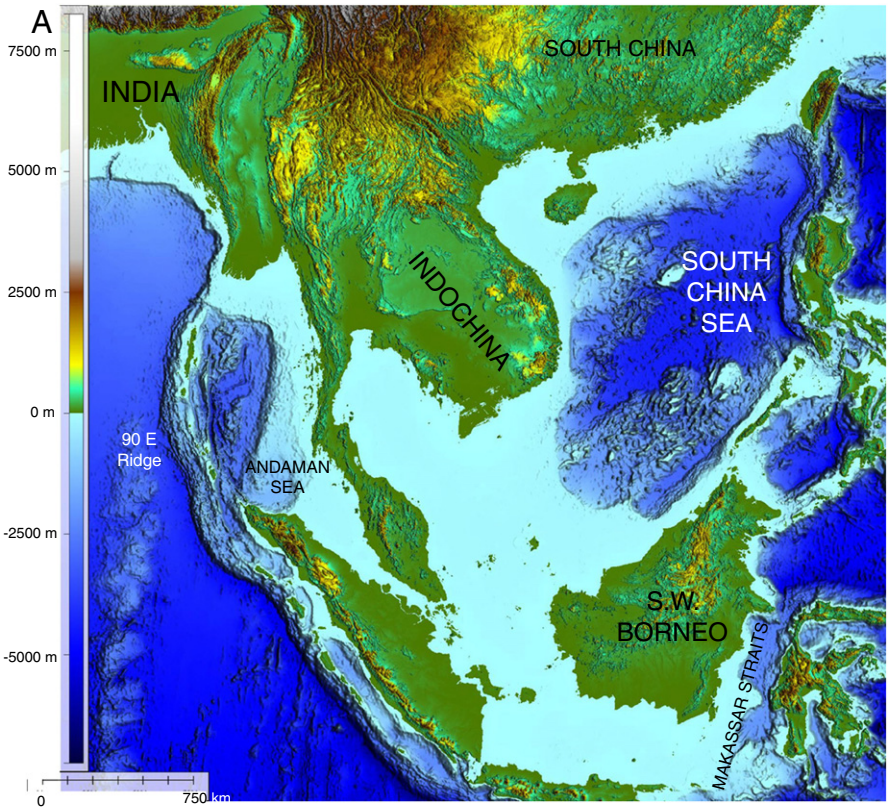
1. Introduction

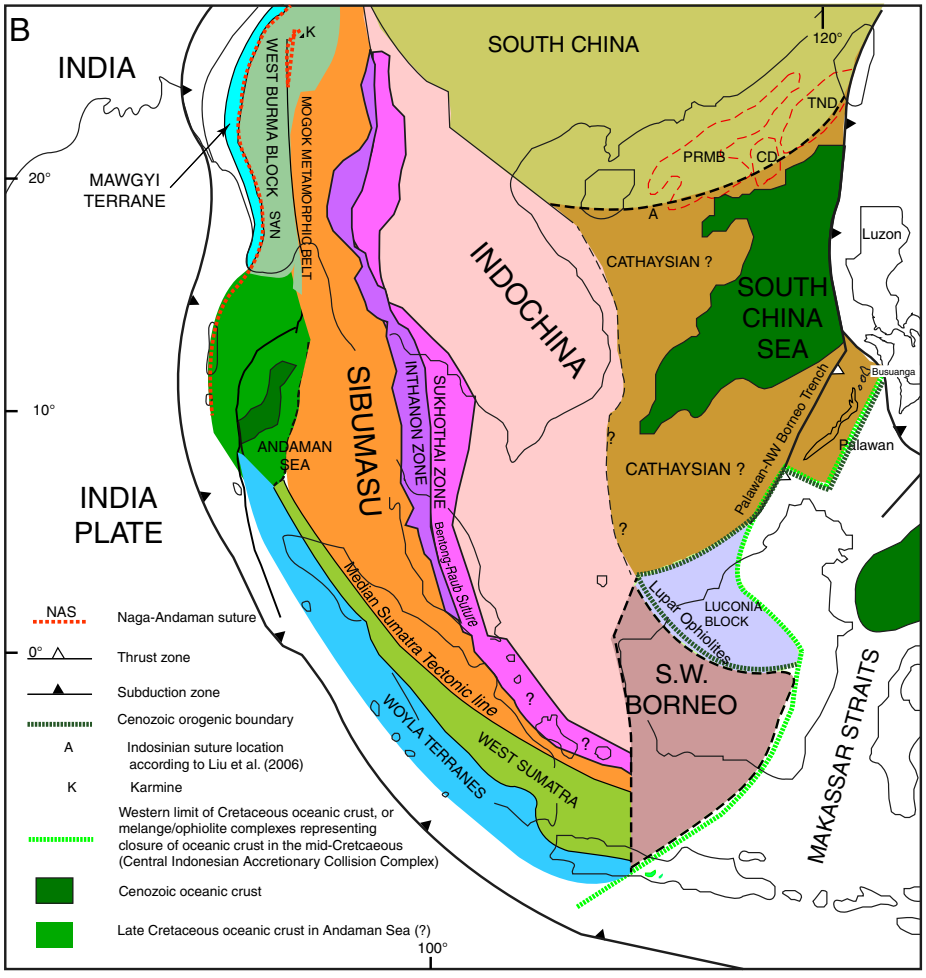
Late Cretaceous–Palaeogene tectonic events within, and surrounding the Gulf of Thailand mark an important period in the development of the continental core of SE Asia (Sundaland) continental crust (Fig. 1). Yet the story remains fragmentary and dating of events is commonly ambiguous. Recently a number of different geological studies and geophysical surveys enable a more coherent picture of the tectonic setting and events affecting the area to be established. This paper reviews the regional publications, publically available data, and presents new data to develop an up-dated tectonic model for the Late Cretaceous–Palaeogene history of the circum-Gulf of Thailand region up to about 40 Ma. This time period is one of significant tectonic activity in central Indochina (Thailand, Laos, Cambodia, Malaysia), which at present is poorly described. The Late Cretaceous–Early Palaeogene period is bracketed by episodes of extensive basin development: i.e. deposition of the Khorat Group, and Phuqoc Group during the Jurassic–Cretaceous (Racey et al., 1996; Racey and Goodall, 2009; Fyhn et al., 2010), and the regional development of rift basins in SE Asia from the Late Eocene onwards (Hall and Morley, 2004).

To understand the Late Cretaceous–Paleogene tectonic development of Sundaland, also requires knowledge of the tectonic development in adjacent areas. The history, in general, is one of subduction of the Tethys and Palaeo-Pacific oceans and associated accretionary events on the western, southern, and eastern margins of Sundaland (Hall et al., 2009; Metcalfe, 2009, 2011). But the details of collision events involving island arcs, continental blocks and smaller fragments remains highly uncertain, and a number models with some contradictory elements have been proposed (e.g. Charusiri et al., 1993; Barber and Crow, 2003, 2009; Hall, 2009; Hall et al., 2009; Fyhn et al., 2010; Wang et al., 2008, 2011). This paper discusses the main plate tectonic models and their compatibility with our present understanding of the Late Cretaceous–Early Palaeogene stratigraphic, sedimentary, structural and tectonic development of Sundaland. The tectonic development is discussed in the context of three geographical regions: 1) the western and southern boundaries of Sundaland, which comprises the story of the subduction of the Tethyan realm beneath Eurasia, the approach of India with Eurasia, and the collision of continental fragments rifted from Australia and island arcs with Sundaland. 2) The continental area of Sundaland, (i.e. Thailand, Laos, Cambodia, Vietnam, Malaysia, Borneo, Myanmar, Sumatra, western Java, Western Sulawesi), and the tectonics of the plate interior, and 3) the eastern boundary of Sundaland, which involves the subduction of the Palaeo-Pacific oceanic crust beneath Southern China, Vietnam and southern Borneo, and the putative collision of continental fragments during the late Cretaceous and Palaeogene (e.g. Hall et al., 2009; Hall, 2012; Fyhn et al., 2010).

The Late Cretaceous to early Palaeogene period is also important for understanding the subsequent development of Cenozoic tectonics during the Himalayan stage of deformation. There is a widespread impression that the significant Cenozoic tectonics of Indochina began with the effects of escape tectonics during the Oligocene (e.g. Tapponnier et al., 1986; Leloup et al., 2001; Replumaz and Tapponnier, 2003; Fyhn et al., 2009). However, is this really the case? For example it has been proposed by many workers (e.g. Leloup et al., 2001; Replumaz and Tapponnier, 2003; Fyhn et al., 2009) that opening of the South China Sea spreading centre (Fig. 1) is linked with displacement on the Ailao Shan-Red River Fault zone. Yet, as reviewed by Morley (2002) the development of extension

Fig. 1. A) DEM of the SE Asia region, ETOPO1 data (Amante and Eakins, 2009). B) Regional location map showing Terranes accreted in the Late Palaeozoic, in part based on Sone and Metcalfe 2008b, Barber et al. (2011), and Metcalfe (2011).


A
7500 m
5000 m
2500 m
0 m
-2500 m
-5000 m
INDIA
SOUTH CHINA
INDOCHINA
SOUTH CHINA SEA
90 E Ridge
ANDAMAN SEA
S.W. BORNEO
MAKASSAR STRAITS
0
750 km


B
INDIA
SOUTH CHINA
120°
TND
PRMB
CD
A
MAWGYI TERRANE
20°
WEST BURMA BLOCK
SYN
MOGOK METAMORPHIC BELT
K
INDOCHINA
CATHAYSIAN ?
SOUTH CHINA SEA
Luzon
Busuanga
Palawan
Palawan-NW Borneo Trench
SIBUMASU
INTHANON ZONE
SUKHOTHAI ZONE
Bentong-Raub Suture
ANDAMAN SEA
10°
INDIA PLATE
Median Sumatra Tectonic line
WOYLA TERRANES
WEST SUMATRA
LUCONIA BLOCK
Lupar Ophiolites
S.W. BORNEO
MAKASSAR STRAITS
NAS
Naga-Andaman suture
0°
Thrust zone
Subduction zone
Cenozoic orogenic boundary
A
Indosinian suture location according to Liu et al. (2006)
K
Karmine
Western limit of Cretaceous oceanic crust, or melange/ophiolite complexes representing closure of oceanic crust in the mid-Cretaceous (Central Indonesian Accretionary Collision Complex)
Cenozoic oceanic crust
Late Cretaceous oceanic crust in Andaman Sea (?)
100°

along the South China Sea margin during the Late Cretaceous and Palaeogene indicates the Oligocene development of the Ailao Shan-Red River Fault zone was superimposed on a margin that was already undergoing extension. Hence the impact of the fault zone on the margin may be a secondary effect, and not the main driver of extension (e.g. Morley, 2002). Similarly in Thailand there are suggestions that some strike–slip fault strands associated with major Oligocene-age escape tectonics-related faults, were actually active prior to the Oligocene (e.g. Morley, 2004; Morley et al., 2007c, Watkinson et al., 2008).

As discussed by Morley and Westaway (2006), Morley (2009) and Morley et al. (2011) the Late Eocene–Neogene sedimentary basins of Thailand demonstrate a number of unusual features for rift and post-rift basins including: 1) numerous low-angle normal faults, 2) high heat flows and geothermal gradients in post rift basins, 3) very deep, rapid subsidence post-rift basins, 4) a pronounced south to north propagation of rift initiation and rift termination, and 5) alternation of episodes of inversion and strike–slip motion with extension. An improved understanding of the geological events preceding rift initiation is needed to assess how these unusual characteristics were influenced by pre-existing crustal conditions.

2. The western and southern boundary of Sundaland during the Late Cretaceous–Palaeogene

2.1. Tectonic models for Myanmar–Sumatra

The major tectonic units of Mynamar comprise an eastern area in the Shan Plateau, which is part of the Sibumasu Block, and an area west of the Shan Scarp called the West Burma Block (e.g. Metcalfe, 1996; Fig. 1). The Sibumasu Block is one of the terranes derived from Gondwana in the late Early Permian (Metcalfe, 1996, 2011). As reviewed by Metcalfe (2011) our understanding of the West Burma Block has undergone a checkered history of interpretation with regards to its timing of its rifting from Australian Gondwana, timing of collision with Indochina, and nature of the crust. Accretion of the West Burma Block during the Late Cretaceous–Cenozoic was invoked to explain the Late Cretaceous–Eocene, S-type, tin-bearing granitic plutons that run through Thailand and Myanmar (Charusiri et al., 1993, Hutchison, 1994) and has also been used to explain Palaeogene deformation in the Indo-Burma ranges (e.g. Socquet et al., 2002; Acharyya, 2007). Recently the origin of the block (and the West Sumatra Block to the south) has been explained by Early Carboniferous accretion to South China as part of Indochina, followed by strike–slip emplacement of the combined West Burma–West Sumatra block on the western margin of the Sibumasu Terrane during the Triassic (Barber et al., 2005; Barber and Crow, 2009). In contrast Mitchell (1993) does not identify a West Burma continental block, but instead thrusts the Mawgyi island arc over the Indochina margin in the late Lower Cretaceous, and only establishes a West Burma Block in the Mid Eocene as a consequence of movement along the Sagaing Fault. Hence a strike–slip fault, not a suture zone is used to separate the West Burma Block from eastern Myanmar in Mitchell's (1993) model.

At present the West Burma Block and West Sumatra Blocks are regarded as part of the terranes derived from Gondawana in the Devonian (Barber and Crow, 2009; Metcalfe, 2011). According to Metcalfe (2011) the West Burma Block is bounded to the west by a belt of ophiolites (including the Mount Victoria metamorphics), and to the east by the Mogok Metamorphic belt. West of the Burma Block is the Mawgyi Terrane (Fig. 1), whose origin is discussed in this section. The Cathaysian derivation of the West Burma Block is suggested by the presence of Middle Permian rocks near Karmine with Cathaysian fusulinids (Oo et al., 2002; location K, Fig. 1) similar to those in the West Sumatra Block and different from the Sibumasu Block (Barber and Crow, 2009).

The late Early Cretaceous to early Late Cretaceous period is marked regionally by a period of island arc collision with Eurasia. According to Barber and Crow (2003) and Barber et al. (2005) the Woyla Group in Sumatra (Fig. 1) represents collision of an east-facing, Late Jurassic–Early Cretaceous, intra-oceanic volcanic arc and accretionary complex, as part of the Mesotethys Triassic–Cretaceous oceanic crust was subducted. To explain the Late Jurassic–Early Cretaceous granites in Sumatra a west-facing subduction zone beneath Sundaland is also required, i.e. a double subduction zone similar to the Molucca Sea is envisioned (Barber et al., 2005, 2011). Subsequently the Woyla Terrane (or nappe) was thrust over West Sumatra causing metamorphism up to amphibolite grade. Albian–Aptian age carbonates are involved in the overthrusting, and Late Cretaceous and younger granites intrude the Woyla Terrane, bracketing the age of collision between about 100 and 90 Ma (Barber and Crow, 2009).

2.2. Stratigraphic, structural, magmatic and metamorphic evidence for the tectonic models for Myanmar

The strong overprint of Cenozoic deformation, poor exposures, and inaccessibility of many areas for modern studies makes attempts to critically evaluate the general tectonic models for western Myanmar, difficult. This section discusses the key elements that have been identified by previous studies. These elements are discussed in four categories: 1) evidence for the age of formation of the oceanic crust that forms the ophiolites, 2) stratigraphic and structural evidence for the timing of ophiolite emplacement events; 3) structural relationships, and 4) timing of magmatism and metamorphism.

2.2.1. Evidence for the age of formation of oceanic crust

Within the Western Belt of the Indo-Burma Ranges ophiolites occur as blocks and dismembered lenses within Upper Cretaceous–Palaeogene deep-marine flysch deposits, as well as low-angle overthrust sheets that are preserved in large synclines (Acharyya, 2007). The ages of these ophiolites are not well constrained, however, Upper Jurassic radiolarians from the Naga Ophiolite in NE India (Baxter et al., 2011) and recent U–Pb dating of zircons (oldest zircon groups ages are 163.2 ± 3.3 Ma) from the Jade Mines belt of northern Myanmar (Guanghai et al., 2008) supports the long held contention of Mitchell (1993) that the ophiolites are of Jurassic age.

The age of oceanic crust in the Andaman Sea is poorly known, except for the Pliocene–Holocene area of the young spreading centre (Curray, 2005). The oceanic crust of the Andaman Sea intervenes between the onshore outcrops of the Woyla and Mawgyi terranes, (Curray, 2005; Fig. 2). Two prominent areas of rugged sea floor topography called the Alcock and Sewell Rises lie adjacent to the active spreading centre. The eastern part of the Alcock Rise area, which comprises oceanic crust as mapped by Curray (2005). 2D seismic reflection data that extends to the Mergui Basin indicates this eastern Alcock Rise crust is covered by sediments of probable Late Eocene–Early Miocene age, (Fig. 2). Hence the oceanic crust is overlain by early syn-rift sediments, which indicates that the oceanic crust did not form during Miocene extension, and must be of Palaeogene age or older. Another possibility is that the rises are actually highly attenuated continental crust, but for the following discussion it is assumed that the identification of oceanic crust by Curray (2005) is correct. Dating of the Andaman Ophiolite in the Andaman Islands suggests the oceanic crust formed around 95 Ma (Pedersen et al., 2010). If the oceanic crust of the Sewell and Alcock Rises is contiguous with the Andaman Islands, then it probably began to develop during the early Late Cretaceous. This age is notably younger than the Jurassic ages for the ophiolite belt to the north, onshore in the Chin Hills and Naga Ophiolite (Baxter et al., 2011; Fig. 3) and indicates the Naga-Andaman suture (Fig. 1) cannot be considered as developing as a single entity. Miocene lavas that were dredged from the Alcock Rise (Curray, 2005) are regarded as a later rifting or arc related

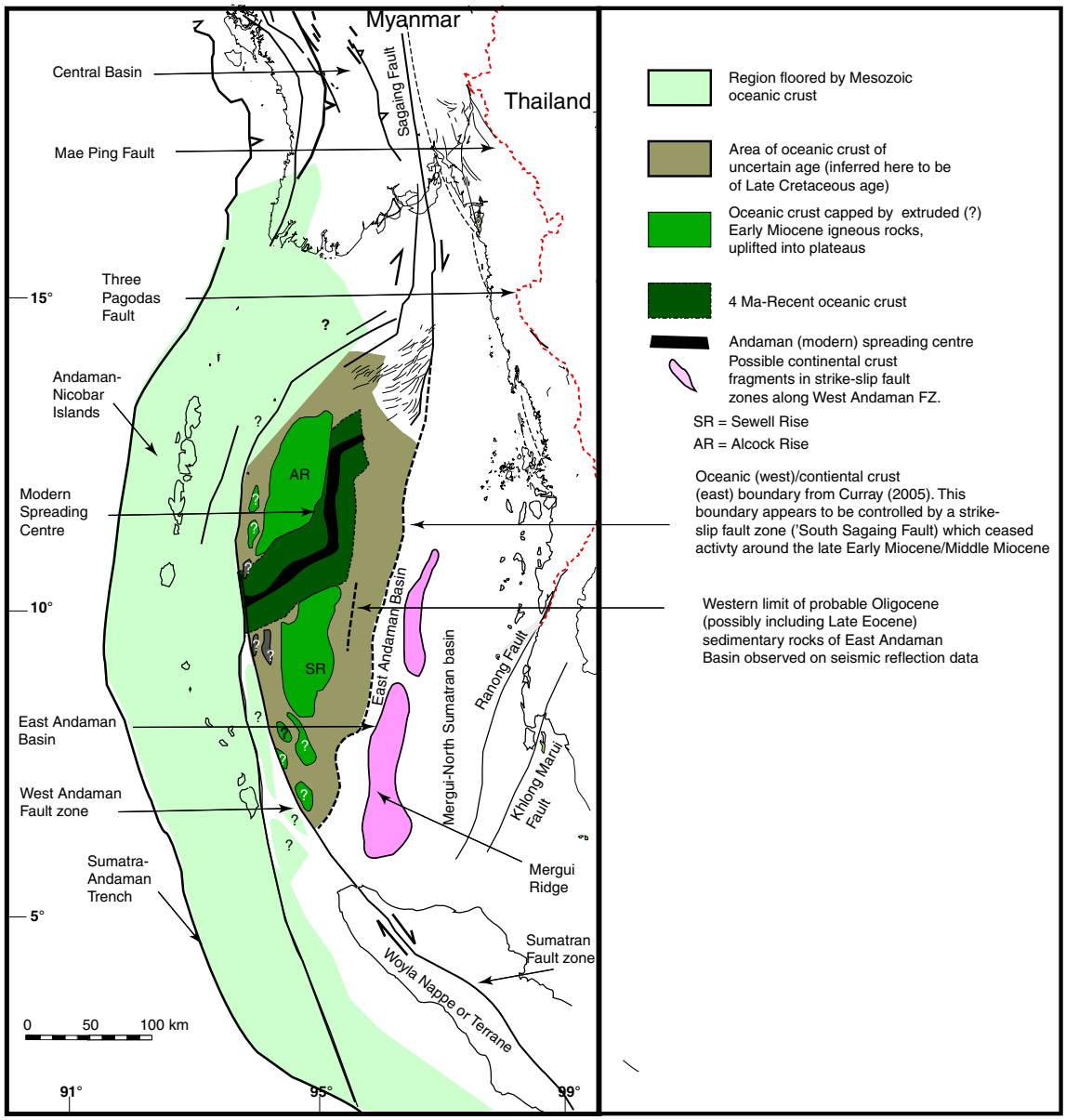


Fig. 2. Geological setting of the Andaman Sea area (modified from Curray, 2005; Searle and Morley, 2011).

phenomena extruded over Late Cretaceous crust, and do not indicate the age of crust formation. The rugged nature of the rises is probably a later feature caused by a mantle hotspot or plume prior to Pliocene seafloor spreading, as the area formed a basaltic plateau that was dissected by Cenozoic normal and strike–slip faults.

2.2.2. Stratigraphic and structural evidence for the timing of deformation

Late Cretaceous sedimentary rocks occur extensively in western Myanmar both within the Indo-Burma Ranges to the west, and the Central Basin to the east (Fig. 3x). The Indo-Burma Ranges can be sub-divided into Western and Eastern belts (Mitchell 1993; Mitchell et al., 2010; Fig. 3x). In the Eastern Belt there is a strong overlap between its stratigraphy and that of the Central Basin to the east during the latest Cretaceous and Cenozoic.

The Eastern belt contains two key areas for assessing the tectonic models, the Chin and Naga Hills (Fig. 3x), in both areas a core of island arc or continental metamorphic rocks are unconformably overlain by *Orbitolina*-bearing mid-Cretaceous limestones (Paung Chaung and Nimi formations) and this metamorphic unit is juxtaposed with ophiolites. The relationship between the basement, ophiolites and mid-Cretaceous limestones is critical for understanding the tectonic development of the area, yet these relationships are not fully understood and appear to differ between the two areas (e.g. Acharyya, 2007).

The Chin Hills comprises a core of muscovite and biotite schists, with minor meta-pillow basalts (Kanpetlet Schists, Mt Victoria area, Brunnschweiler, 1966). Triassic flysch overlies the schists. The flysch contains, and is overlain by, bodies of partly serpentinised harzburgite, dunite, chromitite and gabbros. The Paung Chaung Limesone is the basal unit for the Chindwin and Minbu Basins east of the Indo-Burma Ranges. This unconformable relationship of the limestone with underlying ophiolites emplaced on metamorphic rocks and Triassic flysch is of critical importance for demonstrating the existence of the Mawgyi Terrane, and also its timing. Mitchell et al. (2007) proposed an arc (Mergui island arc) collision scenario for Myanmar, similar to that proposed by Barber and Crow (2003, 2009; Fig. 4) for the Mawgyi-Woyla Terrane. The key difference lies in the timing of collision proposed by Mitchell et al. (2007), which is earlier (pre ~110 Ma) than the 90–100 Ma collision of the Woyla Arc (Barber and Crow, 2009), based on the constraint of the Albian

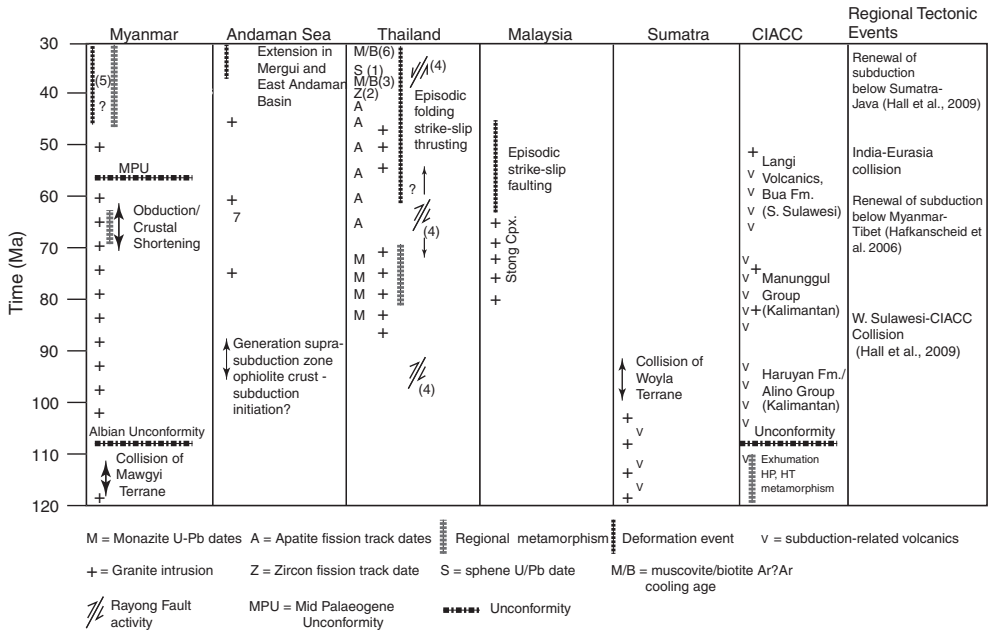


Fig. 3. Timing of tectonic events during the Late Cretaceous–Palaeogene in western SE Asia. 1 = Geard (2008), 2 = Zircon apatite fission track cooling age, Umphang Gneiss, Upton (1999), 3) Muscovite and biotite Ar/Ar cooling ages reported in Morley et al. (2007c) 4 = Watkinson et al. (2011), 5 = U–Pb ages from Mogok Metamorphic belt, Searle et al. (2007), 6 = muscovite and biotite Ar/Ar cooling ages for Mae Ping Fault Zone reported in Lacassin et al. (1997).

mid-Cretaceous unconformity. A group of zircons identified by Guanghai et al. (2008) from the Jade Mines belt yielded a mean U–Pb age of 122.2 ± 4.8 Ma, which they called a later unknown thermal event. This age coincides reasonably well with the timing of arc collision proposed by Mitchell et al. (2007).

Passing northwards to the Naga Hills the Nimi Formation appears to be the temporal and lithological equivalent of the Paung Chaung Limestone. The *Orbitolina*-bearing Nimi Formation is dominated by a low-grade metamorphic, regionally prominent, neretic limestone that interfingers with phyllites, quartzites and quartz-sericite schists (Agrawal and Ghose, 1986; Ghose et al., 2010). The limestone is strongly folded and contains tectonic slices of serpentinite and volcanics. Fossil evidence, and basement-derived arkosic quarztites suggests it may be a Mid-Cretaceous cover sequence to the Naga Metamorphics, which include mica schists and granitoid gneisses (Ghose et al., 2010). There is no evidence for the unconformable relationship of the Nimi Formation overlying older ophiolites that is seen in the Chin Hills.

The Western Belt sediments are predominantly deep-marine sediments of Upper Cretaceous–Eocene age, they commonly contain turbidite fan complexes, debris flows, melanges and sheared up lenses of ophiolites. The oldest rocks found in outcrop are thin sandstone turbidites and varicoloured micritic limestones, with lenses of reddish radiolarian cherts, and silicified radiolarian limestones, which comprise flysch deposits of predominantly Campanian and Maastrichtian age (Brunnschweiler, 1966). Small single-keeled *Globotruncana* suggest the oldest part of the succession is Cenomanian or possibly Albian (Gramann, 1974; Stokes, 1988). Contemporaneous mafic–ultramafic sills or submarine extrusions, tuffaceous beds and agglomerates occur within the succession. These deposits are called the Falam Formation (United Nations, 1979). The Palaeocene–Early Eocene Chungsung Mudstone–Turbidite Formation overlies the Falam Formation (Fig. 5x), and is more than 2 km thick.

Considerable re-working of Cretaceous deposits occurred during the Upper Cretaceous and Cenozoic (Brunnschweiler, 1966). For example on Ramri Island in the Bay of Bengal, Early Miocene clays contain blocks of Cretaceous rocks with the following lithologies: Cenomanian–Maastrictian Globotruncana limestone, red and green radiolarites and unfossiliferous cherts, and dark, hard limestones with belemnites and ammonites (probable Turonian age).

In the Eastern Belt the sedimentary rocks overlying the Paung Chaung Limestone lack the accretionary prism characteristics of the sediments found in the Western Belt. The sediments were deposited on the western margin of the Central Basin, which has a history of deposition in a broad, rapidly subsiding synformal basin during the Cenozoic. The central and southern parts of the basin show little structural deformation until the Miocene (Pivnik et al., 1998), and depositional environments range predominantly between shelf, outer shelf and slope. Overlying the Paung Chaung Limestone is the Kabaw Formation, which is primarily a shale-prone formation that contains glauconitic sandstones and shallow marine fossils of Campanian, Maastrichtian and Danian age (Gramann, 1974; Mitchell et al., 2010; Fig. 5). Late Middle Palaeocene to early Lower Eocene clastics and minor limestones of the Paunggyi Formation (Theingi Kyaw, 2005) lie unconformably on the Kabaw Formation, pillow basalts and Triassic turbidites (United Nations, 1979; Socquet et al., 2002). In parts of the Paunggyi Formation east of the Chin Hills cobble-size clasts of calc-alkaline volcanic and plutonic rocks are present, indicating the Chindwin-Minbu basins lay in a fore-arc location (Win Swe et al., 1972). The Lower Eocene Laungshe Shales overlie the Paunggyi Formation.

2.2.3. Structural relationships

Metamorphic units in both the Chin and Naga Hills have been thrust westwards over western-belt deep marine sediments (Fig. 6). In places, particularly in the Naga Hills, the metamorphics have been thrust over ophiolites, which overlie Cretaceous–Eocene Western Belt sedimentary rocks (Brunnschweiler, 1966; Acharyya, 2007). The structural relationships strongly suggest that the oceanic crust lies west of the Kanpetlet metamorphics and Naga Metamorphics, and not to the east of the metamorphics as interpreted by Socquet et al. (2002). In the Naga Hills there is no ophiolite

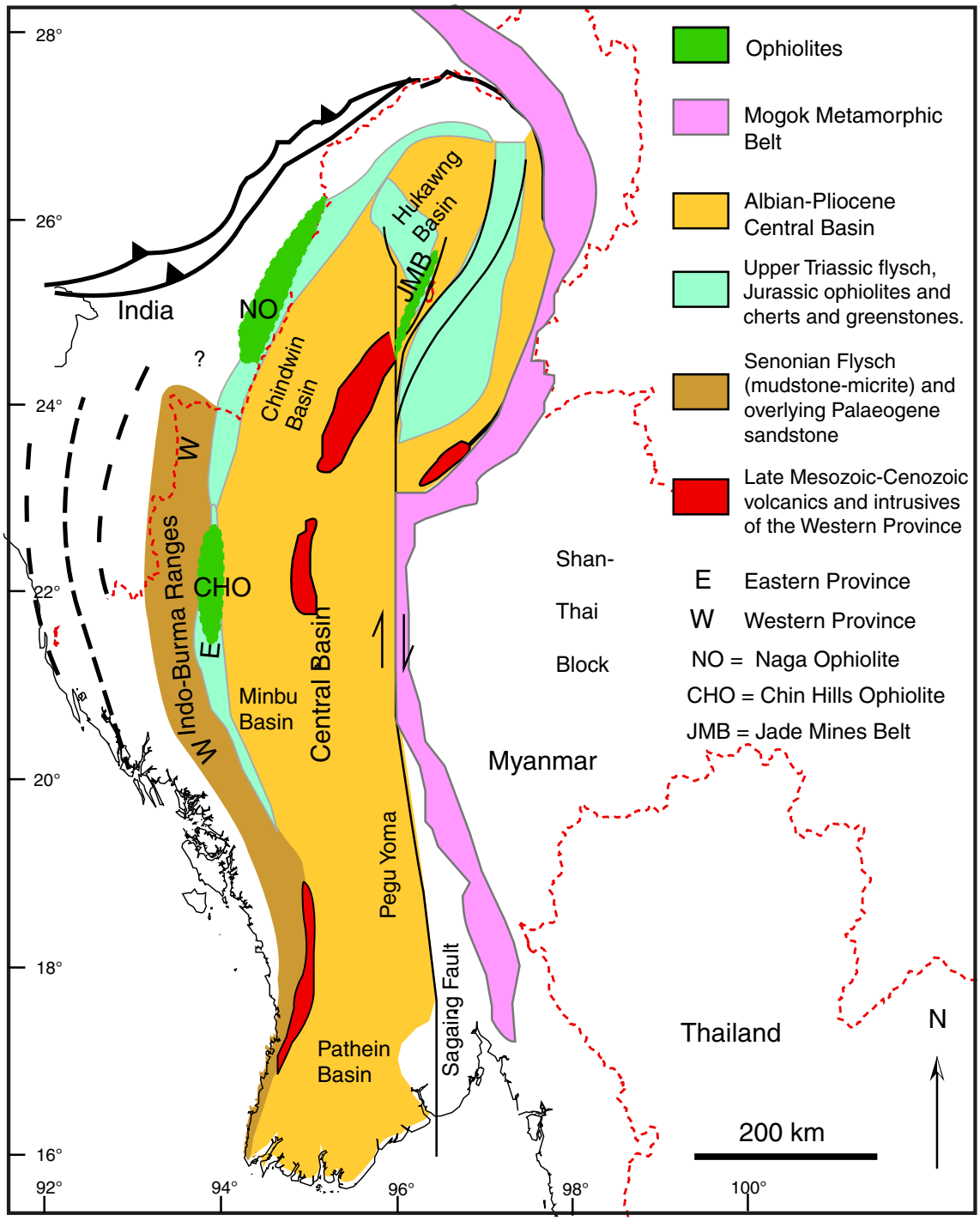


Fig. 4. Key geological elements for the Late Cretaceous–Palaeogene history of Myanmar, based mostly on Mitchell et al. (2007).

recorded as lying unconformably below the Nimi Formation, however, further east on the eastern side of the (Neogene) Chindwin Basin Brunnschweiler (1966) shows in a regional cross-section Orbitolina limestones unconformably overlying metamorphics and jade-bearing ophiolites (Fig. 6). These ophiolites have been interpreted as being related to the accretion of the Mawgyi Arc (Barber and Crow, 2009).

East of Mount Victoria the Kanpetlet metamorphics are overlain by the Late Cretaceous–Palaeogene sequence of sediments that are not strongly deformed and dip off to the east (Fig. 6). It seems impossible that the Central Basin sediments could have been deposited in shallow marine conditions and be so little deformed if they were close (Present Day distance about 10 km) to a Palaeogene collision zone as required by Socquet et al. (2002) and Acharyya (2007).

The term ophiolite has been used somewhat liberally in the literature of Myanmar, and appears to be applied to all serpentinite bodies. One issue is how to interpret the serpentinite sheets that occur locally within and at the base and top the Paung Chaung Limestone (Brunnschweiler, 1966; Mitchell et al., 2010). Brunnschweiler (1966) interpreted these sheets as representing serpentinised sills that were contemporaneous feeders to basaltic and peridotitic submarine pillow lavas found within the Late Cretaceous Naga Hills flysch. Later thrusting utilized the sills as slip planes and destroyed the original continuity of the igneous intrusions. A similar

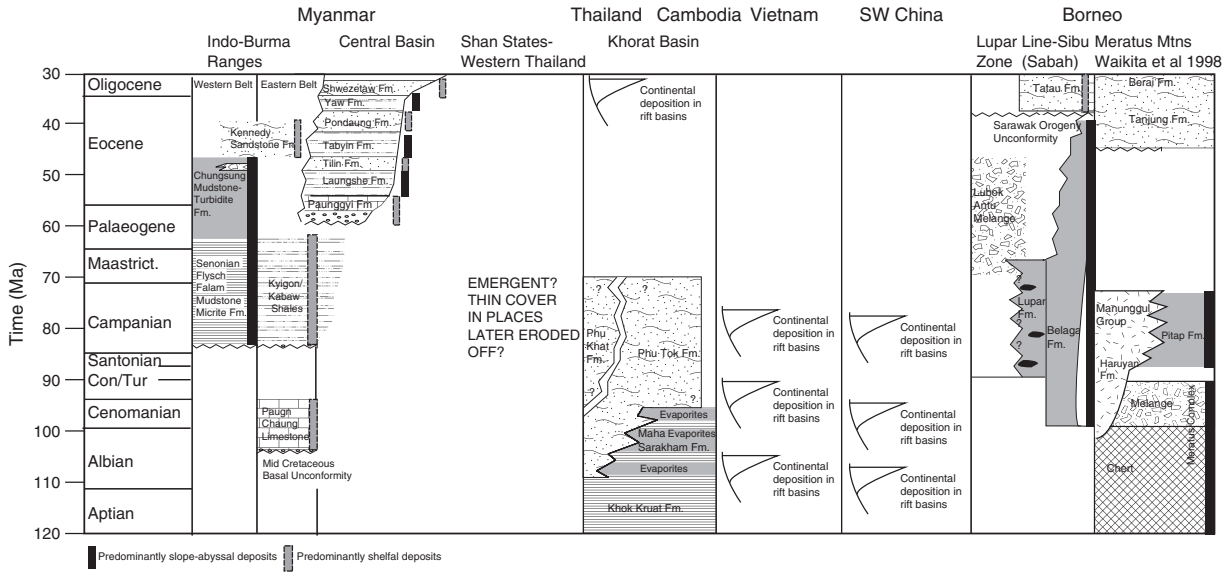


Fig. 5. Comparison of stratigraphic events in SE Asia, based on Sattayarak (1985), Stokes (1988), Mitchell et al. (2010), Racey et al. (1996), Hutchison (2005), Waikita et al. (1998). The Phu Khat Formation of the Nakhon Thai basin appears to be the lateral equivalent of the Phu Tok Formation found in the Khorat Plateau area.

relationship is described for the Nimi Formation in the Naga Hills, where the tectonic setting as interpreted as an arc-trench gap (Agrawal and Ghose, 1986). However, the Nimi Formation is described as a neritic deposit, not deep marine (Ghose et al., 2010). The Paung Chaung and Nimi formation limestones overlie continental or island arc crust in a post-collisional setting, the buoyancy of the thickened crust post-collisional setting, and westward extent of the newly accreted Mawgyi Terrane would preclude the Paung Chung Limestone residing in an arc–trench location. More likely the intrusions were serpentinised during Late Cretaceous–Palaeogene deformation, and they are indicative of a later tectonic setting than their depositional one.

2.2.4. Late Cretaceous–Palaeogene granites of Thailand and Myanmar

A geochemical and U–Pb dating study of granitoid plutons along the western margin of the Shan Plateau, in eastern Myanmar

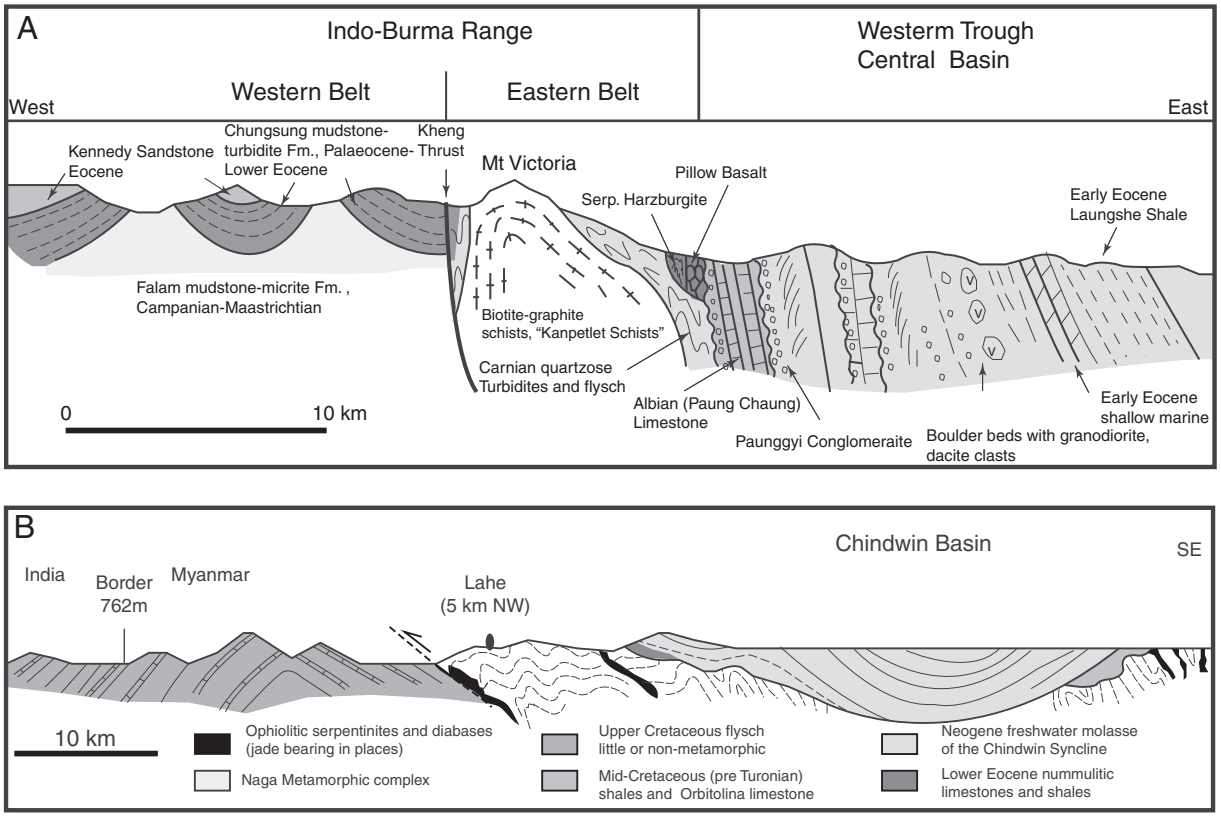


Fig. 6. A) Cross-section through the Chin Hills area modified after Mitchell (1993), B) Cross-section through the Naga Hills area, modified from Brunnschweiler (1966).

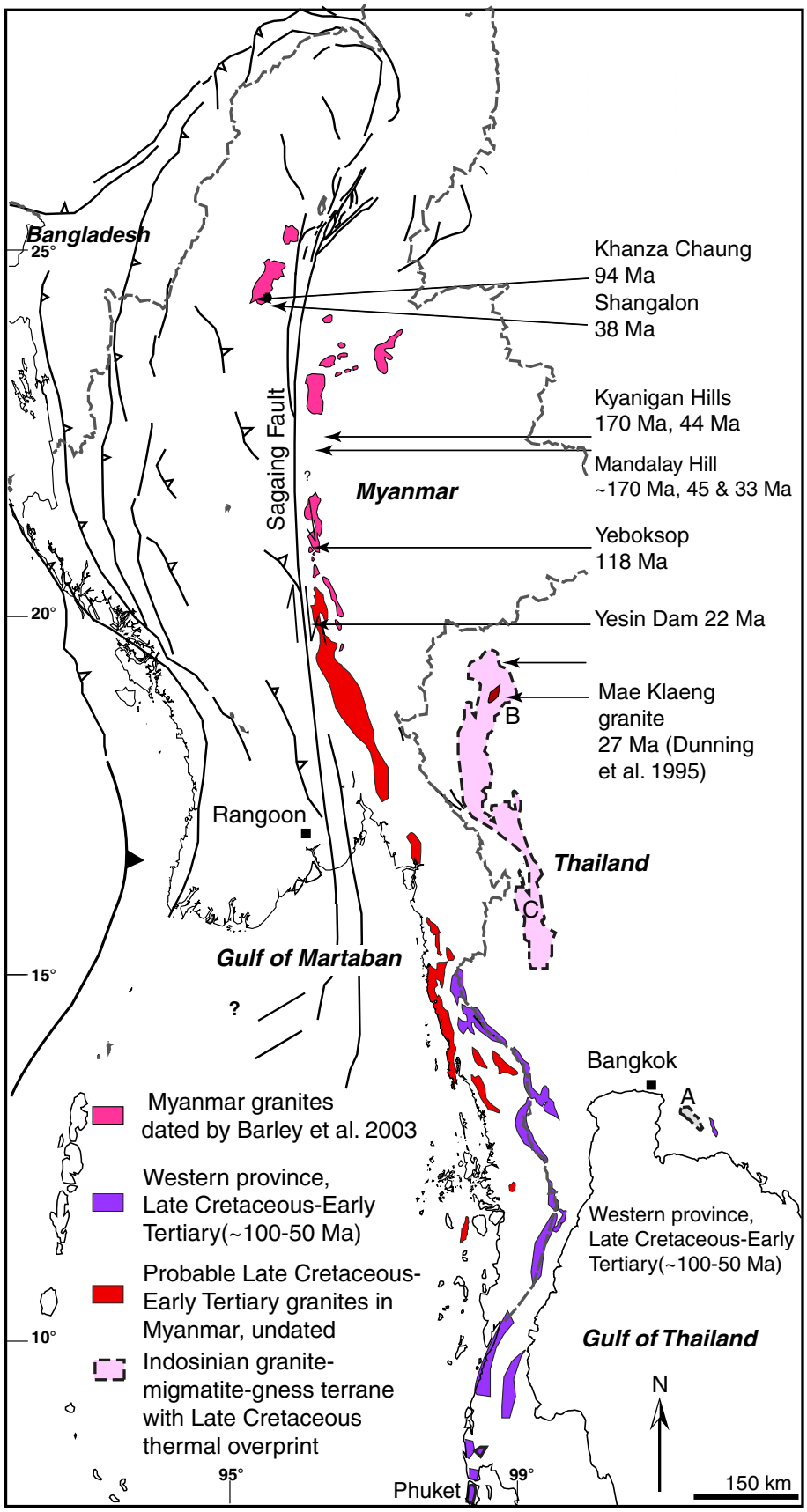


Fig. 7. Distribution of Cretaceous granites in Thailand and Myanmar. U–Pb Monazite ages from Indosinian (Late Triassic–Early Jurassic) granite-migmatite-gneiss complexes, which indicate Late Cretaceous thermal overprint: A = 79.1 ± 2.2 Ma and 76.8 ± 2.6 Ma,; Geard, 2008; B = 80–60 Ma partial lead loss from zircons (Hansen and Wemmer, 2011), C = 84 Ma and 72 Ma, Doi Inthanon, (Macdonald et al., 2010).

(Fig. 7), was conducted by Barley et al. (2003). They concluded the data was consistent with the presence of a mid-Cretaceous to earliest Eocene magmatic arc. This arc, up to 200 km wide, was part of a much larger Andean-type margin that extended along the Eurasian margin from Pakistan to at least Thailand, and more controversially to Sumatra (Pakistan–Ladakh–Myanmar–Sumatra arc or PLMS Arc; for a more detailed discussion see Section 2.3).

In western Thailand granites (including those around Phuket) have yielded $^{40}Ar/^{39}Ar$ ages that range between 50 and 88 Ma (Charusiri et al., 1993). Recent U–Pb and $^{40}Ar/^{39}Ar$ dating by Watkinson et al. (2011) has shown that $^{40}Ar/^{39}Ar$ ages of micas in undeformed granites in the peninsular in some cases are associated with cooling shortly after pluton emplacement, while in other cases may indicate exhumation ages associated with Eocene deformation, or possibly the effects of hydrothermal activity. Consequently the implied Eocene emplacement ages of granites from $^{40}Ar/^{39}Ar$ dating from older studies can be regarded as unreliable. Nevertheless LA-ICPMS zircon U–Pb dates of three samples from Phuket collected by Charusiri (pers. Comm., 2010) yielded mean ages of 54.9 ± 1.4 Ma, 58.17 ± 0.97 Ma, and 82 ± 2.4 Ma (Table 1), which support the previously determined ages. Watkinson et al. (2011) obtained U–Pb SHRIMP ages for zircons obtained from four granites associated with the Ranong Fault zone, which indicate the large plutons were emplaced between about 80 and 71 Ma, with more minor granite intrusion occurring between 48 and 37 Ma (some were emplaced during dextral shear along the Khlong Marui and Ranong Fault zones). One $^{40}Ar/^{39}Ar$ age from hornblende in a migmatite suggests anatexis around the Ranong Fault zone was occurring around 88 Ma (Watkinson et al., 2011).

Charusiri et al. (1993) emphasized the S-type nature of the granites, and considered a continent–continent collision was necessary to generate them, whereas Barley et al. (2003) noted that in Myanmar I-type granitoids are dominant, and they could be explained by an Andean-type margin setting (i.e. the PLMS arc). The Phuket granites have mixed I and S type characteristics, which Pollard et al. (1995) considered could be derived by extensive crystal fractionation of an I-type melt, rather than implying partial melting of a sedimentary protolith.

In addition to the intrusion of Cretaceous granites, a thermal event has also been identified within Late Triassic–Early Jurassic (Indosinian) granite-migmatite-gneiss complexes in Thailand. In older radiometric dating campaigns this event (placed between 80 and 60 Ma) has been observed as partial lead loss from zircons and a consequent lowering of apparent ages (Darbyshire, 1988; Charusiri et al., 1992; Ahrendt et al., 1997; Hansen and Wemmer, 2011). More recently these observations have been corroborated by U–Pb dating of monazite crystals, which indicates Late Cretaceous metamorphism to upper amphibolite facies (84 Ma and 72 Ma, Doi Inthanon, Macdonald et al., 2010; 79.1 ± 2.2 Ma and 76.8 ± 2.6 Ma, Chonburi-Rayong area, Geard, 2008, see Fig. 7 for locations). In the Mogok Metamorphic belt in eastern part of Myanmar, Searle et al. (2007) found evidence for migmatisation, ductile deformation and biotite granite sill injection associated with Early Palaeogene regional metamorphism.

2.2.5. Preferred tectonic model

The model of a Cenozoic West Burma Block collision with Indochina has largely been discounted in regional plate reconstructions (e.g. Hall et al., 2009; Hall 2012; Metcalfe, 2011), and instead the block seems to have been part of Sundaland since the Triassic, as part of the Indochina Block (e.g. Barber et al., 2005; Barber and Crow, 2009). The general theme of the Barber and Crow (2009) model is followed in this paper. However, from the review in this section (2) there are several details that do not fit their model: 1) the Andaman sea does not appear to show evidence for the continuity of the arc in this region, 2) the Aptian Unconformity in the Chin Hills suggests collision was older (following the Mergui Arc model of Mitchell et al., 2007) than in Sumatra.

The Andaman Sea ophiolites contain ophiolitic chromites, and high Mg olivine and orthopyroxenes, which suggest a boninite source, in a supra-subduction zone setting (Ghost et al., 2009; Pal, 2011). Consequently the formation of oceanic crust above an eastwards subduction zone appears to have occurred in the Andaman Sea region during the mid Cretaceous. The strong topography of the Sewell and Alcock Rises probably developed when Cretaceous oceanic crust was deformed by Miocene strike–slip and extensional tectonics, accompanied by volcanic eruptions. are here inferred to overlie Cretaceous oceanic crust. The likely presence of Cretaceous oceanic crust between the Woyla and Mawgyi terranes, and the difference in timing of deformation (Fig. 4) suggests instead of being a single continuous terrane, they were two separate island arcs that underwent diachronous collision with the Eurasian margin during the Cretaceous (following the timing for the Mergui Arc of Mitchell et al., 2007, and the Woyla Nappe, Barber et al., 2005; Barber and Crow, 2009).

A widespread period of ophiolite emplacement is seen during the Palaeogene: ophiolites were thrust within Palaeogene sediment in the Andaman-Nicobar Islands (Pal, 2011), throughout the Western Belt, including the Naga Ophiolite (Socquet et al., 2002; Acharyya, 2007). In the Naga Hills eclogites and blueschists (bulk compositions plot in the ocean floor tholeiite field of a Ti-Zr-Y diagram) are present in sheared lenses and layers intercalated with basaltic and ultramafic units (Chatterjee and Ghose, 2010). The Naga Ophiolite is thrust westward over the deep marine, Eocene Disang Formation, shallow marine deposits of the mid-Eocene Phokphur formation overlie the ophiolite. On this evidence Acharyya (2007) infers that the Naga Ophiolites emplacement ended just prior to the mid Eocene.

Fig. 8 presents a geodynamic model for the Myanmar area during the Cretaceous and Palaeogene based on the preferred models in this paper. The model incorporates the Mid Cretaceous arc collision model discussed by Barber and Crow (2009), with the timing discussed by Mitchell et al. (2007; Fig. 8A, B). Following collision of the Mawgyi Terrane the margin may have undergone extensional collapse, bringing the western part of the continental margin below sea level and allowing deposition of the Paung Chaung Limestone across eroded remnants of the suture zone (Fig. 8C). Also at this time (~95 Ma) subduction was renewed with oceanic crust, and led to the generation of Andaman Sea age oceanic crust in a supra-subduction zone setting. Modern understanding of ophiolite emplacement indicates most ophiolites developed in forearc lithosphere during subduction initiation in a supra-subduction zone setting (Whattam and Stern, 2011). This setting seems geochemically appropriate for the Andaman Sea ophiolite, and could be extended northwards, to cover subduction initiation following collision of the Mawgyi arc with the West Burma Block. As the subducted oceanic slab deepened the renewed subduction hydrated the upper Asthenosphere, causing upper Mantle convection, high heat flow, migmatites were generated in the lower crust, and arc granitic plutions were emplaced (Fig. 8, Section 2.2.4). Subduction led to accretionary prism deformation within the Western Belt sediments.

Both Socquet et al. (2002) and Acharyya (2007) inferred a Palaeogene West Burma Block (or its equivalent) collision with eastern Myanmar to explain the Palaeogene ophiolite obduction. This seems the simplest explanation, except as discussed above, there is no evidence for the Western Burma Block existing as a separate

Table 1
Summary table for unpublished U–Pb dates from Phuket granites, Thailand, P. Charusiri pers. comm. (2010), dating conducted by Univeristy of Tasmania.

T4	Toan Sai 1–1,Phuket , 54.9 ± 1.4 Ma	08° 01′ 37″ N	98° 21′ 51″ E
T5	KT-1,Phuket, 82.3 ± 1.0 Ma	07° 48′ 53″ N	98° 18′ 14″ E
T6	KR-PT2-1,Phuket, 80.8 ± 2.4 Ma	07° 52′ 50″ N	98° 17′ 14″ E

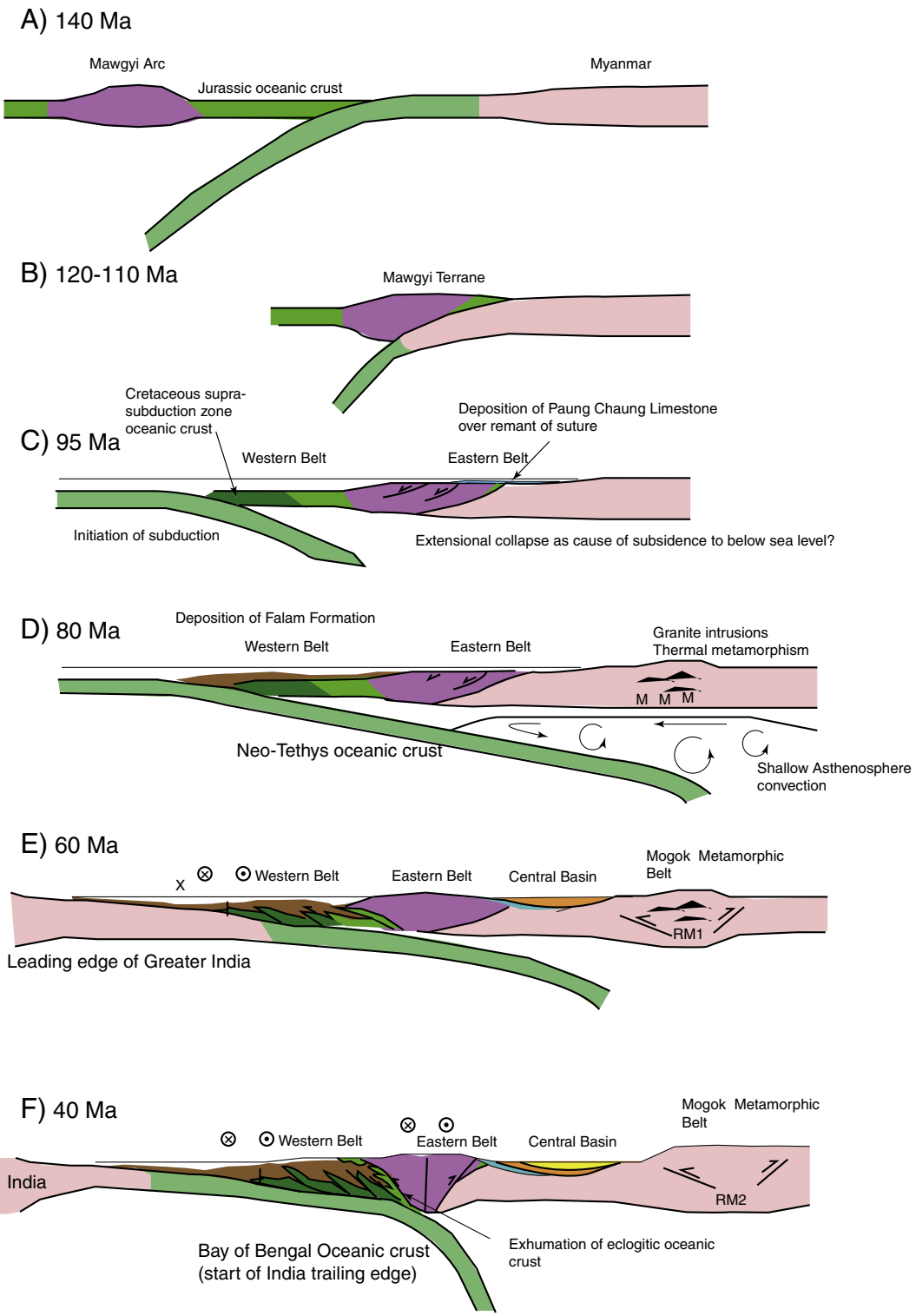


Fig. 8. Geodynamic model for the evolution of Myanmar during the Late Cretaceous and Palaeogene. The section is schematic, but is appropriate for a section approximately through the Chin Hills. RM1 = regional metamorphic event 1 (Searle et al., 2007), RM2 = regional metamorphic event 2 (Searle et al., 2007).

continental fragment at this time, and the present data, while far from comprehensive, suggests emplacement of the West Burma Block west of Sibumasu occurred along strike–slip faults in the Triassic (Barber and Crow, 2009). The tectonic models of Socquet et al. (2002) and Acharyya (2007) only show cartoon representations of the Western Burma Block collision. No properly scaled, balanced cross-sections at a regional scale have been constructed across the Indo-Burma Ranges. The most comprehensive properly scaled regional cross-section is provided by Maurin and Rangin (2009). In this cross-section it is apparent that it is impossible to include a thick (20–30 km) region of continental crust below the Western Belt sediments of the Indo-Burma Ranges, there is no space available. Consequently no microcontinental block collision is shown in Fig. 8. Instead it is suggested that the leading edge of Greater India began interacting

with the margin in the Palaeogene. Convergence was highly oblique, and consequently obducted oceanic crust emplaced onto the leading edge of India could have been sliced up by strike slip faults, and transported northwards, and lost in the Himalayan collision zone.

Plate reconstructions by Van Hinsbergen et al. (2011) indicate the key period when the leading edge of India passed by Mynamar was between 50 and 60 Ma. The glancing collision may have been responsible for the early Palaeogene metamorphic event idenfied by Searle et al. (2007) in the Mogok Metamorphics (indicated as RM1, Fig. 8E). After this time oceanic crust which forms the Bay of Bengal began to subduct beneath Myanmar and the continental edge had moved further north. It is suggested that the oblique convergence was accommodated by a progressively broadening zone of strike–slip deformation across Myanmar with time, as shown in Fig. 8E and F (see Maurin and Rangin, 2009 for a more detailed history of the oblique-slip development).

2.3. Central Indonesian Accretionary Collision Complex

Following the PLMS arc eastwards a complex area of deformation is reached in South Java, where the Cretaceous subduction zone trended E–W to NE–SW. This area is known as the Central Indonesian Accretionary Collision Complex (CIACC). The CIACC extends from southwestern Java (Ciletuh complex) through eastern Borneo (Meratus Complex) and Western and Central Sulawesi (Pompangeo Complex) (e.g. Waikita et al., 1998; Parkinson et al., 1998; Kadarusman et al., 2007; Fig. 9). It comprises scattered outcrops of mélanges, pillow basalts, dismembered ophiolites, high-pressure metamorphic rocks, cherts, and occasionally garnet-lherzolite ecologites and blueschists, more typical continental crust schists and gneisses are also present (Waikita et al., 1998; Parkinson et al., 1998; Kadarusman et al., 2007; Fig. 9). High pressure-low temperature metamorphism occurred between 110 and 120 Ma (Waikita et al., 1998; Parkinson et al., 1998). The accretionary complexes are characteristically unconformably overlain by Late Cretaceous deepwater sedimentary and volcanic sequences (Heryanto and Sanyoto, 1994; Heryanto et al., 1994; Waikita et al. 1998; Parkinson et al., 1998; Fig. 5).

The mid-Cretaceous stage of tectonism marks exhumation and erosion of ultramafic rocks, mélanges and high-pressure schists from depths in excess of 60 km, (Parkinson et al., 1998; Kadarusman et al., 2007). Parkinson et al. (1998) interpreted this event as the collision of a Gondwana-related crustal fragment, which led to rapid exhumation of the accretionary complexes. The subduction zone consequently migrated oceanward (southward) to facilitate continued plate convergence, and the magmatic front and forearc basin migrated ~200–300 km southwards. In the Meratus Complex this southwards migration is marked by the unconformably overlying Late Cretaceous volcanic Haruyan Formation (Late Aptian to Cenomanian, Yuwono et al., 1988) and the turbiditic Pitap Group (Waikita et al., 1998). The overlying Manunggul Group Formation

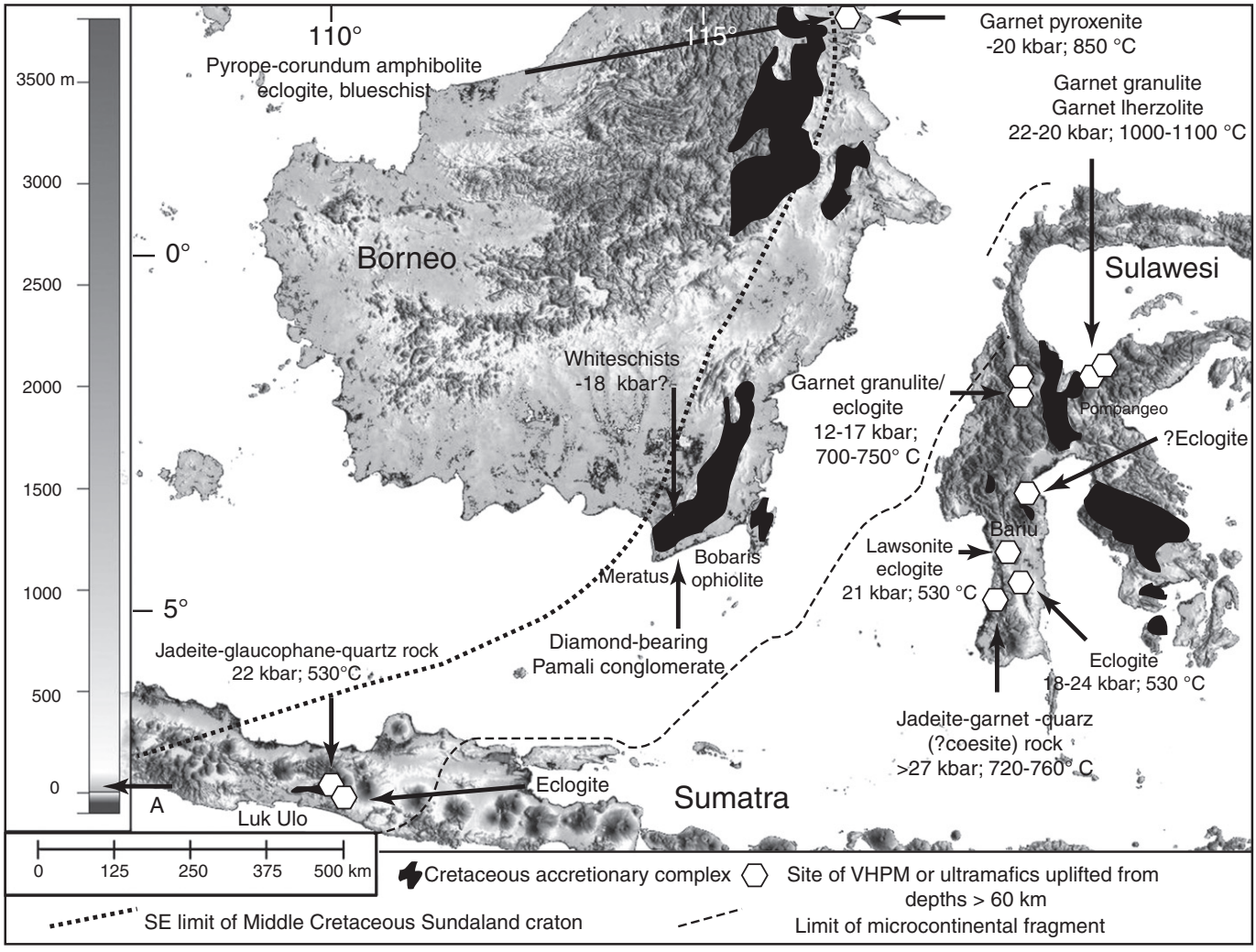


Fig. 9. Key tectonic elements of the Central Indonesian Accretionary Collision Complex (CIACC), modified from Parkinson et al. (1998). A = to the Ciletuh Complex (~100 km west).

comprises extrusive volcanics, with intrusive dykes of basaltic to dacitic composition, and gabbroic to granitoid stocks, K–Ar dates range between 87 ± 4 Ma and 72 ± 4 Ma (Sikumbang and Heryanto, 1994; Yuwono et al., 1988). Note Waikita et al. (1998) lump the Manunggul Group with the Haruyan Formation. Geochemistry indicates these volcanics are subduction related (Yuwono et al., 1988). More recent work indicates the high pressure assemblages were exhumed in subduction channels, and that significant differences in P–T paths and conditions between the HP and UHP complexes suggest there was not a single, simple subduction-collision event (Kadarusman et al., 2007). Hall et al. (2009) and Hall (2012) place the termination of subduction by collision with West Sulawesi between 92 and 80 Ma, although Waikita et al. (1998) suggest final collision was during the Palaeocene. There was also a short-lived Late Palaeocene to Early Eocene subduction system in West Sulawesi and Sumba (see review in Clements and Hall, 2011; Langi Volcanics and Bua Formation, Fig. 4).

2D industrial seismic reflection data across the Makassar Straits (e.g. Hall et al., 2009) occasionally images a thick reflective sequence below the base of the syn-rift section (probable Middle Eocene age). These pre-rift reflections can be over 2 s two way travel time thick (i.e. 4 + km thick), they appear to be mildly folded and thrusted. Hence they can be interpreted as representing thick Late Cretaceous forearc deposits (following the model of Parkinson et al., 1998). Some of the Eocene half-graben bounding normal faults are low-angled (~20°) and appear to reactivate older thrusts within the inferred Late Cretaceous–Early Palaeogene deposits.

2.4. Tectonic evolution of the western and southern margins of Sundaland

Sedimentary rocks of Late Cretaceous–Early Palaeogene age are very scarce along the western and southern margins of Sundaland, except for the predominantly deepwater deposits of the CIACC and Myanmar. Much of the Late Cretaceous–Early Palaeogene tectonic history is based on the geochemistry and radiometric dating of granitoid intrusions. Hall (2009) notes that of 299 radiometric ages from Cretaceous and Palaeogene granites, 215 are older than 80 Ma. He conceded that granites older than about 80 Ma could be explained as a product of subduction of the oceanic crust of the India Plate under SE Asia. However, the decline in plutonic activity, and virtual absence of volcanic activity in Java, Sumatra and minor occurrence in SE Borneo suggested to Hall (2009), that the younger granites may have resulted from collisional thickening not subduction. This thickening was a consequence of the mid-Cretaceous accretionary events described in Section 2.1. As discussed by Hall et al (2009) and Hall (2012) there exists considerable diversity in models for the Late Cretaceous–Palaeogene development of the area. Some authors assume subduction along the Sundaland active margin was continuous into the Late Mesozoic and Cenozoic (e.g. Yuwono et al., 1988; Metcalfe, 1996; Soeria-Atmadja et al., 1994, 1998; Abdullah et al., 2000; Sribudiyani et al., 2003). However, others have either explicitly or implicitly suggested tectonic quiescence along parts of the Sundaland margin from the Late Cretaceous–Eocene, and a renewal of subduction during the Eocene (Hamilton, 1979; Hasan 1990, 1991; Daly et al., 1991; Bon et al., 1996; Carlile and Mitchell, 1994; Waikita et al., 1998; Parkinson et al., 1998; Hall et al., 2009; Hall 2012).

Hall et al. (2009) and Hall (2012) proposed that between 70 Ma and 90 Ma tectonic activity east and west of the 90E ridge became significantly different. East of the ridge (i.e. the margin from Sumatra to southern Borneo and Western Sulawesi) subduction was slow, or had ceased. West of the ridge subduction was very active and facilitated the rapid northwards movement of India. Hence in Myanmar the Late Cretaceous plutonic activity could still be associated with subduction in agreement with the conclusions of Barley et al. (2003), whereas to the SE plutonic activity is associated with collisional crustal thickening. As discussed by Hall et al. (2009) the absence of a volcanic arc, and reduced plutonic activity during the Late Cretaceous provides a compelling argument that subduction ceased or slowed for a considerable period following mid-Cretaceous accretion. This appears to be particularly the case for Sumatra and Java. Subduction, perhaps at a significantly reduced rate may, however, have continued in the vicinity of the Borneo–West Sulawesi margin, where there is some evidence for Late Cretaceous arc activity and deposition in a foredeep basin setting (e.g. Parkinson et al., 1998, see Section 2.3).

3. Tectonics of the interior of Sundaland during the Late Cretaceous–Palaeogene

3.1. Thailand

3.1.1. Khorat Plateau

The largest occurrence of Cretaceous continental-shallow marine sedimentary rocks in Indochina is found in the Khorat Plateau, which lies within NE Thailand and adjacent areas of Laos and Cambodia (Figs. 10 and 11). The plateau is composed of flat-lying to tilted and folded continental red-bed sediments of predominantly Late Jurassic–Early Cretaceous age (e.g. Racey et al., 1994, 1996; Racey and Goodall, 2009; Buffetaut et al., 2006, 2009). Passing to the west, north and south there are patchy occurrences of Mesozoic sediments throughout Thailand and the Gulf of Thailand (e.g. Saengsrichan et al., 2009, 2011; Fig. 10). To the west and south these sediments tend to be more marine, and include lower Jurassic age sediments (e.g. Saengsrichan et al., 2009, 2011; Fyhn et al., 2010). On the western fringes of the Khorat Plateau there are indications of occasional marine incursions (e.g. Hahn, 1982). Further north is the thick succession of non-marine red beds of Jurassic to early Cenozoic age in the Lanping-Simao region of western Yunnan (Bureau of Geology and Mineral Resources of Yunnan Province, 1990; Fig. 10). Hence there is good reason to suspect that prior to deformation, uplift and erosion, Mesozoic rocks were present over a very wide area of Thailand, Laos, Myanmar, Cambodia, Vietnam, the Gulf of Thailand and Malaysia (Fig. 10) and deposition within the basin passed from continental into shallow marine conditions at least to the west. Consequently it can be inferred the basin was connected, at least on occasions, to the Tethys ocean to the west, and that the region lay near sea level. The time of deposition of the Jurassic–Early Cretaceous is earlier than the period considered in this paper, but the continued deposition during the Late Cretaceous, and episodes of uplift, erosion, and deformation that dismembered this great sedimentary basin are a key part of the Late Cretaceous–Palaeogene history.

The depositional setting and stratigraphy of the Khorat Group has been reviewed by Racey (2009) and Racey and Goodall (2009). The Khorat Group comprises a 3–4 km thick sequence of Late Jurassic–Cretaceous continental red beds (Carter and Bristow, 2003; Racey and Goodall, 2009). The E–W trend to isopach contours for the Khorat Group (Phu Phan Formation to the base of the Khorat Group, Fig. 10) follows the E–W trend of the underlying Permian and Triassic basins, and maybe a relatively local effect related the pre-existing crustal architecture (Booth and Sattayarak, 2011). The overlying Khok Kruat Formation displays more NNE–SSW trending isopach contours that thicken eastwards. This reconfiguration suggests a significant reconfiguration in the regional tectonics during the Aptian, with regional thickening (and tilting?) towards Vietnam.

The age of the youngest Cretaceous section in the Khorat Group is uncertain. The two youngest formations are the Maha Sarakham Formation (Thangkon Formation in Laos) and the Phu Tok Formation (Saysomboun Formation in Laos). Sattayarak et al. (1991) proposed the Maha Sarakham Formation was deposited sometime between the Albian and Cenomanian, this dating is supported by K/Ar (93.7 ± 2.7 Ma) and Rb/Sr (95.8 ± 15 Ma) ages of potash and rock

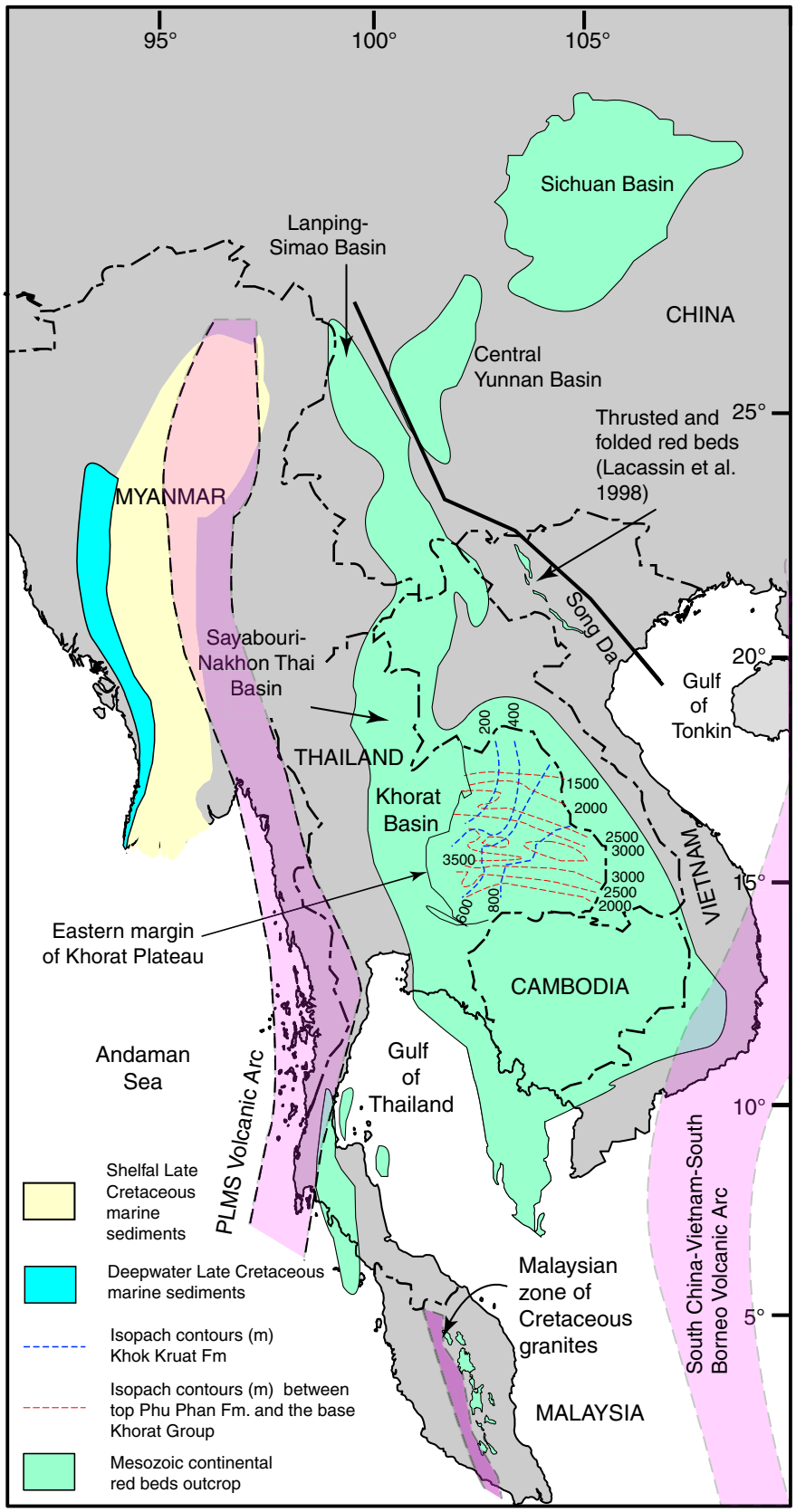


Fig. 10. Regional setting of the Khorat Basin and related Mesozoic red bed occurrences. The approximate location of Cretaceous volcanic arcs is also shown. The occurrence of marine Cretaceous sedimentary rocks in Myanmar (Fig. 3) is also shown. Partly based on Racey (2009) and Fyhn et al. (2010). PLMS = Pakistan-Ladakh-Myanmar–Sumatra arc. Isopach contours from Sattayarak and Polachan (1990).

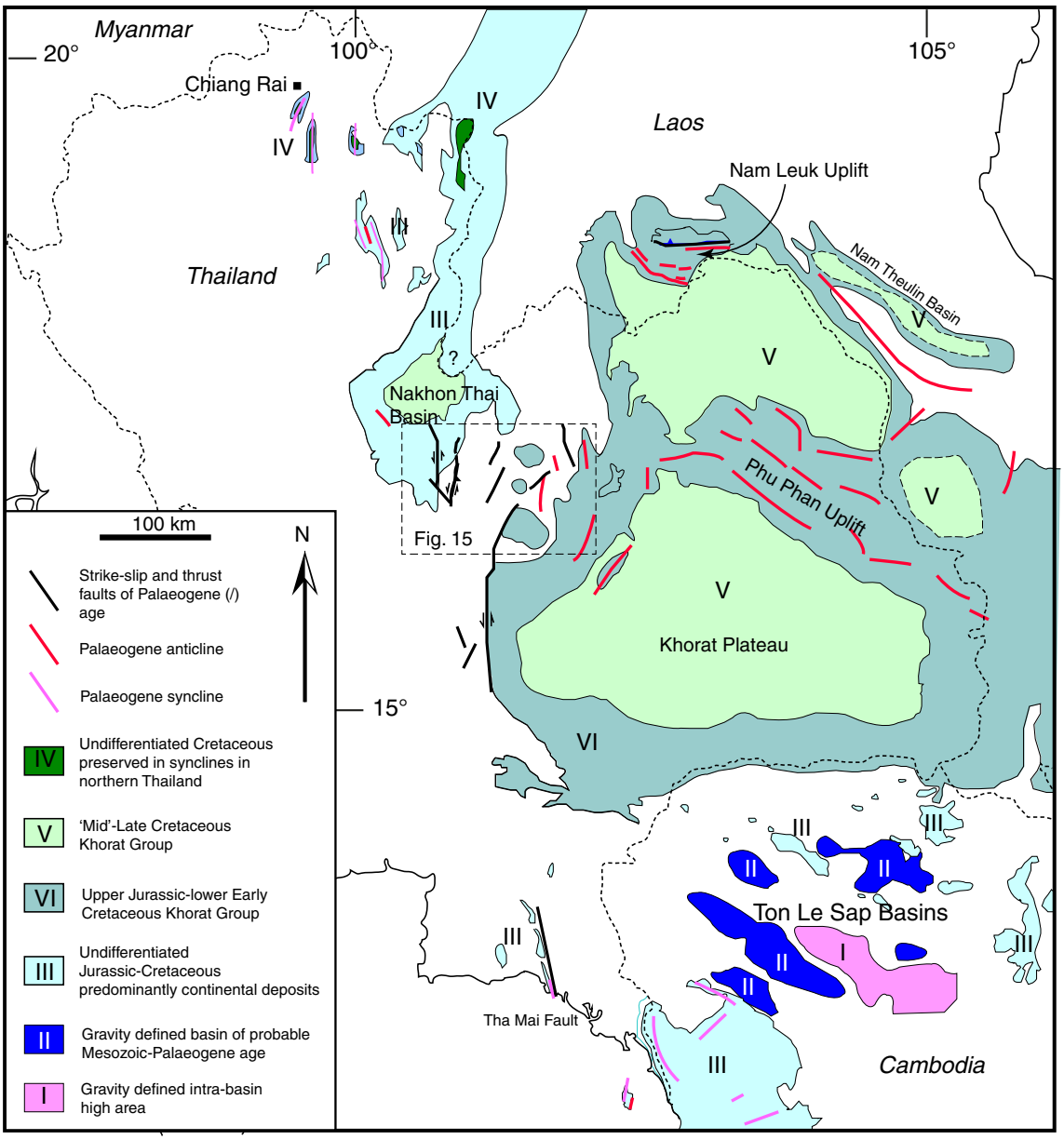


Fig. 11. The occurrence of Cretaceous–Palaeogene sedimentary rocks and structures in Thailand, Laos and Cambodia. Based partly on Lovatt Smith et al. (1996), and Department of Mineral Resources (1999).

salt deposits (Hansen et al., 2002; Fig. 5). The Phu Tok Formation is generally regarded as being of Late Cretaceous age (Lovatt Smith et al., 1996). In the Nakhon Thai sub-basin the Phu Tok Formation directly overlies the Khok Kurat Formation (Sattayarak, 1985). However, a drill core has established that it overlies the Maha Sarakham Formation (Charusiri et al., 2006). This drill core, and the outcrop relationships indicate an interfingering relationship of the Phu Tok Formation with the Maha Sarakham Formation (Hasegawa et al., 2010). Based on these observations a palaeomagnetic study suggests the Phu Tok Formation ranges from 126 Ma to 84 Ma (Hasegawa et al., 2010). However, the presence of mammal footprints of probable Cenozoic age within the Phu Tok Formation (Le Loeuff et al., 2005) needs to be addressed before the formation is definitively assigned to the Cretaceous.

The Maha Sarakham Formation (Thangon Formation in Laos) is notable for containing a considerable thickness of evaporites deposited in three main cycles. In Laos these evaporites horizontally onlap the Nam Leuk Uplift (Fig. 11), suggesting the uplift marks a mid-Cretaceous deformation event that caused restriction of the basin and favoured evaporite forming conditions (Lovatt Smith et al., 1996). The mid-Cretaceous event in the Khorat Plateau (at ~100 Ma) produced uplift in the western area, which also helped restrict the basin (Booth, 1998). However, there is little indication on seismic reflection data of the mid-Cretaceous event within the uplifted and folded areas of the Khorat Plateau (Booth and Sattayarak, 2011). Within the southern part of the Khorat Plateau there are a few areas where the Phu Tok Formation is present. However, probably some 1000–1250 m of Late Cretaceous section once covered the whole of the Khorat Plateau area, and has subsequently been eroded (Booth and Sattayarak, 2011).

Modelling of apatite fission track data to determine the timing of maximum burial temperatures of various Khorat Group Formations by Upton (1999) yielded the following results: Phu Kradung Formation (~70–45 Ma, 3 samples). Phra Wihan Formation (~70–60 Ma, 6

samples), Phu Phan Formation (50–48 Ma, 1 sample), Khok Kruat Formation (~70–65 Ma, 1 sample). It is uncertain whether the ages represent a thermal event post-burial, or the time of maximum burial or a combination of the two. If it is burial, then deposition until the latest Cretaceous and probably into the early Palaeogene is implied by the results. Fig. 12C shows modeled apatite fission tracks for the Khorat Group by Upton (1999), and illustrates that the samples arrive at maximum burial around the Cretaceous–Cenozoic boundary. Fig. 13 is a burial history plot using maximum formation thicknesses, the plot shows a significant reduction in subsidence rate around the 'mid' Cretaceous, with subsidence continuing into the Late Cretaceous.

The Lower Nam Phong Formation (Late Triassic), Upper Nam Phong Formation (Late Jurassic) and Khorat Group were all deposited in a broad basin with a sag-like geometry (Booth and Sattayarak, 2011). The formations appear to be conformable on seismic reflection data. However, apatite fission track provenance ages and palynology (Carter and Bristow 2003; Racey and Goodall, 2009) indicate the Early Jurassic is absent. Prior to these works it was thought that the Nam Phong Formation and Khorat Group were deposited in a thermal subsidence basin that developed in response to Late Triassic extension (e.g. Lovatt Smith et al., 1996). However, the absence of the Early Jurassic would not be predicted for a classic post rift basin. Consequently it has been proposed that the Late Jurassic–Cretaceous basin formed in response to flexural subsidence induced by loading from an Andean-type orogenic belt to the east (e.g. Carter and Moss, 1999; Carter and Bristow, 2003), or a combination of initial thermal subsidence followed by foreland flexuring (Cooper, et al. 1989). From regional mapping of trends (Fig. 10) it appears that the Cretaceous red bed basins are broadly bound by the PLMS arc to the west and the South China–Vietnam–South Borneo arc to the east, hence the Andean-type orogenic belt model of Carter and Moss (1999), and Carter and Bristow (2003), should be modified to consider the effects of two bounding arcs.

The main objection to flexural subsidence due to a mountain belt load is that the geometry of the foreland basin would produce an asymmetric basin in cross-section that thickens towards the mountain front, whereas the geometry of the Khorat Group basin is more of a symmetric saucer-shaped depression that thickens towards the middle of the basin. However, in Andean-type foreland basins subsidence is not necessarily just related to flexural loading by thickened crust, but also to large-scale geodynamic forcing. Mantle flow in response to subduction geometry and density contrasts can considerably modify the foreland basin geometry producing a much broader, more saucer shaped basin (Dávila et al., 2010), as these authors discuss for the Pampean foreland in Argentina. Their model appears to be applicable to the Khorat Basin.

The thick, extensive Late Triassic deposits and Khorat Group of NE Thailand are deformed by a number of broad folds (Kozar et al. 1992; Chaviroj, 1997). The folds trend N–S and WNW–ESE (Figs. 11 and 14), and are thick-skinned features associated with inversion of Permian

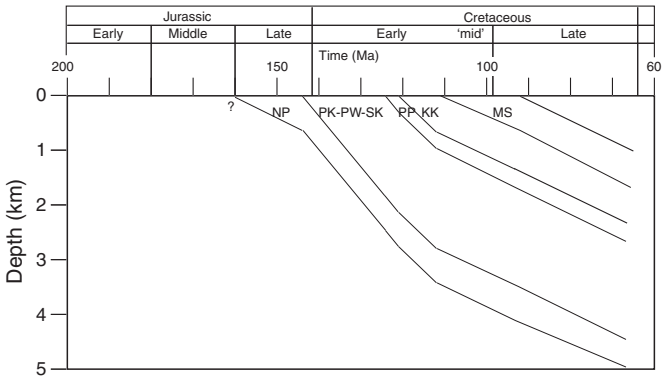


Fig. 13. Burial history for the Khorat Plateau using maximum formation thicknesses from Racey et al. (1996), and ages from Racey and Goodall (2009). NP = Nam Phong Formation (1,465 m thick), PK = Phu Kradung Formation (1,100 m thick), PW = Phra Wiang Formation (136 m thick), SK = Sao Khua Formation (720 m thick), PP = Phu Phan Formation (183 m thick), KK = Khok Kruat Formation (709 m thick), MS = Maha Sarakham Formation (1000 m thick).

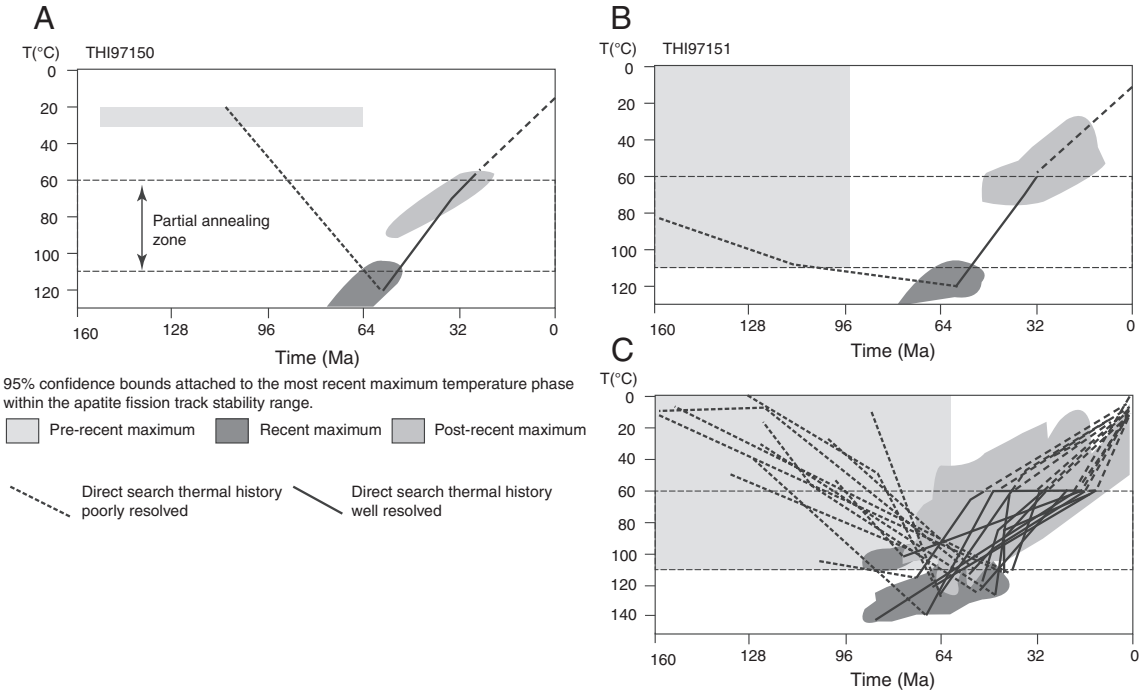


Fig. 12. Results of apatite fission track analysis for the Khorat Plateau area (samples from the Khorat Group) by Upton (1999). A) Sample TH97150, B) sample TH97151, C) all modelled samples for the area. The location of A) and B) is shown in Fig. 15.

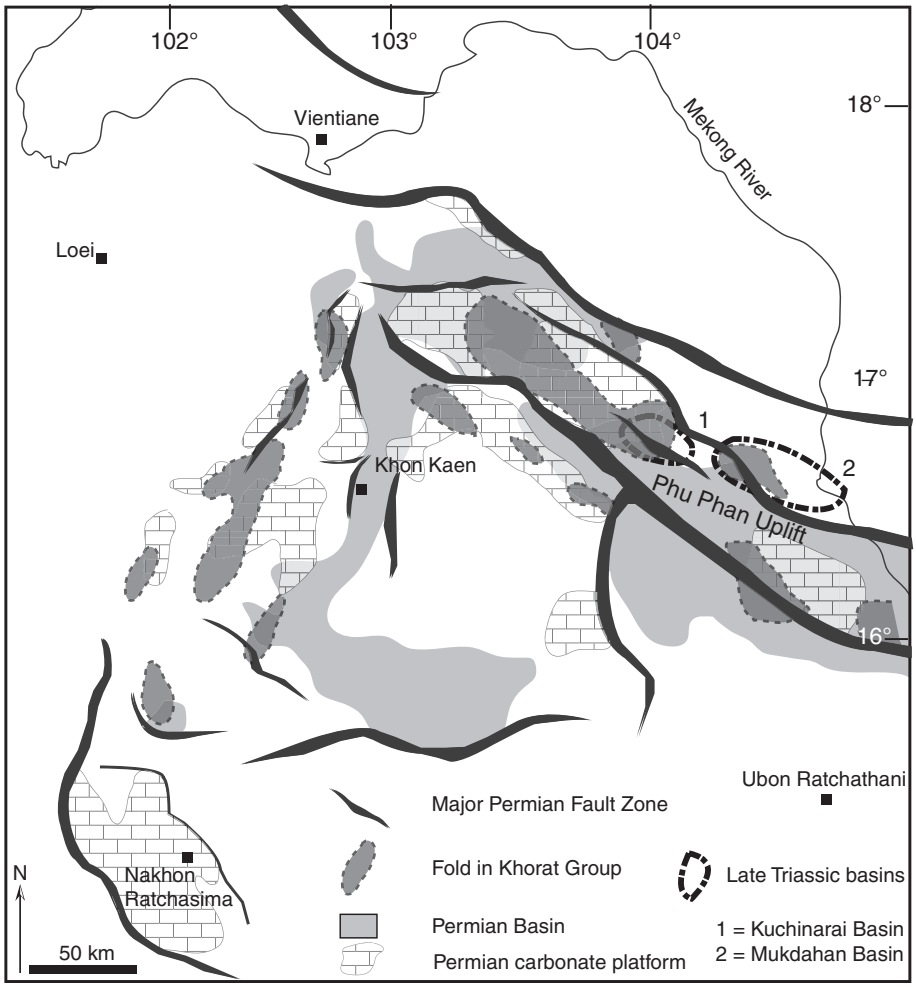


Fig. 14. Map showing the subsurface trends of Permian Basins and related faults, overlain by folds in the Khorat Group (based on Booth and Sattayarak, 2011). Just two of the numerous Late Triassic rift basins are also shown as examples of their control on the location of Cenozoic anticlines.

or Triassic extensional basins, and/or reactivated Indosinian thrusts (e.g. Cooper et al., 1989; Kozar et al., 1992; Booth 1998; Morley et al., 2011; Booth and Sattayarak, 2011). Consequently the variable fold trends reflect the orientation of older faults that controlled the inversion (e.g. Kozar et al., 1992). A map with the distribution of Permian basins and associated faults and the location of folds in the Khorat Group is shown in Fig. 14. The figure demonstrates a strong relationship between the location and orientation of the folds, and Permian basins with their associated faults. The Permian basins in places underwent strong folding, thrusting and inversion, sometime between the latest Permian and Late Triassic (Booth and Sattayarak, 2011), consequently the Cenozoic folds are not located only over normal faults inverted during the Cenozoic, but also over older compressional features. The WNW–ESE trend is best developed in the Phu Phan Uplift (Figs. 11 and 14) and coincides with the dominant strike direction of underlying Permian and Triassic rift basin boundary faults (Kozar et al., 1992).

3.1.2. Central and Western Thailand

The Upper Jurassic–Cretaceous Khorat Group red beds are up to 4000 m thick (Racey et al., 1996; Racey, 2009) and thin gradually approaching the northern and southern margins of the plateau, and to a lesser degree towards the western margin (Fig. 10). Hence the Khorat Group must have extended westwards over a very large area and then been subject to extensive erosion during the Cenozoic (Racey et al., 1997; Upton, 1999). Isolated remnants of Khorat Group, other Mesozoic sedimentary rocks and marine Jurassic rocks of the Mae Moei Group are found scattered in synclines in northern Thailand, around Chainat in central Thailand and in southern Peninsular Thailand (Fig. 11). In northern Thailand folding was probably more intense than in the Khorat Plateau since only small erosional remnants of Mesozoic units remain. For example south of Chiang Rai (Fig. 11) a prominent isolated syncline on satellite images has a ~5–8 km wavelength with 45° west-dipping limbs. This fold is part of a belt of folded Mesozoic outliers, which form the link between the Khorat Plateau and the folded Mesozoic red beds of the Lanping-Simao Fold Belt in Yunnan.

In eastern central Thailand the Nakhon Thai Basin (Fig. 11) contains a Late Cretaceous section, which involves the upper part of the Phu Tok Formation (Cenomanian-Turonian), and Phu Khat Formation (Coniacian–Maastrichtian?; Hasegawa et al., 2010). Some gentle folding and faulting of the Khorat Group affects the Nakhon Thai Basin, and may have arisen as the result of inversion of Late Carboniferous–Permian basins (J. Booth Pers. Comm. 2008). Thrusted and folded Khorat Group also subcrops in parts of some Cenozoic rift basins, as seen on seismic data and a few well penetrations in the Gulf of Thailand and the (onshore) Phitsanulok Basin (Morley et al., 2004, 2007a,b).

West of the Khorat Plateau is a region of hilly country of outstanding natural beauty. In this region hills of Permian limestone are juxtaposed with hills composed of Khorat Group of similar elevation (Fig. 15, locations 3–7). On the basis of satellite image, and geological map interpretation a number of major faults can be identified that

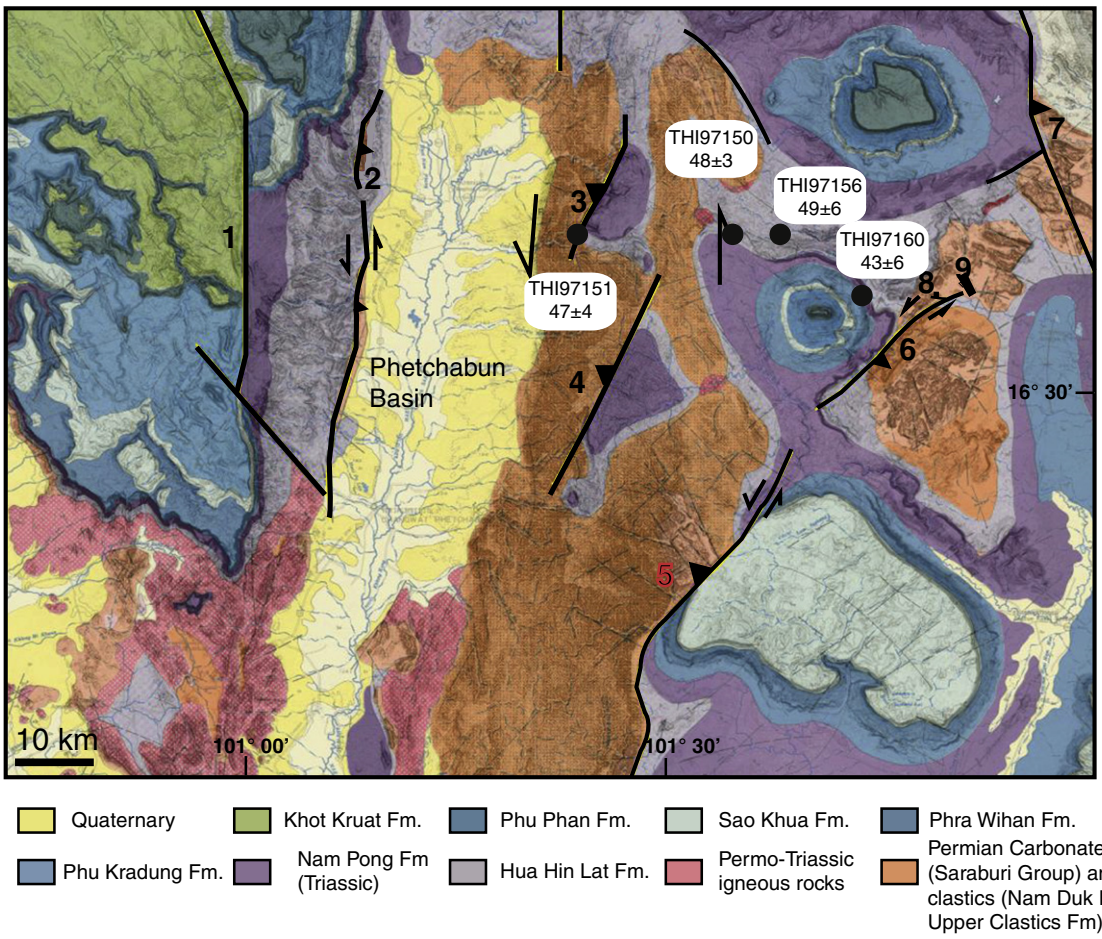


Fig. 15. Geological map draped on slope shader relief for the northern Phetchabun Basin area. The image illustrates the nature of the Palaeogene deformation that caused uplift and erosion of the Khorat Group at the margins of the Khorat Plateau. 1 = strike slip fault (probably sinistral motion) that juxtaposes Lower Cretaceous Khot Kruat Formation with Triassic Nam Pong Formation (which implies several kilometers vertical component of motion). 2 = Sinistral strike–slip fault trend which emplaces Permian Carbonates over Triassic Hua Hin Lat Formation (there is a thrust component to the deformation). 3, 4 and 5 = transpressional en-echelon faults at tip of large N–S strike–slip fault trend where Permian carbonates or clastics are juxtaposed with Triassic and Khorat Group rocks at similar topographic heights. 6 and 7, other transpressional faults (some where beds approaching the fault are sub-vertical to overturned), which juxtapose Permian carbonates with Triassic units. 8 = splaying fault strand to fault 6 exposed in quarry of Permian carbonates (see Fig. 16D). Faults 1–8 are probably all of early Cenozoic age, and helped exhume the western margin of the Khorat Plateau. The timing of exhumation (Paleogene) is indicated by the THI locations, which are sample numbers with central ages for apatite fission track data (Upton, 1999).

were responsible for the exhumation (Figs. 11 and 15). Some faults have map view patterns that indicate a strong component of strike–slip motion (Fig. 15). Rarely quarries and road cuts exhibit steeply-dipping fault zones with well developed low-angle striations (Fig. 15). The fault patterns suggest a sinistral component of motion on the N–S to NNE–SSW trending faults (Fig. 15). Although typically the Khorat Group exhibits gentle dips in much of the area, approaching major faults, bedding dips can become vertical to overturned. In Fig. 15 (location 2) Permian limestones are thrust over Triassic sedimentary rocks, perhaps associated with sinistral strike–slip displacement along a N–S trending fault. The occurrence of overturned beds and thrusts indicates that the vertical component of motion on the faults is reverse, not normal, with magnitudes on the largest faults in the order of kilometres of throw.

Apatite fission track and stratigraphic data (Racey et al., 1997; Upton 1999) indicate that about 3.5 km thickness of Khorat Group has been eroded west of the Khorat Plateau. Both Racey et al. (1997) and Upton (1999) used apatite fission track data to identify a two-stage cooling history for the Khorat Plateau, with early cooling from palaeo-temperatures above 120 °C during the Palaeogene–Oligocene and final cooling through the 105–55 °C isotherms during the Middle Tertiary (between ~35 and 25 Ma). Fig. 12A,B shows two examples (from the area of Fig. 15) of modelling apatite samples from the Khorat Group by Upton (1999), which are characteristic of the general pattern of exhumation. The early cooling event between ~65 Ma and 35 Ma (Racey et al., 1997; Upton, 1999) is probably related to folding, thrusting and uplift of the Khorat Group. The onset of folding in the Khorat Plateau appears to be older than in the Lanping-Simao Belt to the north, where deformation is of Oligocene age (e.g. Akciz et al., 2008). Racey et al. (1997) concluded that about half the thickness was removed during Palaeogene folding, and about half by more regional uplift which commenced in the Oligocene–Early Miocene. This later uplift and erosion might reflect the effects of rifting, hot mantle or more regional isostatic adjustments (Carter et al., 2000; Morley et al., 2011).

3.1.3. Role of the major Cenozoic strike–slip faults, during the Late Cretaceous–Palaeogene

The major strike–slip faults of Thailand (Mae Ping, Three Pagodas, Ranong, Khlong Marui faults; Fig. 17) as have been described largely in the context of Oligocene–Recent movements related to escape tectonics (e.g. Tapponnier et al., 1986; Polachan et al., 1991; Huchon et al., 1994; Lacassin et al., 1997; Replumaz and Tapponnier, 2003). However, more recently Morley (2004), Morley et al. (2007c) suggested the possibility of an older Palaeogene, pre-escape tectonics component to the strike–slip activity (as discussed in Section 3.1.2.).

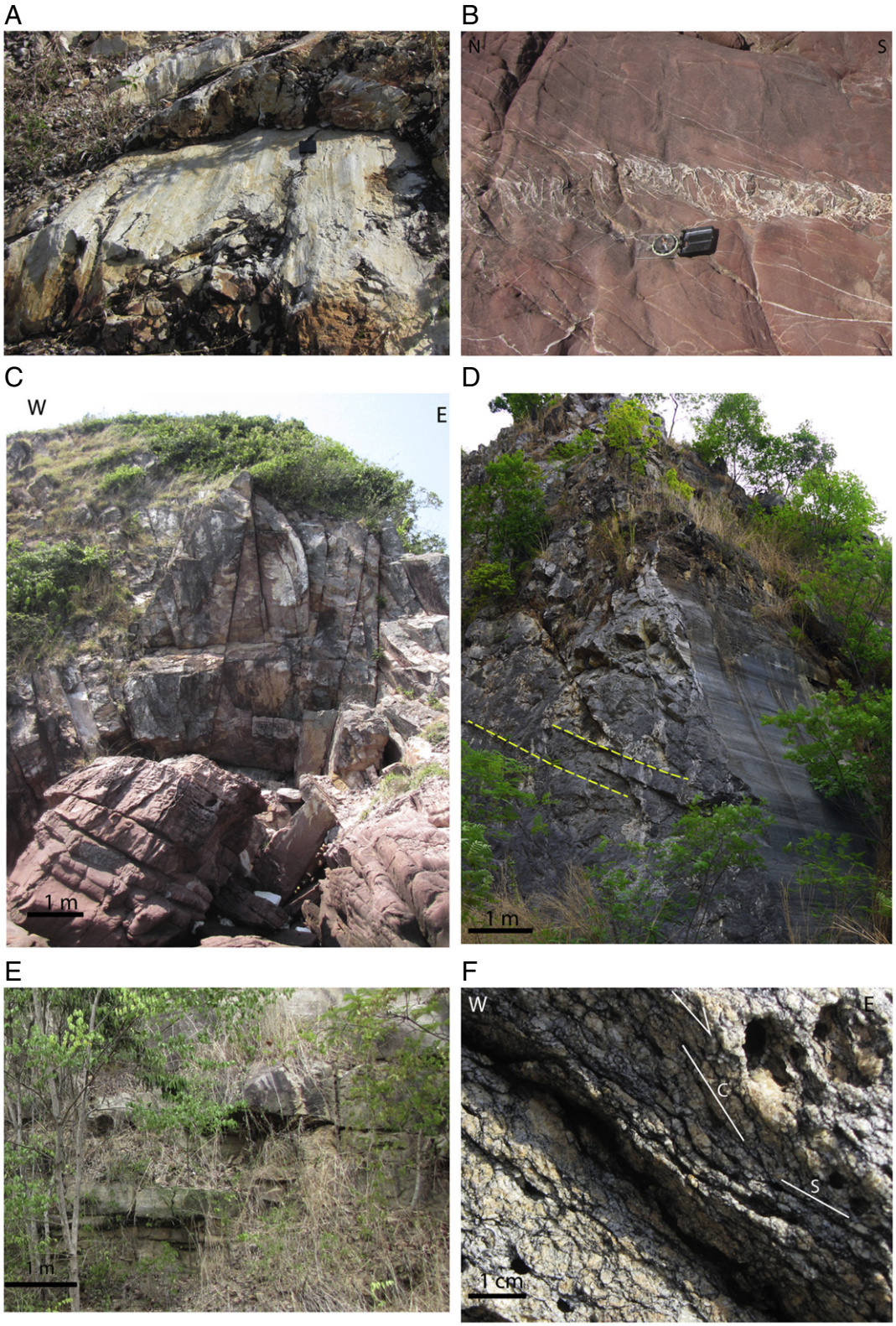


Fig. 16. Outcrop examples of the Palaeogene deformation. A, B and C from Laem Thai Ran Dok Mai Peninsula (Fig. 18), A) low angle thrust striations (steronet A, location A, Fig. 18), B) R (Riedel) and R' shears either associated with post-folding strike–slip fault, or pre-folding normal fault, in Khorat Group sedimentary rocks, (location C, stereonet C, Fig. 18), C, sub-vertical, N–S striking, west-younging Khorat Group Beds deformed by motion along the Tha Mai Fault, (location B, Fig. 18). D) Permian carbonates affected by 100° trending, S-dipping strike–slip fault, and large-sub-horizontal striations, bedding indicated by yellow dashed lines. Note that if bedding is restored to horizontal the strike–slip fault would dip less steeply. Hence it is inferred that the strike–slip fault post-dates folding (which in turn appears to be related to folding of the Khorat Group), which suggests the fault is of Cenozoic age. This inference is supported by the relationship of the fault to a larger fault (Fig. 15, fault 6), which is clearly of post-Khorat Group age. E) Low-angle (~20°), E to NE dipping, top to the east mylonites, about 100 m thick, capping a Triassic granite, and underlying schists and Khorat Group. They form part of the Ranong low-angle normal fault (Fig. 17), probably of Palaeogene age. F) S-C fabric in Indosinian-age amphibolites, with schistosity dipping 335° 30° E and a higher-angle C-surface that is interpreted to be related to Eocene normal faulting. ^{40}Ar–^{39}Ar ages of micas indicate Eocene cooling, which can be explained by top to the east unroofing of the Khanom metamorphic core complex, see Fig. 17 for location.

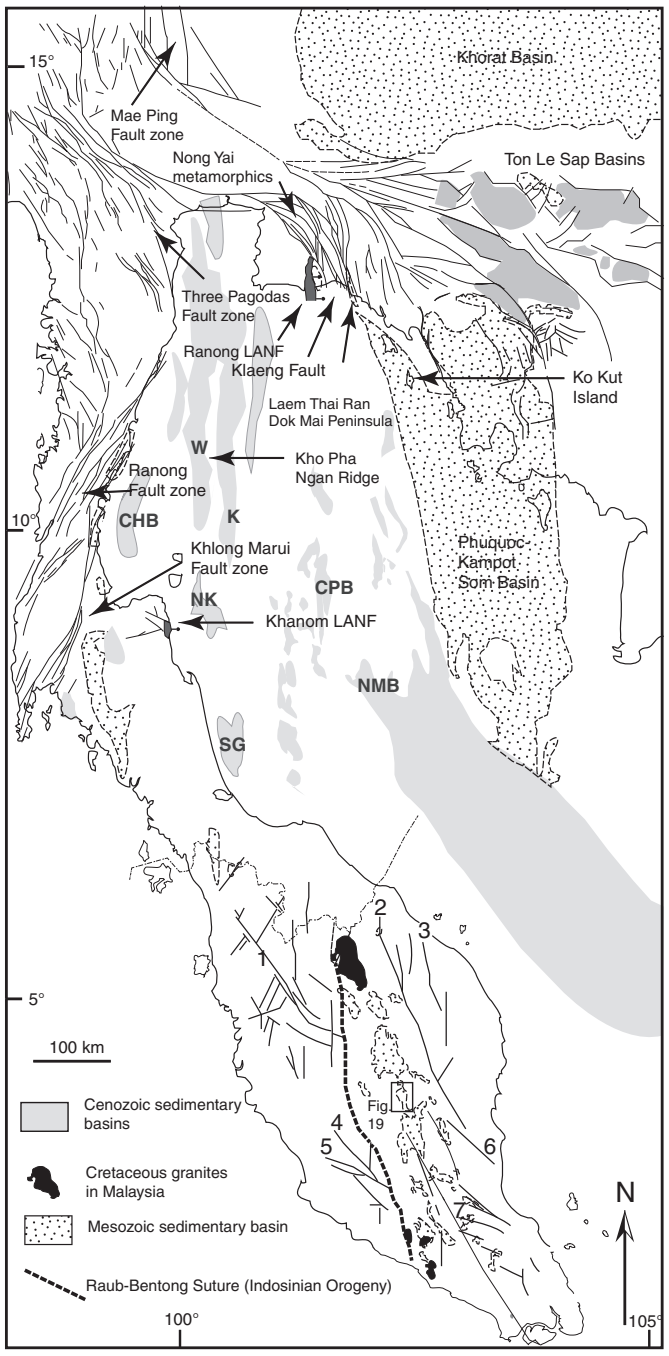


Fig. 17. Regional map of the Gulf of Thailand and surrounding areas. Basins where Mesozoic sediments underlie the Cenozoic rift section, CHP = Chumphon Basin, K = Kra Basin, W = Western Basin, NK = Nakhon Basin, CPB = Central Pattani Basin, NMB = North Malay Basin. 1–7 Faults in Malaysia thought to have been active during the Late Cretaceous and Palaeogene. 1 = Bok Bak Fault zone, most faults in the up to 10 km wide zone are sinistral, some with reverse sense of offset (Zaiton and Basir, 1999; Zaiton, 2002). Possibly initiated during Upper Triassic, transpressive movements attributed to Late Cretaceous (Zaiton, 2002). Tertiary strata are affected by the fault zone. Mixed dextral and sinistral reactivation. 2 = Lebir Fault zone (inferred by Tija, 1996 to have controlled Mesozoic Basin formation), 3 = Ka Buloh Fault zone, 4 = Bukit Tinggi Fault zone, (initiated in early Mesozoic, displays early ductile dextral shear and later sinistral motion, 83.6 Ma date from sinistral mylonite zone (Zaiton, 2002)). 5, Kuala Lumpur Fault zone, Sinistral, active until Eocene (?), Shuib (2009), 6 = Lepar Fault zone, initiated in Late Cretaceous and reactivated during Cenozoic (Shuib, 2009), 7 = Mersing Fault zone, initiated in early Mesozoic , active until Cenozoic, (Shuib, 2009).

This history was linked with the structures associated with exhumation of the western part of the Khorat Group, to suggest a significant Palaeogene transpressional belt developed prior to the escape tectonics phase. Watkinson et al. (2008, 2011) studied the Khlong Marui and Ranong faults, and on the basis of cross-cutting relationships between Late Cretaceous–Palaeogene granites and ductile fault strands, $^{40}Ar/^{39}Ar$ and U–Pb SHRIMP ages concluded that multiple stages of ductile, dextral strike–slip deformation could be identified as follows: D1, low grade ductile dextral strike–slip prior to 81 Ma, and between 59 and 49 Ma (D2). A major period of medium to high-grade ductile strike–slip deformation at temperatures between 300 °C and 500 °C occurred during the Middle Eocene (48–40 Ma, D3). Brittle sinistral and sinistral-reverse oblique strike–slip occurred after 37 Ma (D4), and brittle dextral strike–slip occurred around 23 Ma (D5). The D2 and D3 events fit with the regional Palaeogene transpressional deformation seen elsewhere in Thailand (Section 3.1.1.). Between the Ranong and Khlong Marui Faults, and west of the Khlong Marui Fault folded Mesozoic rocks are present predominantly in synclinal areas. The deformation in some areas produced broad, open folds, while elsewhere it is more intense with rotation of beds to near vertical. This deformation may have occurred in stages during the Mesozoic, but at least one phase probably occurred between the Late Cretaceous and deposition of the Late Eocene deposits in the Krabi Basin.

In the northern Gulf of Thailand a N–S trending, strike–slip fault bounded ridge of deformed Khorat Group is present at Laem Thai Ran Dok Mai Peninsula (Figs. 17 and 18). It is one of the few areas in the region where a detailed structural study has been conducted (Tuan, 2010). The Peninsula is composed of two linear ridges. The eastern ridge is composed of red beds, predominantly sandstones and conglomerates. The youngest U–Pb age from detrital grains of zircon was 196 ± 4 Ma (Early Jurassic, Geard, 2008), which confirmed the red beds could be correlated with the Khorat Group. The ridge is impressively linear, trends NNE–SSE for about 35 km and is 1–2 km wide (Ridd and Morley, 2011). It is one of a swarm of NW–SE to N–S trending ridges in the area, which are splays of the Three Pagodas and Mae Ping fault zones (Ridd and Morley, 2011). In thin section the sandstones show evidence for early-stage dynamic recrystallization. Numerous minor faults, many with striations are present in the area. However, despite the good coastal exposure it is rare to find NNW–SSE trending faults parallel to the Tha Mai Fault. The Laem Hin Khan Fault zone is exposed, but lacks good fault kinematic data. Most of the faults outcropping west of Tha Mai fault are WNW–ESE and ENE–WSW trending minor conjugate faults (Fig. 18, location C, stereonet C) exposed in vertical beds that young westwards. The offsets on the faults are compatible either with pre-folding, E–W trending normal faults (Fig. 18D), or post-folding conjugate strike–slip faults, the correct interpretation remains ambiguous. However, the large-scale fault geometries and a limited amount of palaeostress data suggests the Laem Thai Ran Dok Mai Peninsula was first subject to an E–W to WNW–ESE Shmax direction (Fig. 18B). Superimposed on the NNW–SSE Lae Hin Khan Fault zone are later WNW–ESE trending thrusts (Fig. 18 location A and stereonet A). These observations are compatible with a change from sinistral transpressional deformation to dextral transpression. Based on regional considerations (Morley, 2004) the sinistral transpression could be of Late Cretaceous or Palaeogene age, while the dextral transpression is likely to be of Late Oligocene to Neogene age.

3.1.4. Palaeogene metamorphic core complexes fringing the Gulf of Thailand

In Thailand only the Doi Inthanon-Doi Suthep metamorphic core complex (Fig. 7, location B) has been described in any detail (e.g. Macdonald et al., 1993, 2010; Dunning et al., 1995; Rhodes et al., 2000, 2005; Morley, 2009). This complex appears to be primarily of Early Miocene age, although the history of top to the east displacement on the detachment may be as old as 40 Ma (Macdonald et al., 2010). However, recent investigations indicate there maybe older metamorphic core complexes on the edge of the Gulf of Thailand.

About 40 km west of the Laem Thai Ran Dok Mai Peninsula, within the splaying Three Pagodas Fault zone is a top to the east to northeast, low-angle mylonite zone at least 100 m thick. It can be traced for about 30 km in a N–S direction, and extends offshore into the island of Ko Samut. This shear zone dips to the east between 20° and 30° and is referred to here as the Rayong LANF (Fig. 17). It can be interpreted as part of an extensional metamorphic core complex that comprises Triassic granites and amphibolites in the footwall, mylonites and schists within the shear zone, and Mesozoic sedimentary rocks in the hangingwall. The timing of low-angle normal faulting remains unclear. A thermal event in the Late Cretaceous is indicated by two U–Pb LA-ICP-MS Monazite ages (76.8 ± 2.6 Ma and 79.1 ± 2.2 Ma; Geard, 2008). This event preceded cooling and uplift associated with the core complex and so provides a maximum age for core complex development. Brittle faults strands of the Three Pagodas Fault zone have uplifted mid-crustal level rocks in a NW–SE trending zone centred around Nong Yai (Fig. 7 area A; Fig. 17) that terminates the northern extent of the Rayong LANF. U–Pb LA-ICP-MS dating of sphene from the Nong Yai metamorphic complex has yielded ages of 37.8 ± and 35.5 ± 3.1 Ma (Geard, 2008). These ages are interpreted here to indicate the time of exhumation associated with the brittle faulting, not the time of ductile shearing as interpreted by Geard (2008). This is because there is no indication of consistent, through-going ductile strike–slip mylonitic shear zones within the Nong Yai complex. The complex is a mixture of migmatites, gneisses and only some narrow (10's cms wide), steep shear zones. Development of the Three Pagodas Fault strand at Nong Yai probably marks the minimum age of the Rayong LANF since it appears to be cross-cut by the strike–slip fault system at its northern end (Fig. 17).

In the southern Gulf of Thailand at Khanom (Fig. 17), there is also a small metamorphic core complex present where a low-angle (25°–30°) east-dipping, top to the east mylonitic shear zone is imposed on foliations and dykes with a range of orientations, interpreted to represent Indosinian orogeny-age fabrics. Unpublished ^{40}Ar–^{39}Ar ages of muscovites and biotites from eight samples from the gneisses at Khanom have yielded step heating dates between 35–43 Ma and total fusion dates between 37–43 Ma (P. Charusiri pers. comm., 2010). These cooling ages suggest the timing of onset of fault-related exhumation, although further refinement of the cooling history is required.

3.2. Deformation within the Gulf of Thailand

Some wells in the western and eastern Gulf of Thailand have encountered Mesozoic sedimentary rocks (e.g. W-B-1, Kra Basin, Wollands and Haw, 1976; Katiya-1, Chumphon Basin, Heward et al., 2000). A Cretaceous granite was exhumed in the Ko Pha Ngan Ridge before being buried by the Oligocene–Neogene Western and Kra Basins (Pradidtan and Dook, 1992; Fig. 17). Some islands also display outcrops of Mesozoic sedimentary rocks, such as the marine Early Cretaceous deposits of Ko Kut island (Cuny et al., 2010). The deposits continue south into Cambodia and Vietnam (Phu Quoc, Bokor, Cam Pong Formations, Tien 1991; Vysotsky et al., 1994; Fyhn et al., 2010), and into Malaysian waters (Hutchison, 1989; Ngah, 2000). Seismic reflection data also images areas of reflective section below Cenozoic basins, which probably represent Mesozoic section (Fig. 19). In a number of places the reflective pre-rift section is deformed by folding and thrusting (Fig. 19; Morley et al., 2004 their Fig. 16). This deformation has been documented by Fyhn et al. (2010) as being widespread in the Phuquoc-Kampot Som Basin, which straddles the western coasts of Vietnam and Cambodia, and can be seen onshore in western Cambodia. These authors document the timing of deformation as Palaeocene–Early Eocene, and suggest linkage with the deformation of the Laem Thai Ran Dok Mai Peninsula (Fig. 17) and further north the Khorat Plateau.

In the SW Gulf of Thailand beneath the Nakhon and Chumphon Basins in places there are thick (locally greater than 1 section TWT, i.e. ~2 + km), reflective sequences that are tilted, contain faults with a normal component of offset (they may be transtensional) that have rotated the section. In some localities the beds are folded, and exhibit multiple unconformities that developed during folding, and they are unconformably overlain by syn-rift section that can be as old as the Eocene. The age of the sub-unconformity sequence is unknown, it maybe of Palaeogene or Mesozoic age, or both. However, given the occurrence of these deposits close to onshore outcrops of Mesozoic deposits in the Peninsula, the presence of the Phuquoc-Kampot Som Basin to the east, and the fault-bounded basins of the Malaysia (see Section 3.5) it seems reasonable to infer these sequences are fault-bounded Mesozoic basins.

3.3. Cambodia: Ton Le Sap Basin

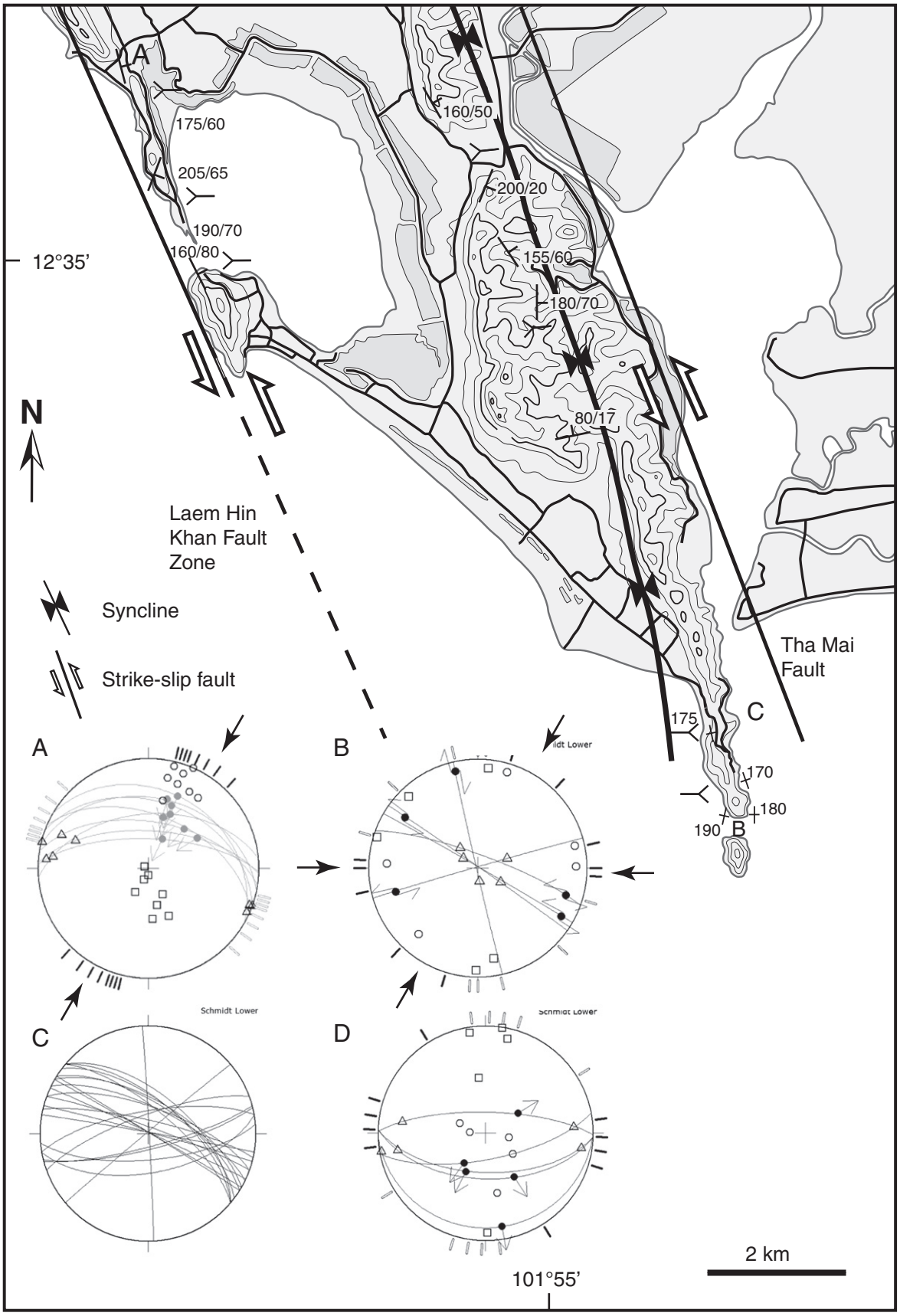
Until recently virtually nothing was known about the structure of the Ton Le Sap area, although there was speculation that a branch of the Mae Ping Fault zone ran through the Lake to pass offshore in the vicinity of the Mekong Delta, and Cuu Long Basin (see review in Morley, 2002). Recently a gravity survey and a reconnaissance accelerated weight drop seismic reflection survey (Hendrick et al., 2009) have provided some general information about the basin configuration, which can be viewed at the Cambodian Nation Petroleum Authority website (http://cnpa-cambodia.com/pages/petroleum-exploration/seismic-data.php). Some 2D seismic lines are presented on the website, but they are only stated to be from within the basin, and no exact locations are given. In the absence of well data, and specific seismic line locations only some general comments can be made. Nevertheless the data represents an important glimpse into the structural development of the area. A number of basins, defined by negative gravity anomalies can be defined, and suggest that the fault pattern is a continuation of the strongly splaying geometry seen in the Mae Ping Fault zone around the Thailand–Cambodian border (Fig. 17). In the absence of drilling, the age of the basins remain uncertain. A number of angular unconformities can be seen on the seismic reflection data, and from regional considerations it can be suggested that these unconformities occur at the top of Mesozoic, Eocene–Oligocene, and Neogene sequences (Fig. 20). But these ages are, at best, educated guesses at this time, and possibly the entire basinal sequence is of Cenozoic age. Alternatively CHplus Resources interpret the section as predominantly of Mesozoic age (http://www.slideshare.net/chris_newport/block-xi-information-memorandum-march-2010-c). Given that the Mesozoic sequence is well imaged offshore (e.g. Fyhn et al., 2010) it seems reasonable to expect that the Mesozoic sequence is also imaged onshore, particularly since it is present in hills immediately adjacent to the basins (Fig. 11). The seismic data at least point to a prolonged, episodic development of basins, with phases of extension/transtension and inversion/transpression (Fig. 20). If Mesozoic rocks are imaged on the seismic data, as seems likely, then the Ton Le Sap area differs from the Khorat Plateau, in that the Mesozoic basins are fault controlled.

The data show that the Ton Le Sap area has a more complex fault pattern than just a single, simple fault strand of the Mae Ping Fault zone running through the area. If a Cretaceous history of basin development is present in the area, then either: a) the faulting is unrelated to the Mae Ping Fault, and it just became linked with the fault during the Cenozoic, or b) it points to a very early stage of the fault development. A multi-stage history (b, above) is supported by the evolution of the Ranong Fault zone described by Watkinson et al. (2008, 2011).

3.4. Vietnam

Lacassin et al. (1998) reported that in the Song Da area Cretaceous red sandstones and conglomerates unconformably overlie schistose,

epimetamorphic middle-Triassic sediments (see Fig. 10 for location). The unconformity is, in places, strongly deformed, with folded Triassic limestones thrust onto the Cretaceous rocks. The authors preferred a Cenozoic age for the deformation, and inferred an India–Eurasia collision origin. Considering the widespread nature of early Palaeogene deformation affecting the Khorat Group, and the Phuquoc Group in



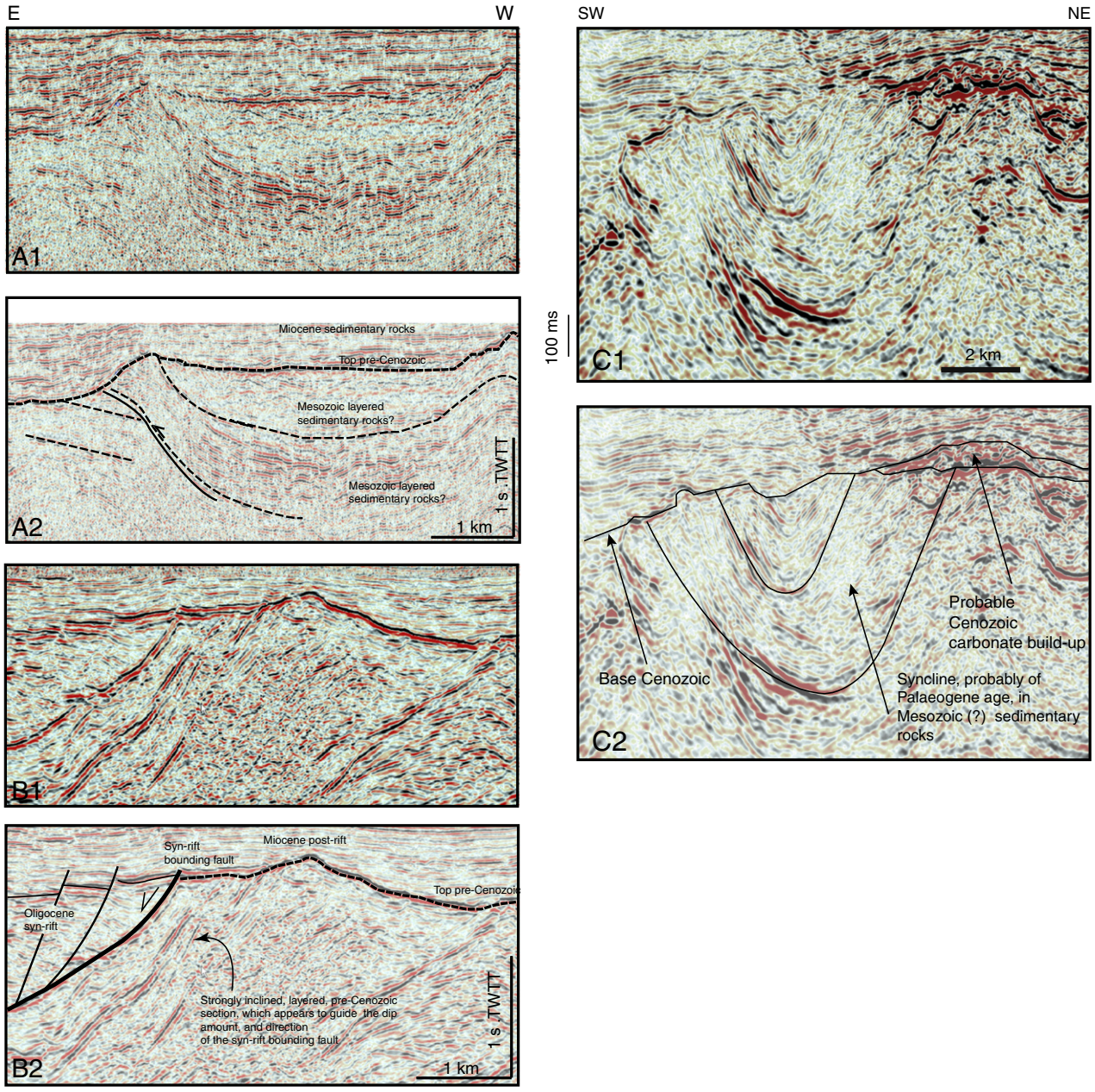


Fig. 19. Seismic examples of inferred Palaeogene deformation in the Gulf of Thailand. A1, uninterpreted and A2, interpreted section of a seismic line showing below the Cenozoic section is a layered, folded and thrusted reflective unit interpreted to be Jurassic–Cretaceous sedimentary rocks. B1 uninterpreted and B2 interpreted seismic reflection line showing that on the western edge of the layered Mesozoic?) reflections there are more steeply inclined reflective layers. They may have been thrust over the Mesozoic sedimentary rocks. In places later Cenozoic faults seem to follow these steeply dipping layers (which may represent thrusts, or rotated layered sediments, or meta-sediments). Clearly there is considerable speculation as to the interpretation of the pre-Cenozoic reflections at this early stage in our understanding. C1 Uninterpreted and C2 interpreted 2D seismic reflection line from the western Gulf of Thailand showing a large syncline below Cenozoic syn-rift section. To the south, onshore Peninsular Thailand Mesozoic rocks are preserved in synclines of similar dimensions. There are few pre Cenozoic rock units other than Mesozoic sedimentary rocks that are likely to produce such continuous reflections on seismic data. Hence a Mesozoic age for the section is inferred. Locations in the Gulf of Thailand withheld for confidentiality reasons.

Cambodia and Vietnam discussed above (also see Morley, 2004; Morley et al., 2007c, 2011; Fyhn et al., 2010,) it is also possible that the deformation identified by Lacassin et al. (1998) is of early Palaeogene age, and is too old to be simply explained by India–Eurasia collision. If the red beds are of Lower Cretaceous age then the deformation could be of Cretaceous age, since Lepvrier et al. (1997) document reactivation of NW trending ductile, partly mylonitic shear zones in the same general region between 90 and

Fig. 18. Map of the Laem Thai Ran Dok Mai Peninsula showing two NNE–SSE trending linear ridges associated with inferred Palaeogene faults and folds. The early transpressional deformation which produced the fault parallel syncline to the Tha Mai Fault is inferred to be the product of approximately E–W compression and sinistral strike–slip deformation. Later dextral transpression also affected the area, and is associated with WNW–ESE trending thrust faults (stereonet A). A–D stereonets showing structural orientations and calculated palaeo-stress directions. A) Relatively late stage thrust faults (location A on map), B) faults near the Tha Mai Fault (location B on map) with inconsistent fault-striation orientations suggesting two phases of deformation (NNE–SSE and E–W directions maximum horizontal compressional stress. C) Conjugate fault orientations found in the sub-vertical beds on the west side of the Tha Mai Fault (location C on the map). It is uncertain whether these faults are post-folding conjugate strike–slip faults, or pre-folding normal faults. A few limited striations associated with these faults, show that when the faults are rotated to horizontal they have a normal sense of displacement (D).

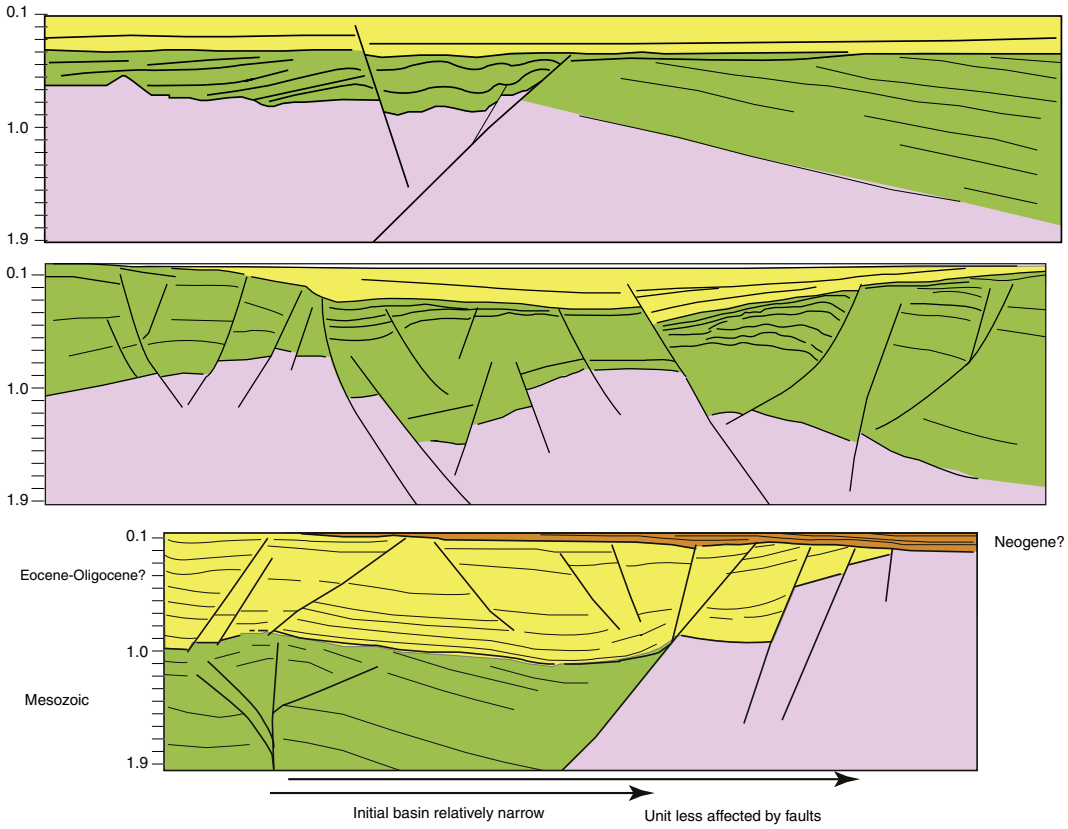


Fig. 20. Cross-sections based on line drawings of seismic lines across the Ton Le Sap Basin, using seismic data available online at http://cnpa-cambodia.com/pages/petroleum-exploration/seismic-data.php. There are no wells drilled in the area, hence the age of the section is uncertain and ages indicated are speculative, and no locations for the seismic lines were provided on the website.

120 Ma, from Ar–Ar dating of syn-kinematic muscovites. This deformation was accompanied by thermal event.

One other feature of the Cretaceous sedimentary rocks in NW Vietnam is their use in palaeomagnetic studies to estimate displacement magnitude along the Ailao Shan-Red River fault zone. Takemoto et al. (2005) and Geissman et al. (2008) both conclude there is no evidence for significant latitudinal translation (or rotation) of the Song Da terrane with respect to the South China Block from the Late Cretaceous-Present, i.e. the ASRR fault zone does not define the extruded Indochina Peninsula.

3.5. Malaysia

Cretaceous sedimentary rocks and plutons are concentrated in a N–S trending strip running through central Malaysia (Fig. 17). The most important plutonic complex is the Stong complex in northern Malaysia . There are several areas where poorly dated Jurassic–Early Cretaceous (?) continental red beds crop out in central Malaysia (Fig. 10; Abdullah, 2009). Rapid lateral facies changes to coarse clastic deposits and palaeocurrent patterns have been used to suggest the red beds were controlled by strike–slip faults and deposited in pull-apart basins (e.g. Harbury et al., 1990; Tija, 1996; Abdullah, 2009; Shuib, 2009). A number of predominantly N–S to NW–SE trending faults are also thought to have a Late Cretaceous–Palaeogene history (Zaiton, 2002; Fig. 17).

3.5.1. Stong migmatite complex

The Stong migmatite complex lies in mountainous terrane and comprises three plutonic granitoid components, the older Berangkat Tonalite and Kenerong Leucogranite, which are locally strongly deformed, and the younger, undeformed, pink, biotite-hornblende Noring Granite (Singh et al., 1984). The Noring Granite has been dated as Late Cretaceous (Rb–Sr, 79 ± 3 Ma Cobbing et al., 1992; K–Ar 65 ± 2 Ma for muscovite and 70 ± 2 Ma for biotite Bignell and Snelling, 1977), while the Kenerong Leucogranite has yielded a K–Ar biotite age of 69 ± 2 Ma (Bignell and Snelling, 1977).

East of the Stong complex lies the predominantly pelitic Taku Schists, which also include layers of amphibolite grade orthogneiss. The schists are deformed into a broad simple N–S trending antiformal dome, and K–Ar mica ages suggest the metamorphism developed during the later part of the Indosinian orogeny (219 ± 8 Ma, 212 ± 8 Ma, 215 ± 8 Ma, Bignell and Snelling, 1977). The northern part of the Taku Schists is intruded by the Kemahang Granite, which has yielded K–Ar biotite ages of 107 ± 3 Ma and 124 ± 4 Ma (Bignell and Snelling, 1977; Ishihara et al., 1979).

Abdullah and Setiawan (2003) identified at least three phases of deformation within the Kenerong Leucogranite, which is intruded as sheets and veins into the amphibolite grade pelitic schists. D1 = isoclinal folds, foliation development and sinistral oblique-reverse faulting, veins underwent rotation and extension, forming boudins and pinch-and-swell structures. This deformation occurred during intrusion, and a regional context of sinistral transpressional deformation is suggested. D2 and D3 involved NE–SW compression, formation of conjugate strike–slip faults, boudinage, and ptygmatic folding. Some earlier faults were reactivated as normal faults. It is uncertain whether the deformation is a result of stress-relaxation following granite emplacement or due to Cenozoic deformation (Shuib, 2009).

3.5.2. Late Cretaceous deformation

So far in Malaysia only a few K–Ar ages have been obtained from fault zones by Zaiton (2002), these results suggest that the ENE–WSW to N–S trending faults that cut the Main Range Granite are of Late

Cretaceous–Eocene age. Radiometric dating of the timing of fault zones commonly produces highly controversial results, even when a variety of methods are used (e.g Leloup et al., 2007; Searle, 2007). Consequently the dating of the fault zones in Malaysia based on K–Ar ages only cannot be viewed as particularly reliable, comprehensive, or documenting the full sequence of fault activity. The results have been used to suggest a Late Cretaceous phase of sinistral deformation followed by Cenozoic dextral deformation (e.g. Shuib, 2009).

The Jurassic–Early Cretaceous red-beds of the Tembeling Group and the underlying Kerum Formation in the Sungai Tekai region are strongly folded (Fig. 19; Harbury et al., 1990; Tija, 1996). The overall orientation of the folds, together with the local structures indicative of NE–SW compression and NW–SE extension, indicate they developed in a dextral strike–slip regime (e.g. Harbury et al., 1990; Hutchison, 1994; Tija, 1996). This sense of motion can be used to imply that folding occurred during the Eocene (Tija, 1996). However, observations of sinistral offsets along some faults, and the dating of some faults has been used to suggest that the folding occurred during the Late Cretaceous during sinistral strike–slip (Zaiton, 2002; Shuib, 2009). It is thought that many faults in Malaysia which are oriented NNW–SSE, NW–SE and WNW–ESE were reactivated under sinistral strike–slip or sinistral transpression during the Late Cretaceous (Zaiton and Basir, 2000; Zaiton, 2002; see review in Shuib, 2009; Fig. 16).

3.5.3. Discussion

The evidence for Late Cretaceous–Palaeogene deformation in Malaysia is fascinating, but very incomplete, and many questions remain to be resolved. For example it is hard to understand how the NNW–SSE trending folds in the Sungai Tekai region, could develop within a N–S striking, sinistral strike–slip setting, which contains the implication that the maximum horizontal stress was subparallel to the fold axes (Fig. 21C). Maybe some faults display sinistral motion reflecting local strains, but sinistral motion on the major faults cannot explain the folding, which is compatible with dextral motion (Fig. 21B). The nature of the argument about the origin of the folds in the Sungai Tekai region illustrates that much more work is required to understand the timing and displacement of the faults. Understanding of the Stong Migmatite Complex would benefit from U–Pb dating to better establish the sequence of plutonic events, accompanied by Ar–Ar and apatite fission track dating to determine the cooling history. Current descriptions of the Stong complex emphasize the link between sinistral strike–slip deformation and emplacement of the complex (Shuib, 2009). However, perhaps like the Doi Inthanon area in Thailand, there is a later extensional story to the unroofing and uplift of the complex to be discovered?

4. The Eastern margin of Sundaland and South China during the Late Cretaceous–Palaeogene

4.1. South China and South China Seas region

For the purposes of this paper the questions that need to be resolved from the South China area are: a) is there evidence for an Andean-type mountain belts that bordered the Khorat Basin to support the foredeep basin interpretation of Carter and Bristow (2003) and Fyhn et al. (2010), b) are the granitic plutons in southern Borneo and Vietnam related to the tectonics of SE China? c) Why did plutonism and extension cease in the Late Cretaceous and can that

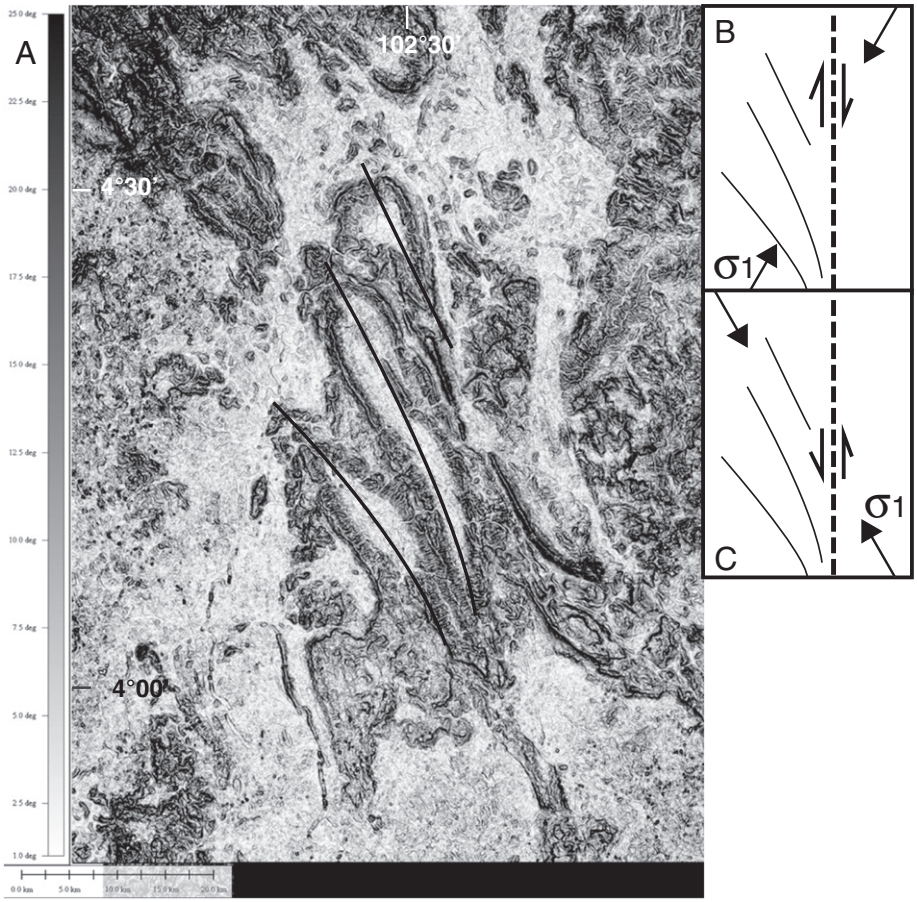


Fig. 21. A) Slope shader image of digital elevation model of the Sungai Tekai region showing the geometry of folded Cretaceous red beds in central Peninsular Malaysia. B) Likely stress orientation to explain fold development in a strike–slip context. C) Orientation of folds and stress direction for a Late Cretaceous sinistral strike–slip scenario (incompatible). See Fig. 15 for location.

be related to a collision event in the South China Sea area as proposed by Hall et al. (2009), Hall (2012) and Fyhn et al. (2010)?

Along southern China, Vietnam and southern Borneo is a chain of I-type Jurassic–Late Cretaceous granitic rocks (age spans ~180–90 Ma in SE China, Jiang et al., 2009; 112–88 Ma in Southern Vietnam, Nguyen et al., 2004; ~153–75 Ma in Borneo, Haile et al., 1977; Fig. 22). In SE China this plutonic activity is focused on the Yangtze and Cathaysian blocks, which formed a single terrane following collision in the Neoproterozoic (e.g. Li et al., 2002; Fig. 20). Analysis of xenoliths support the theory that the Cathaysian block is a fragment of a Precambrian craton, rather than a Late Palaeozoic–Mesozoic collision-type orogeny (Zheng et al., 2004). The present day crustal thickness is about 30–40 km, with gradual thinning towards the coast (Zhang and Wang, 2007).

Two primary models for the development of the Mesozoic igneous province have been proposed: 1) Indosinian crustal thickening (~257–205 Ma), followed by loss of an overthickened lithospheric mantle keel, and consequent asthenospheric mantle upwelling, which triggered intra-plate rifting (Wang et al., 2008), and 2) Indosinian crustal thickening followed by the establishment of an Andean-type margin (~180–70 Ma) related to subduction of the proto-Pacific plate (e.g. Nguyen et al., 2004; Li and Li, 2007, see review in Wang et al., 2011).

Li and Li (2007) described the SE China province as one of the world's largest Basin and Range-style magmatic provinces. Analysis of the Mesozoic sedimentary basins in the province by Shu et al. (2009) suggests they developed in three stages as: a) Late Triassic–Early Jurassic late orogenic to post orogenic 'molasse' basins, b) rifting during Middle Jurassic bimodal volcanism, and c) regional extension that produced Early Cretaceous volcanic and volcanoclastic-filled, fault bounded depressions, and Late Cretaceous–Palaeogene red-bed sediment filled fault-bounded depressions. This extensional deformation is associated with frequent juxtaposition of Mesozoic plutons in the footwalls of large faults and sedimentary basins in the hangingwall (Shu et al., 2009).

Rhyolite dominated bimodal (rhyolite/basalt) volcanic suites, mafic dykes and A-type granite comprise the 800 km long Southeast Coast Magmatic Belt of China (Chen et al., 2008). These different

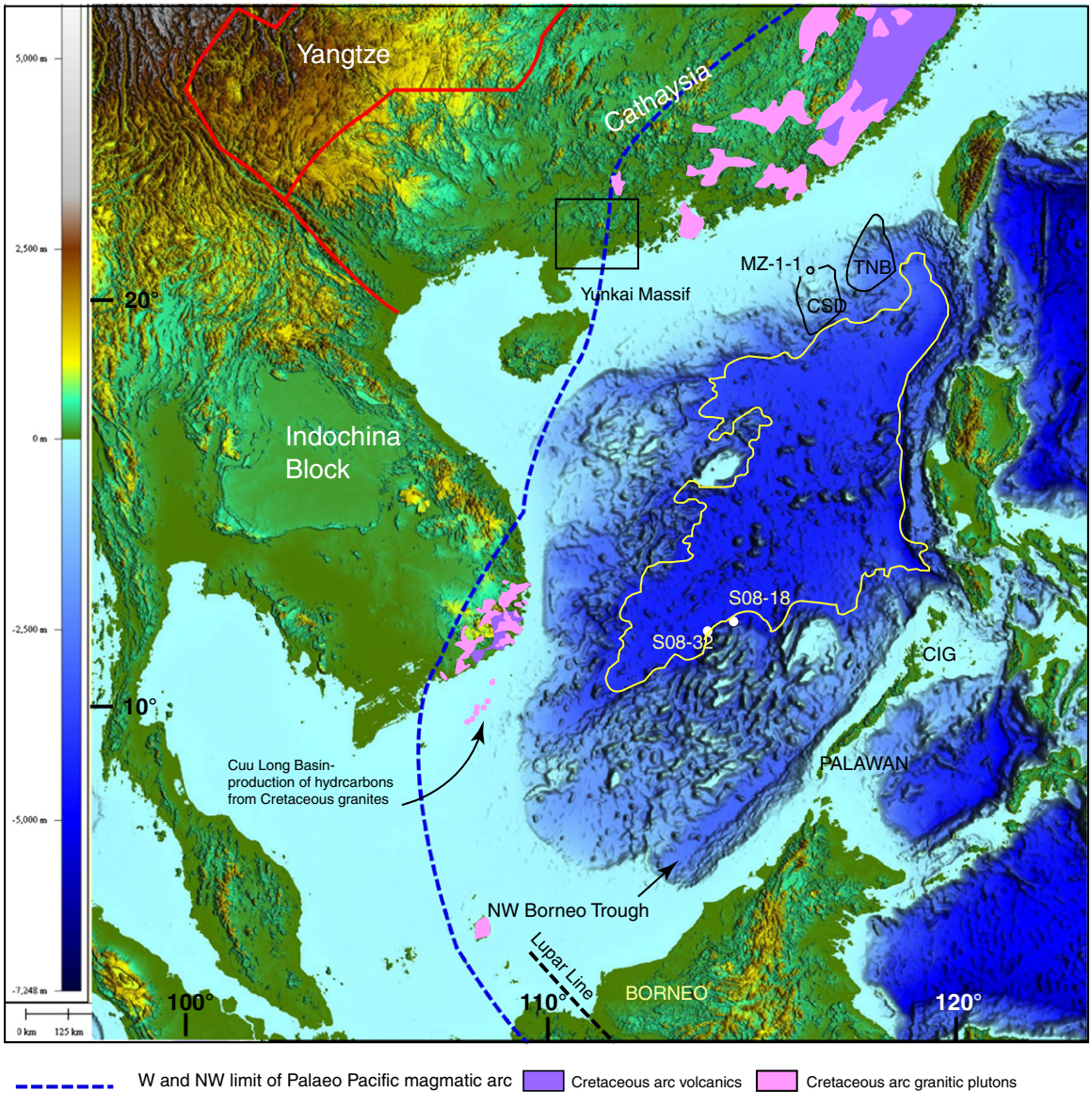


Fig. 22. Regional location map for the South China Seas, showing data from Li et al. (2008), Yan et al. (2010), Wang et al. (2008), and Nguyen et al. (2004). CSD = Chaoshan Depression, TNB = Tainan Basin, MZ1-1 = drilling site of well MZ 1-1. S08-18 = plagiogranite samples, magmatic zircons dated by LA-ICPMS U–Pb 159.1 ± 1.6 Ma, 157.8 ± 1.0 Ma, S08-32 adamellite samples 153.6 ± 0.3 Ma, and 127.2 ± 0.2 Ma (Yan et al., 2008).

igneous suites have been dated using 40Ar/39Ar, K–Ar whole rock ages and LA-ICPMS U–Pb zircon ages, and the results indicate rhyolitic volcanism (101–72 Ma), was contemporaneous with A-type granite intrusions (100–90 Ma), and mafic dyke intrusion (94–77 Ma). The rhyolites are a mixture of strongly peraluminous and peralkaline to mildly peraluminous types. Chen et al. (2008) explain this chemistry as a consequence of low-pressure melting of felsic granite, and the granodioritic, tonalitic and trondhjemitic members of core complex assemblages, plus other possible sources of contamination. Numerical simulations suggest the magmas developed during rapid thinning of the lithosphere, and were rapidly exhumed (Chen et al., 2008).

Despite the descriptions above that imply considerable exhumation of plutons and their surrounding country rock, a fission track study by Yi et al. (2009) of one part of the margin shows surprisingly little exhumation during the Late Cretaceous. Exhumation accelerated at different times during the Palaeogene in different locations along the margin and was followed by a final stage of cooling in the Middle Miocene. A palaeostress study by Chan et al. (2010) for the greater Pearl River Delta Region determined the extension direction during the Late Mesozoic to early Cenozoic was primarily in a NW–SE direction (activating NE–SW and secondarily E–W trending faults), while during the early to mid Cenozoic extension was primarily N–S (occurring primarily along E–W trending faults, with strong reactivation of NE–SW trends). Early Palaeogene extension appears to be a feature of both the South China Sea and East China Sea margin, and west of Taiwan it is associated with 65–59 Ma andesitic volcanism (e.g. Li et al., 2009; Chen et al., 2010).

The precise tectonic evolution of the margin during the Mesozoic remains, at present, unclear, with contradictory interpretation of subduction zone dip direction based on igneous and metallogenic younging trends (e.g. NW slab dip Li and Li, 2007; SW slab dip Wang et al., 2011). The SW subduction and intra-plate extension models do not address the more regional aspects of Cretaceous pluton distribution, and so fail to explain the apparent continuation of the SE China plutonic province into southern Vietnam and Borneo. The NW subduction model provides the simplest explanation for the regional distribution of Mesozoic granitoids. At its western end the subduction zone curves around to the south and runs through to the Lupar line area of Borneo (Fig. 22; Nguyen et al., 2004; Hall et al. 2009; Hall, 2012). At present the consensus of opinion favours the NW subduction model for the Cretaceous period of SE China (e.g. Honza and Fujioka, 2004; Metcalfe, 2009; Zhang et al., 2010; Wang et al., 2011).

Mantle tomography has the potential to resolve the disputes about subduction zone geometry, which as discussed above are based primarily on surface criteria derived from the surface geology. Li and van der Hilst (2010) presented tomographic images of the mantle beneath East Asia, they identified mantle structures between depths of 400 and 800 km beneath the Yangtze craton and South China fold belt, that could be related to late Mesozoic to early Cenozoic subduction. This requires the stagnant slabs to have resided for tens of millions of years in the middle mantle. Yet in the cross-section presented across southern China (their sections g, h in Figs. 5 and 6, Li and van der Hilst, 2010) the Mesozoic slab graveyard is not clearly defined at depth. This lack of definition can be explained by arrest of late Mesozoic–Early Cenozoic slab descent in the middle Mantle in the region of East China and Japan (as reviewed by Li and van der Hilst, 2010), while further to the SW the slab sunk much deeper in the mantle (e.g. Ricard et al., 1993; Grand et al., 1997; Lithgow-Bertelloni and Richards, 1998). Additionally the upper mantle tomography in SE Asia records the effects of continent–continent collision processes as well as subduction-related effects (Li and van der Hilst, 2010). Consequently the tomographic evidence for long-lived NW subduction under SE Asia is somewhat ambiguous. However, given the clearer picture further to the NE, NW subduction under SE Asia seems generally supported by the tomography.

In the reconstructions used in this paper the W to NW subduction model is followed (Fig. 23). This figure is based on the Hall et al. (2009) reconstructions, but has been significantly modified following the discussion in this paper. In the following sections these modifications are discussed.

4.2. NW Borneo, the Rajang Group and the Sarawak orogeny

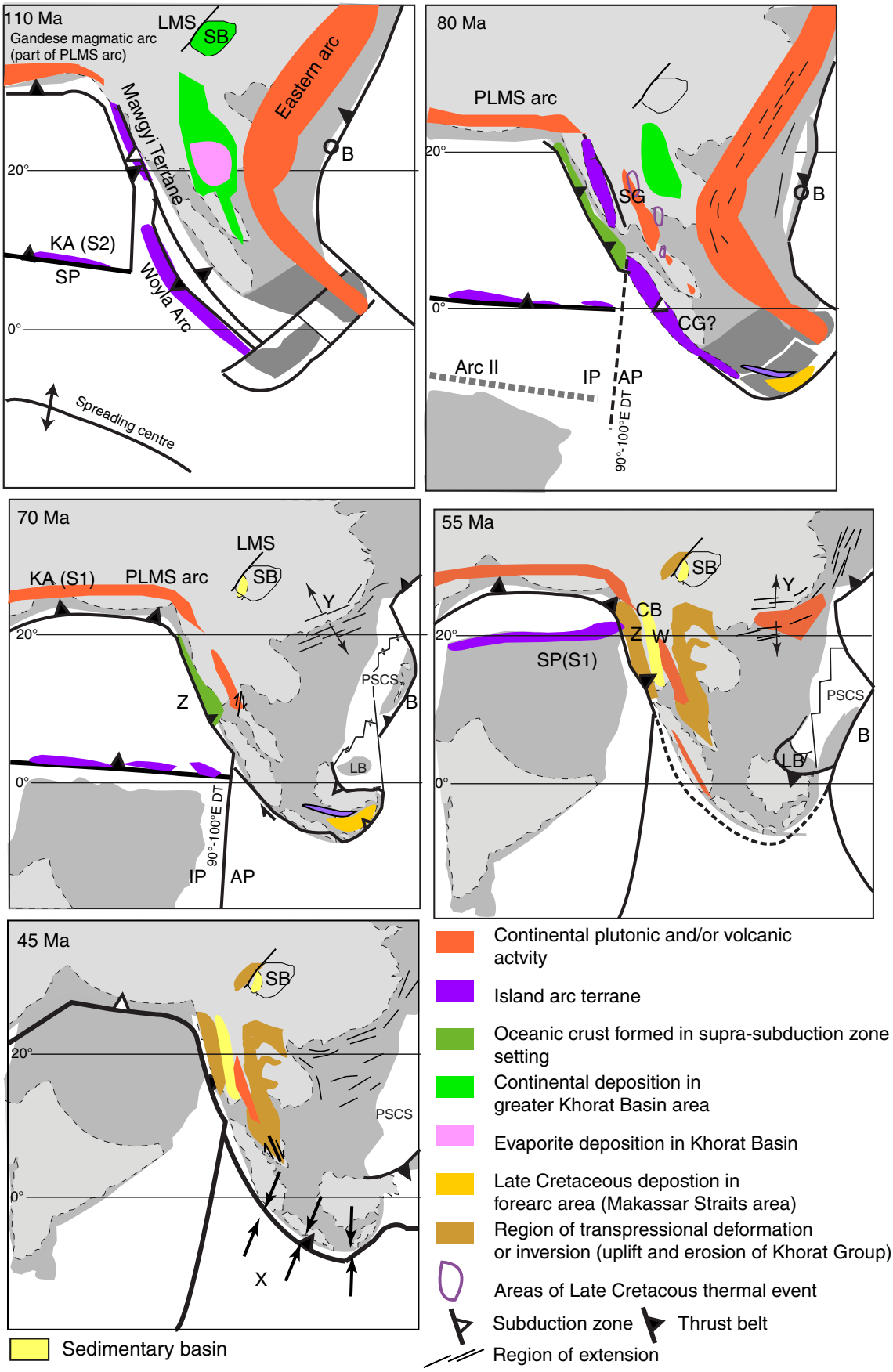
The Rajang Group is a steeply dipping, thick, isoclinally folded flysch sequence, composed of interbedded mudstones and sandstones, locally metamorphosed to lower greenschist facies (e.g. Kirk, 1957; Liechti et al., 1960; Wolfenden, 1960; Hutchison 1996, 2005). Microfossils indicate the oldest part of the Rajang Group in Sarawak is of Santonian–Maastrichtian age, and the entire formation appears to range in age from about 85 Ma to 45 Ma (Hutchison, 1996, 2005). Haile (1974) proposed that the Rajang Group developed in the context of Late Cretaceous–Late Eocene southwards-directed subduction beneath the Kuching Zone, and was deposited in a region he named the Danau Sea (this is part of the Proto South China Sea, which in turn is probably a late-stage fragment of the Palaeo-Pacific). Hutchison (2005) notes that from about 64 Ma onwards there is an absence of subduction-related igneous activity in NW Sarawak (in the Lundu and Tinteng Bedil areas), and concludes that the Palaeogene age section of the Rajang Group does not represent accretionary prism material, but the sedimentary fill of the Proto South China Sea Basin, compressed between the Balingian–Luconia continental fragment and Sundaland (Hutchison, 1996; Moss, 1998). The Rajang Group was subject to compression and strong folding by Middle Eocene times, and by the Late Eocene the shallow marine Tatau Formation was deposited unconformably on the deformed and eroded Rajang Group (Hutchison, 2005).

4.3. Termination of Cretaceous subduction

As reviewed in Section 4.1 the south China-South Vietnam margin experienced considerable volcanism, accompanied by crustal extension, and in places loss of lithospheric mantle during the Late Cretaceous. If this tectonic environment occurred in the upper plate of a north-dipping subduction zone, then sometime between the Late Cretaceous and Palaeogene the subduction zone was terminated, and was replaced by a subduction zone or zones further to the south and east (see 55 Ma reconstruction of Hall, 2002).

Recently, Hall et al. (2009), Hall (2012) and Fyhn et al. (2010) have proposed two different models for collision-related termination of subduction by collision of part of the continental crust of the South China Seas area with Asia (Fig. 24). The model by Hall et al. (2009) proposes collision of the Luconia continental fragment with the Vietnam margin between 110 Ma and 90 Ma. Hall et al. (2009) clearly relate the end of subduction-related granite emplacement to the Luconia collision, although they note that arc-related plutonism may be as young as 80 Ma. A collision around 90 Ma could help explain the restriction of the Khorat Plateau area, which led to deposition of the Maha Sarakham Formation evaporites. Hall et al. (2009) and Hall (2012) do not attribute any major plate-tectonic significance to Palaeogene deformation in the Rajang Group adjacent to the Lupar Line.

Conversely Fyhn et al. (2010) attribute great plate-tectonic importance to the Lupar Line. They proposed the Luconia fragment collided during the Palaeogene as an explanation for Paleogene inversion and folding of the Khorat Group in Cambodia, Vietnam and the Gulf of Thailand as well as deformation of the Rajang Group in Borneo. However, their Palaeogene timing seems incorrect since there is a ~16 my gap between the end of subduction-related granite emplacement at about 80 Ma along much of the margin, and the Luconia–SE Asia collision at around 64 Ma proposed by Moss (1998) and Hutchison (2005). Note that although the absence of magmatic activity suggested to these authors that the Rajang-Embaluh Group was deposited within a closed basin near the beginning of the Cenozoic,


110 Ma
Gandese magmatic arc
(part of PLMS arc)
LMS
SB
Eastern arc
Mawgyi Terrane
B
20°
KA (S2)
SP
Woyla Arc
0°
Spreading centre
80 Ma
PLMS arc
20°
SG
B
0°
CG?
Arc II
IP
AP
90°-100°E DT
70 Ma
LMS
KA (S1)
PLMS arc
SB
Y
20°
PSCS
Z
B
LB
0°
90°-100°E DT
IP
AP
55 Ma
SB
Y
CB
20°
Z
W
SP(S1)
PSCS
B
LB
0°
45 Ma
SB
20°
PSCS
0°
X
Sedimentary basin
Continental plutonic and/or volcanic actvity
Island arc terrane
Oceanic crust formed in supra-subduction zone setting
Continental deposition in greater Khorat Basin area
Evaporite deposition in Khorat Basin
Late Cretaceous depostion in forearc area (Makassar Straits area)
Region of transpressional deformation or inversion (uplift and erosion of Khorat Group)
Areas of Late Cretacous thermal event
Subduction zone
Thrust belt
Region of extension

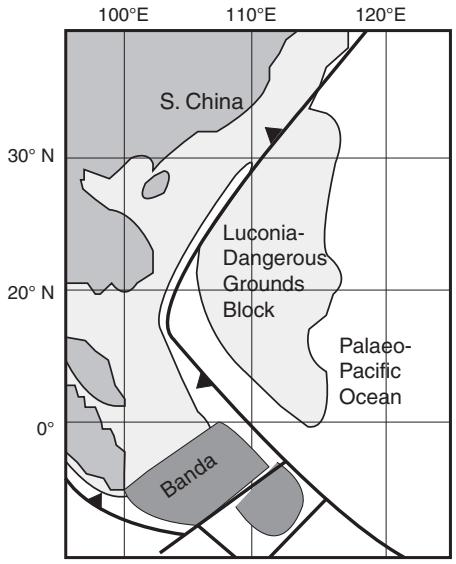


Fig. 24. Collision model for Luconia–Dangerous Grounds Block with SE Asia. Model depicted is from Hall et al. (2009) for 110 Ma, but the model by Fyhn et al. (2010) for Palaeogene collision is similar in depicting the collision of a large continental fragment that represents a large percentage of the offshore South China Sea area.

the unconformity marking the end of deformation is later, during the Middle Eocene, between about 43 and 37 Ma.

The size of the Luconia fragment envisioned by Hall et al. (2009), Hall (2012) and Fyhn et al. (2010) is relatively large and includes the areas that today form the Dangerous Grounds and Reed Bank. There is no evidence for Meratus Mountain-type Late Cretaceous–Palaeogene accretionary event-related lithologies, along the northern South China Seas area (e.g. Zhou et al., 2008). But then evidence for pre-Cenozoic lithologies in the South China Seas is very sparse. The collision of a large Luconia–Dangerous Grounds block places the suture zone approximately along the line of the future South China Sea spreading centre (Hall et al., 2009). Weakly deformed Mesozoic sedimentary rocks are found in the Chaoshan Depression and Tainan Basin just north of the proposed location of the collision zone (Fig. 22; Li et al., 2008). The presence of the sedimentary rocks, and lack of strong deformation and metamorphism is perhaps the clearest evidence against Mesozoic or later collision of a large Luconia–Dangerous Grounds block with southern China and Vietnam. The few metamorphic rocks dredged from the Dangerous Grounds area yielded 146 ± 3 Ma and 122–113 Ma K–Ar ages (Kudrass et al., 1986). The significance of the ages is uncertain and could be related to intrusive events, however, they do not provide support for a Late Cretaceous collisional event. Hutchison (2010) notes the great similarity between basement types in Western Sarawak, the South China Sea, and Vietnam, South China and Peninsular Malaysia, and concludes that the whole region could represent SE Asian continental crust that was rifted in the Cenozoic. This conclusion is supported by U–Pb dates of magmatic zircons from granitoids from sites S08-32 (153.6 ± 0.3 Ma, and 127.2 ± 0.2 Ma) and S08-18 (159.1 ± 1.6 Ma, 157.8 ± 1.0 Ma) in the South China Seas (Fig. 22; Yan et al., 2008), whose ages and geochemistry indicate they were part of the same granite suites that were emplaced across South China, Taiwan, Palawan and Southern Vietnam from the Middle Jurassic to mid-Cretaceous (Yan et al., 2010).

Cretaceous sedimentary rocks have been encountered in six hydrocarbon exploration wells drilled in the Reed Bank plateau in the 1970's and 1980's (Hinz and Schülter, 1985). The wells found Lower Cretaceous clastics, with some limestones, and in Cadlao-1 volcanics were also present. The clastics comprised sub-arkosic and lithic sandstones, siltstones and hematitic shales. Sampaguita-1 found Upper Cretaceous sandstones (Schülter et al., 1996). The depositional environment is predominantly shallow marine.

The southern edge of the Dangerous Grounds block is exposed in the island chain of Palawan and the Calamian Island Group (CIG, Fig. 22). On the island of Busuanga in the CIG there are sedimentary rocks attributed to a Middle Jurassic–Early Cretaceous accretionary prism (Zamoras and Matsuoka, 2004; Waikita and Metcalfe, 2005). This is important evidence for establishing the location of the Izanagi or Proto-Pacific plate subduction zone with respect to the South China Margin. It is the strongest evidence for the subduction zone running along the southeastern margin of the Dangerous Grounds Block, and not the northern margin as proposed by Hall et al. (2009) and Fyhn et al. (2010). Zamoras and Matsuoka (2004) discuss evidence for Jurassic–Early Cretaceous subduction, but do not address why subduction ceased sometime during the Early–Late Cretaceous. The island of Palawan shows no evidence for a major Late Cretaceous–Early Palaeogene collision with a continental fragment (see review by Yumul et al., 2009).

Analysis of the Palawan Ophiolite Complex indicates during the Late Cretaceous to Eocene oceanic crust was forming south of Palawan (Yumul et al., 2009). This evolution also seems to preclude a Late Cretaceous–Palaeogene collision event occurring on the southern margin. Hence possibly numerous small collisions, including swarms of seamounts (e.g. Zamoras and Matsuoka, 2004), may have choked the subduction zone around the mid-Cretaceous, and could be associated with small collisional events in the Central Indonesian Collisional Complex. Another way to satisfy the conclusion of Yumul et al. (2009) is to not to terminate subduction along the South China margin by collision, but by back arc spreading (Fig. 25; see discussion in Section 4.4). The subsequent history of ophiolite development in Palawan appears to be polyphase. Encarnacion et al. (1995) notes that two ophiolites may be exposed on Palawan, a structurally upper one of Eocene age, which overlies one of Cretaceous age (possibly 146–97 Ma, Raschka et al., 1985; Faure et al., 1989; Muller,

Fig. 23. Regional tectonic evolution of SE Asia from 'mid' Cretaceous until the Palaeogene. The basic map is modified from the plate reconstructions of Hall et al. (2009). Modifications are to the Hall et al. (2009) model are based on discussion of the tectonic evolution in this manuscript. KA(S1) = location of Kohistan Arc in scenario 1 – within the Gandese Arc (e.g Petterson, 2010), KA(S2) location of Kohistan Arc in Scenario 2 as Island Arc within the Tethyan oceanic realm (i.e. part of the Incertus Arc of Hall et al., 2009), (Khan et al., 2009). Arc II – required location of the arc if the pre-75 Ma deformation of the Kohistan Arc is related to collision with India. SP = Spontang arc, SP(S1) collision of island arc terrane with India, in tectonic scenario 1 the Spontang (Searle et al., 1997; Pederson et al., 2001; Petterson, 2010). B = location of Busuanga, in the future island chain of Palawan-Calamian Island Group, PSCS = Proto-South China Sea, (opened as a back-arc area in the scenario shown), X = compression arising from the onset of subduction in the Late Eocene (following Hall et al., 2009), Y = extension direction from fault kinematic data (Chan et al., 2010). Timing of granite intrusions Sumatra (McCourt et al., 1996; Cobbing, 2005), 55 Ma volcanism in South China Seas area (Chen et al., 2010), areas of Late Cretaceous thermal events (Macdonald et al., 1993, 2010; Dunning et al., 1995; Geard, 2008; Shuib, 2009; Hansen and Wemmer, 2011). Collision of Mawgyi Terrane from Mitchell et al. (2007), collision of Woyla Terrane or Arc from Barber and Crow (2003, 2009). 70 Ma, Hall et al. (2009) highlight the importance of the 98E ridge in separating the rapidly northwards moving India Plate (IP) from the stalled Australia Plate (AP). Only after about 45 Ma when subduction was renewed along the Sumatra-Java trench (X) did the Australian Plate start to move significantly northwards. PLMS arc = Pakistan-Ladakh-Myanmar–Sumatra arc, timing from Cobbing et al., 1992; Charusiri et al., 1993; Barley et al., 2003; Searle et al., 2007. 80 Ma, Hall et al. (2009) suggested that the Myanmar–Thailand part of the PLMS arc may have subduction related granites (SG), but passing further south the granites along the same apparent trend maybe related to collision of the Woyla Arc (CG) rather than subduction. Timing of eastern arc from Haile et al., 1977; Nguyen et al., 2004; Jiang et al., 2009, Wang et al., 2011. W = Early Palaeogene stage of metamorphism in Mogok metamorphic belt (Searle et al., 2007), explained by renewed subduction along Myanmar margin (Z) between the Late Cretaceous and Early Cenozoic, which also triggered deformation within the Indo-Burma ranges. CB = Central Basin, Myamar. Note: the timing of India–Eurasia collision remains controversial, here soft collision is envisioned to have occurred around 50 Ma. LMS = Longmen Shan Belt, SB = Sichuan Basin, yellow area in Sichuan Basin = region of Late Cretaceous–Cenozoic foredeep (Jia et al., 2006).

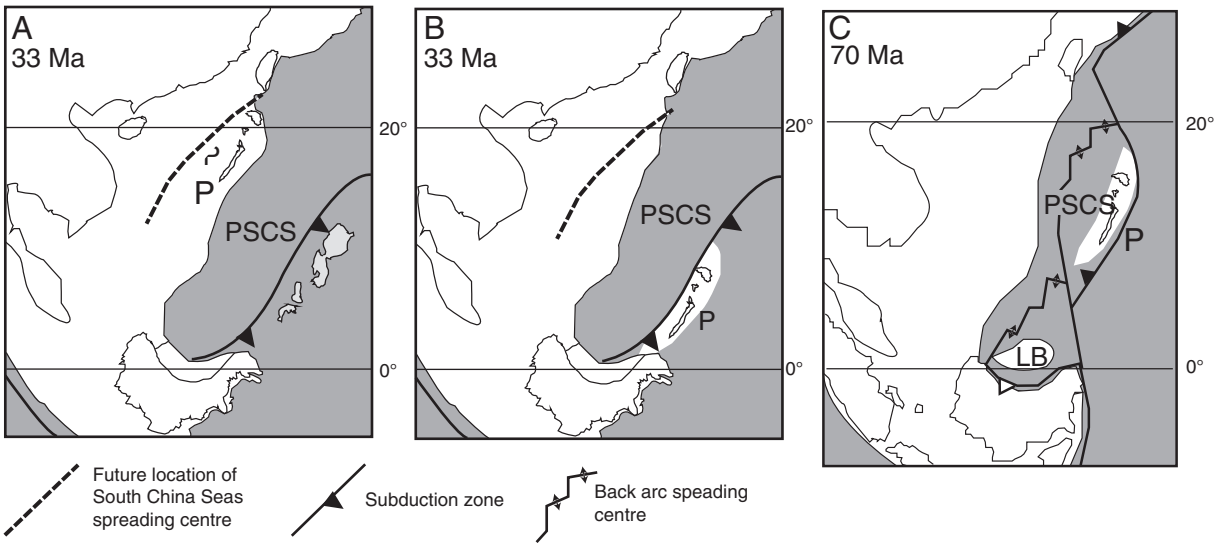


Fig. 25. Models for the development of the Proto-South China Seas. A) Palawan (P) rifted from the SE China margin during the Oligocene, when the South China Seas spreading centre opens. B) Palawan lies on the SE side of the Proto South China Seas (PSCS) oceanic crust prior to opening of the South China Seas. C) One possible solution to explain the situation in B), is shown in C, where back arc spreading rifts Palawan and the Luconia blocks in the Late Cretaceous, terminating subduction along the margin.

1991). According to Encarnacion et al. (1995) the presence of 34 Ma amphibolites from the metamorphic sole indicate an Oligocene subduction initiation event, while the collision of the North Palawan continental crust with the trench and ophiolite obduction occurred around 15 Ma.

4.4. Implications of the Busuanga Accretionary Prism

Today the Busuanga Accretionary Prism lies in the upper plate of the Palawan Trough. However, the Jurassic–Cretaceous accretionary prism is also interpreted as originating on the margin of southern China, in the upper plate of the subducting proto-Pacific plate (e.g. Zamoras and Matsuoka, 2004), which poses a problem explaining how it came to lie south of the Palawan Trough. Two basic tectonic scenarios for the evolution of the Palawan-Busuanga continental sliver (or Calamian micro-continent according to Hutchison, 2010) are considered here:

1) The Palawan Trough is a foredeep basin arising from loading caused by Oligocene–Miocene obduction of the Proto-South China Seas oceanic crust from the south. Rifting of the South China Sea alone moved the Calamian micro-continent away from southern China (Schülter et al., 1996; Hall, 2002; Zamoras and Matsuoka, 2004; Fig. 25A).
2) The Palawan–NW Borneo trough was a site of subduction of the Proto-South China Seas crust. The Proto-South China Sea formed in a back-arc location and split the Calamian micro-continent off from South China and it migrated to the south or southeast with time (Fig. 25B,C).

The difference between models 1 and 2 hinges on whether the NW Borneo-Palawan Trough is considered to be a subduction zone or a foredeep trough. The NW Borneo Trough is generally considered to have been a site of subduction of the Proto-South China Seas oceanic crust that ceased during the late Early Miocene when thin continental crust of the South China Seas area entered the subduction zone (e.g. James, 1984; Sandal, 1996). In model 1, this site of subduction is not followed along strike into the Palawan Trough, instead the site of subduction is located to the south of the Calamian micro-continent, within the Sulu Sea (Hall, 2002). During opening of the South China Sea Palawan lay on the southern edge of the rifted continent and moved into position adjacent to Borneo during the Miocene (Hall, 2002; Fig. 25A). The juxtaposition of the NW Borneo and Palawan Troughs into a contiguous entity is coincidental. The implication of this model is that the deformed 'thrust wedge' offshore NW Palawan should be younger (i.e. of Late Oligocene–Neogene age) than the Palaeogene–Early Miocene accretionary prism of NW Borneo. However, due to post-collisional tectonics and the development of large deltaic complexes the older history of the NW Borneo Trough is somewhat obscured, and the trough has migrated offshore with time as a consequence of delta progradation.

Hutchison (2010) pointed out the similarities in accretionary prism geometry and relationship to South China Sea continental crust in both Palawan and NW Borneo, and leaned towards a common accretionary prism origin for the two troughs, citing the principle of Occam's razor. The consequence of this interpretation leads to a model that requires the presence of Proto-South China Seas crust NW of Palawan, (e.g. model 2). However, Occam's razor also applies to the greater tectonic setting of models 1 and 2. Model 2 invokes a more complicated scenario than model 1, and requires a Cretaceous back-arc spreading event to create the Proto-South China Seas oceanic crust. In this model the Sulu Sea cannot be regarded as a remnant of the Proto-South China sea.

4.5. Preferred interpretation

The weakness of model 1 (Section 4.4) is finding a reason for the termination of subduction along the South China–Vietnam margin. As discussed in Section 4.3 there is no strong evidence in favour of a Late Cretaceous or Palaeogene collision large enough to have stopped subduction (as proposed by Hall et al., 2009; Fyhn et al., 2010). Possibly the cumulative effects of minor collisions on the southern margin of the Dangerous Grounds block could have jammed the subduction zone instead, but in general seamount subduction does not stop subduction. Another option proposed here is that Late Cretaceous back-arc spreading was responsible for cessation of subduction along the margin (Figs. 25C). In this scenario it is possible to drive Luconia–Borneo collision and alignment of Palawan-CIG with Borneo during the spreading event, and explain cessation of subduction along the South China margin, while permitting extension along the margin to continue. Consequently model 2 is the preferred option in this paper, and is the one shown in the regional plate reconstructions

(Fig. 23). However, it is recognized that the data for definitively defining the correct scenario is not yet available.

5. Timing of India–Eurasian collision

The timing of India–Eurasian continent–continent collision is highly significant for the time period studied, because the later part of the period discussed here (55–45 Ma) traditionally encompasses the period of continent–continent collision (Powell and Conaghan, 1973; Tonarini et al., 1993; Clift et al., 2002). However, a good case can be made that collision occurred as late as 34 Ma (Aitchison et al., 2007). Yet, as discussed in Sections 2.1, 2.2 and 3 evidence for Palaeogene tectonic events in Myanmar and Thailand is considerable, so pre-cursor events to the continent–continent collision must be sought to explain them. The Myanmar section discussed the Woyla-Mawgyi terrane collision proposed by Barber et al. (2005), and Barber and Crow (2009). There are, however, a number of competing models for the Cretaceous evolution of the Tethys ocean between India and Eurasia, and these need to be understood both in the context of deformation in SE Asia, as well as in the Himalayas.

The various published models for the Cretaceous evolution of the Tethyan realm tend to emphasise different aspects of the geology in reaching their conclusions; for example Barber et al. (2005) and Barber and Crow (2009) base their conclusions mainly on surface geology of Sumatra and Myanmar; Corfield et al. (2001) are based on the surface geology of the Kohistan and Lhasa; Hall et al. (2009) and Hall (2012) integrate information in the literature with plate reconstructions, and Hafkenscheid et al. (2006) integrated information in the literature with restorations based on mantle tomography and subducted slab volumes. The key similarity between the models is that Cretaceous intra-oceanic subduction zones developed within the Tethyan realm and subdivided the oceanic basin (variously named plates and oceans have been given such as the Ceno-Tethys and northern Meso-Tethys, Hall et al., 2009; or the southern Neo-Tethys and northern Spontang Ocean, Stampfli and Borel, 2002). In these models the intra-oceanic subduction zones gave rise to island arc complexes. The models vary in the timing of deformation and in the fate of the island arc complexes, which either collide with the northern margin of India, or with Sumatra–Myanmar, or both. Mantle tomography modelling favours a scenario of intra-oceanic subduction followed by collision of an arc complex with India during the Late Cretaceous (Hafkenscheid et al., 2006).

The model of Hall et al. (2009) and Hall (2012) is the only one that simultaneously attempts to explain both the SE Asian history and the India–Eurasian collision. A modified version of their reconstructions is presented in Fig. 23. In Hall et al. (2009) and Hall (2012) the island arcs resulting from the intra-oceanic crust subduction were divided into a northwestern arc (Incertus Arc), and a southeastern arc (Woyla Arc). The Woyla Arc collided with Sumatra between 110 and 90 Ma, while the Incertus Arc remained a site of subduction, anchored at its eastern end to the northern tip of the collided Woyla Arc. At about 55 Ma India collided with the Incertus arc, marking the closure of the Ceno-Tethys. Then between 55 Ma and 45 Ma the Meso-Tethys was closed, and the collision between India and Eurasia commenced. The name Incertus was applied to the arc because its timing of development, and tectonic significance are both highly important and uncertain.

Aitchison et al. (2007) argued that the fragments of ophiolite suites in the Ladakh region of NW India (including the Karzog, Nidar, Spontang ophiolites and the Spong Arc), were part of one or more intra-oceanic island arcs that were thrust southward over the northern Indian margin prior to continent–continent collision. U–Pb dating of the Spontang complex by Pederson et al. (2001) indicates the ophiolite is formed from mid-Jurassic oceanic crust ($177 \pm$ Ma), and is overlain by an arc complex dated around 88 ± 5 Ma (providing a minimum age of subduction initiation). The scenario of Palaeogene arc-collision with India is shown as collision of the Incertus Arc in the model of Hall et al. (2009). Both Aitchison et al. (2007) and Hall et al. (2009) favour India-arc collision around 55 Ma.

The amount of complexity attributed to the evolution of the Tethyan realm to a large degree centres upon the tectonic scenarios proposed for the Dras-Kohistan Arc, and whether it developed by colliding first with the Asian margin or the Indian margin (e.g. Aitchison et al., 2007; Khan et al., 2009). As reviewed by Petterson (2010) the arc lasted from about 134 to 90 Ma, the terrane was then strongly deformed by shearing, thrusting and folding. A key relationship within the arc is that of undeformed 75 Ma dykes, which cross-cut penetrative fabrics, including 102 Ma gneissic trondhjemites (Petterson and Windley, 1992). Hence an early deformation stage must be accounted for, and this has been widely attributed to collision of the arc with Asia (see review in Petterson, 2010). However, Khan et al. (2009) on the basis of palaeomagnetic data locate the Kohistan arc near the equator in Late Cretaceous times. This interpretation places the arc close to India but some 3000 km away from Asia, suggesting that collision occurred with India first. Khan et al. (2009) date that collision at about 65 to 61 Ma, on the basis of cessation of arc magmatism, and sole thrust ages at the base of the ophiolite of ca. 65 Ma. Unfortunately Khan et al. (2009) do not address what the very strong evidence for pre 75 Ma deformation represents, if it is not arc-Asia collision, although perhaps the pre 75 Ma deformation can be explained as the initial stage of obduction with India. However, if this is the case then the arc needs to be positioned further south (Arc II location, Fig. 23). Aitchison et al. (2007) also favour an India-first collision, which requires the Spontang Ophiolite-Spong Arc to lie north of the Kohistan Arc.

The arrangement of arcs proposed by Aitchison is opposite to that of Corfield et al. (2001), who argue for three north-dipping Tethyan subduction zones during the Late Cretaceous, (arranged from south to north: the Spontang Arc, Dras-Kohistan Arc and the southern Asian margin). Their arguments are in part based on: 1) the evidence for pre 75 Ma collision of the Kohistan Arc with Asia, 2) palaeomagnetic evidence that placed the Spontang Ophiolite more than 1500 km to the south at this time, and 3) Late Cretaceous–Early Palaeogene collision of the Spontang Ophiolite with northern India. The model of Khan et al. (2009) re-introduces some key similarities between the two arc remnants, i.e. both arcs lay close the equator during the Late Cretaceous, and they both collided with India around the Cretaceous–Palaeogene boundary. Hence the tectonic picture can possibly be simplified into one of a single major Late Cretaceous arc that collided with India during the Late Cretaceous. Petterson (2010) in his review of the tectonic models recognised degrees of uncertainty in all the proposed models that requires further research. However, he considered evidence for a pre 75 Ma arc collision with Asia too strong to ignore and looked to potential weaknesses in the palaeomagnetic results to place doubts on the model of Khan et al. (2009).

One key strand of evidence supporting the Eocene (~54–50 Ma) age of India–Eurasia collision has been the stratigraphic and sedimentary data from the Indus Basin indicating the time when sediment sources from both Eurasia and India entered the same sedimentary basin (e.g. Searle et al., 1990; Clift et al., 2001, 2002). However, recently Henderson et al. (2011) have concluded that Eurasian-derived sediments of 51 Ma do not stratigraphically overlie Indian plate rocks, and hence the time of collision is not strongly constrained, as previously proposed. This conclusion provides some support for the late collision model proposed by Aitchison et al. (2007). Yet there still remains evidence for early collision. For example dating of the final collision between the Kohistan Arc, India and Asia is defined around 50 Ma by the presence of post-collisional intrusions, and the isotopic ages of coesite-bearing ecolgites (Heuberger et al., 2007; Khan et al., 2009).

6. Link between Tethyan history and SE Asia

For much of the literature on Cenozoic deformation in SE Asia there is a tendency to relate deformation to either the Australian–

Indonesia collision, or the India–Eurasian collision. However, as discussed in Section 5 above for the India–Eurasian collision, the actual timing of collision is somewhat controversial, and the discussion of when the early phase of 'soft collision' i.e. subduction of continental crust, turns to 'hard collision', i.e. the onset of widespread shortening within the colliding continent becomes important. It may only be when 'hard collision' occurs, that the effects of the collision zone spread out to cause deformation within SE Asia, as has been proposed for the extrusion of rigid strike–slip fault bound blocks away from the collision zone into SE Asia (e.g. Tapponnier et al., 1986; Leloup et al., 2001; Replumaz and Tapponnier, 2003). The largest displacement occurs on the Red-River Fault and is absorbed by opening of the South China Sea spreading centre.

As the inconclusive discussion in Section 5 above indicates our understanding of the complexities of the pre-collisional history of the Himalayan orogenic belt is fraught with argument and counter-argument, and incomplete data. Despite these problems the evolution of the western SE Asia margin must be compatible with the Himalayan story. The Incertus Arc model of Hall et al. (2009) approximates the model proposed by Khan et al. (2009). The emplacement of the Woyla and Mawgyi Arcs onto Sumatra and Myanmar between about 110 Ma and 80 Ma fits well with the Asia-first collision model for the Kohistan Arc (reviewed by Petterson, 2010), but it is beyond this paper to imply whether this observation has any meaning for the Kohistan Arc collision controversy.

Watkinson et al. (2008) and Hall et al. (2009) also note that during the latest Cretaceous subduction along southern SE Asia slowed, or ceased, while rapid subduction of the Tethyan realm continued to the west. This change in plate motion occurred along a N–S trending transform zone located approximately south of Peninsular Thailand, and may have been at least partly responsible for deformation in Thailand during the Late Cretaceous–Palaeogene, including motion along Ranong and Khlong Marui Faults (Watkinson et al., 2008).

7. Preferred model for the Late Cretaceous–Palaeogene tectonic evolution of SE Asia

This section presents the authors preferred model in the light of the data and models discussed and reviewed above. As such, references are kept to a minimum. The Late Cretaceous to Palaeogene tectonic history of SE Asia has long remained ambiguous, collision events involving continental blocks (West Burma Block, Luconia Block) have been proposed to explain tectonic events and S-type granites in SE Asia. However, the actual origin of the granites remains controversial, the S-type granites are mixed with I-type granites, and may have originated in a subduction zone setting. There is no good evidence for suture zones or fold and thrust belts of the correct age at the expected sites of the collisions. Additionally the tectonic events (such as folding, uplift and erosion in the Khorat Plateau, activation of strike–slip faults such as the Ranong, Khlong Marui and Mae Ping fault zones) observed within SE Asia appear to have occurred episodically over a protracted period of time, and are unsuited to explanations invoking a single collision. Consequently the intepretations preferred here are those related to changes in the configuration and convergence rates of the subduction zones surrounding SE Asia (also see Hall et al., 2009). The young, hot, crust of SE Asia is assumed to be atypically responsive to the changes in stress regime arising from the changing subduction zone history compared with areas of Archean-cored continental crust.

The beginning of the Late Cretaceous appears to be a time when both the Tethyan and Palaeo-Pacific margins were in transition, following major and minor accretionary events involving island arcs, continental fragments and seamounts. These events signal the end of significant collisional deformation affecting SE Asia during the remainder of the Cretaceous. During the Late Cretaceous the South China to South Vietnam margin experienced extensive development of metamorphic core complexes, extensional basin development, uplift of massifs, rift-related volcanics, syn-kinematic plutonism, NE–SW trending dextral strike–slip faults are also widespread (e.g. Lin and Wang, 2006; Lin et al., 2008). In the late Cretaceous subduction along the margin ceased, and was relocated within oceanic crust further to the east. Both Hall et al. (2009) and Fyhn et al. (2010) proposed a significant collision involving the Luconia–Dangerous Grounds Block with Asia either during the Cretaceous or Palaeogene that stopped the subduction. However, such an explanation is here ruled out since the evidence favours the Luconia–Dangerous Grounds area was already part of the southern margin of Vietnam and China during the Mesozoic (Hutchison 2010; Yan et al., 2010). A back-arc spreading model is favoured to explain the cessation of subduction along the margin in the Late Cretaceous, the continuation of extension along the margin, and the collision of the Luconia Block with southern Borneo (Fig. 25). Despite the absence of a major collision, the evolution of plate boundaries during the Late Cretaceous and Palaeogene would have affected the distribution of forces acting on SE Asia as discussed for the Cenozoic by Hall and Morley (2004), and Hall et al. (2009).

The largest depositional basin during the Cretaceous was the Khorat Basin (Fig. 10). During the Jurassic marine sedimentation affected western Thailand, while a depositional hiatus characterizes the Khorat Basin. Then in the Late Jurassic and early Cretaceous subsidence in a predominantly continental setting commenced in the Khorat Basin. Since the basin displays some minor marine incursions it must have lain near sea level, but sedimentation presumably kept pace with subsidence and so avoided more significant marine sedimentation events. Basin development corresponds with the time of active subduction and arc collision on the western margin of SE Asia, and active subduction on the Pacific margin.

Diachronous arc collision events around the subduction margin occurred during the mid-Cretaceous (e.g. Eurasia-Kohistan Arc, Myanmar-Mawgyi Arc, Sumatra-Woyla Arc, Java-Borneo – western Sulawesi + other continental fragments, Palawan-seamounts?). The resulting regional changes in the stress configuration probably led to minor inversion events in Laos, Thailand and Cambodia that caused the restricted depositional environment of the Maha Sarakham evaporites. Subsidence in the Khorat Basin also slowed around 110 Ma (Fig. 13), which coincides with a regional change in tectonics (Fig. 13) including the loss of the Andean type margin to the west with the Mawgyi and Woyla arc collisions.

South of the Khorat Basin the Kampot-Phuquoc, Ton Le Sap area, and Peninsula Malaysia all show evidence for fault-controlled basin subsidence during the Cretaceous. However, dating of the sequences, and the timing and style of the faulting is not well understood.

Reconstructions of Tethyan subduction history for the Indian region by Hafkenscheid et al. (2006) based on estimated volumes of subducted material from seismic tomography suggest the following sequence: ~120 Ma subduction beneath Eurasia; ~100–65 Ma creation of a back-arc basin (Spontang Ocean of Stampfli and Borel, 2002) and ocean–ocean subduction south of Eurasia; around 65 Ma subduction was re-focussed beneath Eurasia. However, as discussed in Sections 2.2.1. and 2.2.5. initiation of subduction under the Myanmar margin appears to have begun considerably earlier around 95 Ma, and was sufficiently advanced by 80 Ma to be the cause of granite pluton intrusion in Myanmar and Thailand.

Subsequent to the Mawgyi and Woyla terrane collisions, Late Cretaceous magmatic activity along western SE Asia from Myamnar to Sumatra was considerably reduced. However, significant plutonism is found in Thailand, and, (based on old, potentially unreliable radiometric ages) the Gulf of Thailand, and the Stong Complex of Malaysia. There is also a thermal event in mid-crustal level rocks in Thailand between about 84–70 Ma. These events are diachronous to deformation associated with the mid Lower Palaeogene unconformity in the Indo-Burma Ranges, and the Early

Palaeogene regional metamorphism, migmatisation, ductile deformation and biotite granite sill injection in the Mogok Metamorphic Belt of eastern Myanmar (Searle et al., 2007).

One possibility is that the Late Cretaceous–Early Palaeogene events in Thailand and Myanmar are related to the onset of renewed subduction beneath Eurasia, which started earliest in the vicinity of the trench opposite Peninsular Thailand, and propagated northwards towards Mynamar. As discussed in Section 2.1 the Andaman ophiolite was formed around 94 Ma (Pedersen et al., 2010) in a supra-subduction zone setting. Recent models for supra-subduction zone ophiolite formation favour development in the fore-arc region during a period of subduction initiation (e.g. Stern, 2004; Whattam and Stern, 2011). Consequently it can be interpreted that following collision of the Mawgyi Terrane, subduction was re-initiated along the northern half of the southwestern SE Asian margin around 94 Ma (Fig. 23). In Thailand the renewal of subduction could explain the subsequent development of migmatites in the Peninsular around 88 Ma, the intrusion of granitic plutons between about 82 Ma and 71 Ma and the 84–70 Ma thermal metamorphic event (see Section 2.1). Subduction initiation would have been limited to the area north of Sumatra, since there was probably overlap between the ophiolite forming events in the Andaman Sea and collision of the Woyla Terrane with Sumatra (Fig. 23). This tectonic division coincides with the location of the putative N–S striking, dextral transform system between 90° and 100° E proposed by Watkinson et al. (2008) and Hall et al. (2009) to separate the rapidly northwards moving India Plate from the slowly moving Australia Plate (Fig. 23).

Some intra-continental strike–slip activity was also occurring during this time and may have controlled the development of basins in the Ton Le Sap area of Cambodia, and offshore western Gulf of Thailand, Peninsular Thailand and Peninsular Malaysia (Fig. 26 locations A, B, C and D).

The period from about 60 to 35 Ma encompasses considerable uplift in Central and eastern Thailand and western Laos and the Gulf of Thailand that resulted in widespread development of folds, thrusts, and transpressional features within the Khorat Group and older section (Fig. 26B). At least some strands of strike–slip faults that were later incorporated into the Oligocene Mae Ping and Three Pagodas Faults were active during this time. Approximately 3–4 km thickness of the Khorat Group was deformed, uplifted and eroded, particularly from the western margin of the Khorat Basin. This deformation is focused along the western margin of the Indochina block, and the Paleo-Tethys suture zone marked by the Inthanon and Sukhothai Zones (Fig. 1). The intensity of Palaeogene deformation appears to increase passing westwards from the Khorat Plateau, culminating in the Late Eocene–Oligocene crustal thickening event in the Mogok Metamorphic Belt, (~47–24 Ma), which caused high-grade sillimanite metamorphism (peak metamorphism at 680 °C and 4.9 kbar), metamorphic zircon overgrowths, and localized partial melting (Barley et al., 2003; Searle et al., 2007).

The early Palaeogene deformation has previously been thought to be too early to be directly related to India–Eurasia collision. Alternatives include a Paleogene collision of the West Burma Block with the Sibumasu Block (Charusiri et al., 1993) and the Luconia Block with the Malaysia–Vietnam margin of the South China Sea (Fyhn et al., 2010). However, the West Burma Block collision is not considered a possibility (the timing of metamorphism in the Mogkok Metamorphic Belt misses the crucial 60–45 Ma period, Searle et al., 2007), and the accretion of the Mawgyi Terrane requires that the area of the West Burma Block was in place prior to the mid-Cretaceous. The

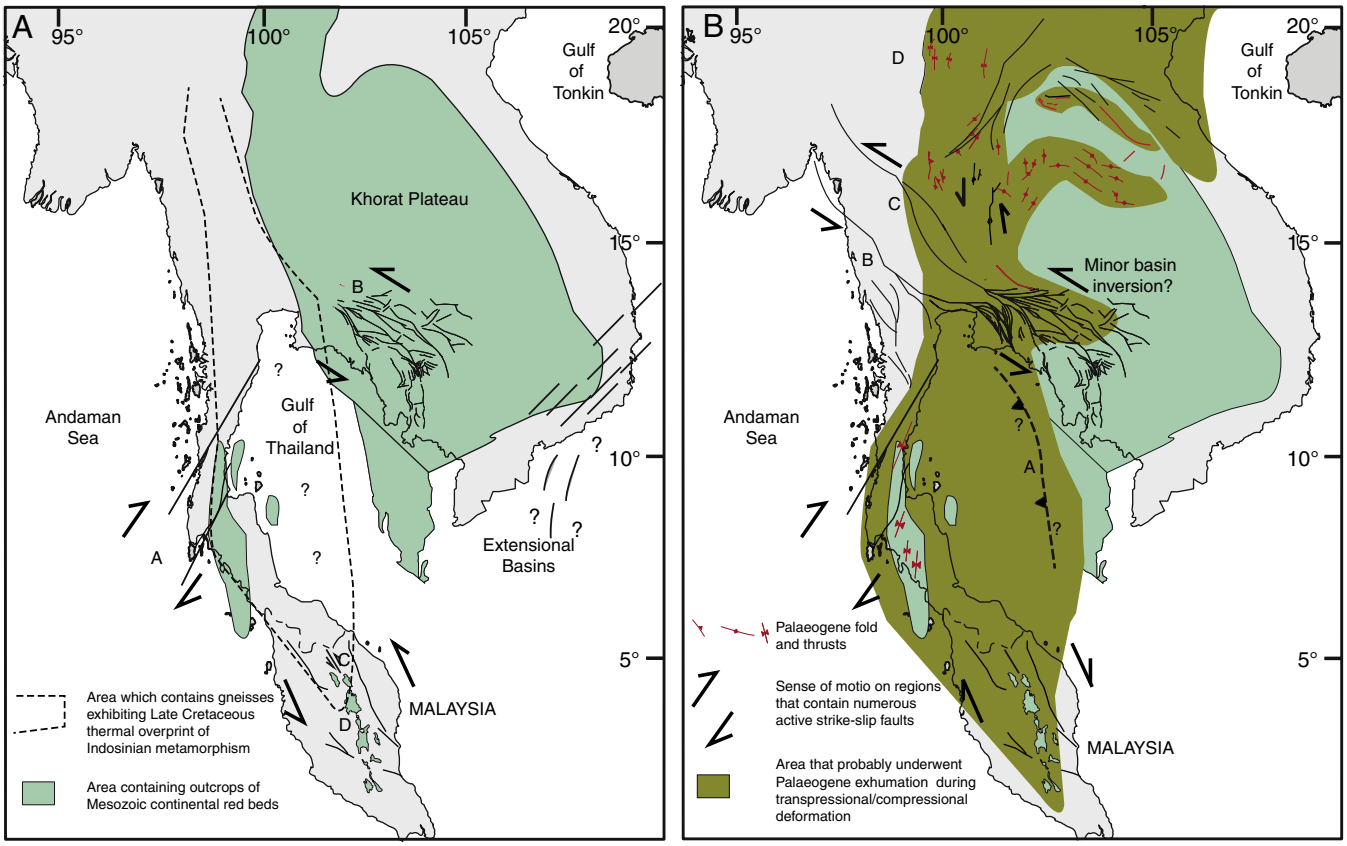


Fig. 26. Distribution of key structural features and basins in Indochina during A) Late Cretaceous and B) Palaeogene. A) A, B, C, D = locations of possible strike–slip fault activity during the Late Cretaceous. B) A = possible offshore fault zone that marks overthrusting of pre-Jurassic section over the western margin of the offshore Mesozoic basin. B = Three Pagodas Fault zone, C = Mae Ping Fault zone, D = folds south of Chiang Rai.

Luconia Block collision is considered here to be too small and locally confined to have had a significant effect.

Van Hinsbergen et al. (2011) have calculated that for a 50 Ma timing of collision of India with Asia the size of Greater India, in a N–S direction, north of present-day India was about 1350 km on the west side, and 2600 km on the east side. The shape of Greater India from Van Hinsbergen et al. (2011) is used in Fig. 23. Using such a reconstruction it is possible to explain the deformation in Myanmar from about 60 Ma onwards as a result of Greater India sweeping past the Indo-Burma ranges. Perhaps the deformation was related to interaction with a small promontory (created by the India-island arc collision?). Given the on–off nature of major deformation pulses in the Indo-Burma ranges, interaction with a passing promontory could explain the initial Palaeogene deformation. The final coupling of Western Myanmar with India appears to have begun in the Late Oligocene–Early Miocene when strike–slip deformation began to rip through the Andaman Sea area (e.g. Curray, 2005; Morley et al., 2011). While the Van Hinsbergen et al. (2011) solution is perhaps the simplest for explaining the observed deformation in Myanmar, it requires a rather extreme length for Greater India, that has yet to be demonstrated to exist from deep geophysical data beneath the Tibetan Plateau. There remain problems with aspects of both Eocene and Oligocene collision models that must be addressed by future research.

Understanding the cause of the early Palaeogene deformation remains problematic, deformation was probably episodic over a protracted time interval (as documented for the Mogok Metamorphic Belt by Searle et al., 2007; the Ranong Fault zone by Watkinson et al., 2011; and indicated by the protracted uplift of the western margin of the Khorat Plateau during the Palaeogene, Fig. 12). Consequently the deformation appears to have accrued as a result of stress changes from a variety of sources rather than one single explanation. The sources include the minor Luconia Block collision, the subducting vs inactive margin configuration of the Myanmar–Sumatra margin (Watkinson et al., 2008; Hall et al., 2009), the variable timing of subduction initiation along the Myanmar–Sumatra margin, ophiolite emplacement in the Andaman Sea and Indo-Burma Ranges, and possibly events to the north, for example the Late Cretaceous–Palaeogene deformation of the southern Longmen Shan belt (Jia et al., 2006), as well as India–Myanmar and India–Asia interactions.

8. Discussion

The Cenozoic rift basins of Thailand exhibit typical syn-rift geometries (e.g. half graben geometries, transfer zones) yet differ in a number of ways from classic syn-rift and post-rift basins (e.g. East Africa, North Sea, Gulf of Suez). The anomalous features of Thailand's Cenozoic basins include (Morley et al., 2011): 1) numerous low-angle normal faults, 2) alternating syn-rift extension and inversion, 3) the diachronous timing of rift initiation and termination, 4) rapid post-rift basin subsidence, and 5) strong interaction of strike slip and extensional faults, without necessarily being simple pull-apart basin relationships. A number of these features (particularly 1, 4 and 5) can, in part, be related to the Indosinian and Cretaceous history of the margin. The nature of the Sundaland lithosphere has been discussed for the Cenozoic by Hall and Morley (2004), Morley and Westaway (2006), and Hall (2011), who characterize it as having a thin, hot, weak lithosphere, with a weak lower crust capable of flow in response to tectonic and topographic forces. This hot weak crust, which owes its characteristics to the long history of subduction zone-related tectonics and deformation in Sundaland, was initially assembled from rifted Gondwana fragments during the Indosinian orogeny (Triassic–Early Jurassic, e.g. Metcalfe, 1996, 2009; Sone and Metcalfe, 2008a). This assembly resulted in regional metamorphism, crustal thickening, and widespread intrusion of highly radioactive Triassic granites. The Cretaceous subduction zones added a considerable volume of granites to the Sundaland crust, increasing the radiogenic component of heat flow within the crust.

In Thailand and Malaysia the Indosinian orogeny developed structural trends that are aligned predominantly N–S, with some local E–W and NE–SW and NW–SE trends. The two main N–S trending continental fragments (Sibumasu and the Indochina blocks) are separated by the oceanic, Palaeo-Tethys suture zone called the Inthanon Zone, and a zone of sea mounts, island arcs and other fragments called the Sukhothai Zone (Sone and Metcalfe, 2008a). Some of these trends are characterized by steep dips, and some of which probably mark early segments of major strike–slip fault zones (e.g. Ridd and Morley, 2011). However, much of the basement fabrics associated with the Indosinian orogeny are low-angle, thrust related (e.g. Jolivet et al., 1999), and these played a significant role during the Cenozoic when they were reactivated as low-angle normal faults (Morley, 2009; Figs. 16F and 19B).

The location of most intense Palaeogene faulting, folding and exhumation in Thailand and the Gulf of Thailand coincides with the Inthanon and Sukhothai zones (compare Figs. 1 and 26B). The region of presumably, thickened crust, would have been an area of increased gravitational potential that collapsed as stresses evolved in response to plate tectonic reconfigurations during the late Eocene–Miocene to form the N–S trending rift basins (e.g. Hall and Morley, 2004; Hall et al., 2009). Although Indochina was not the site of a classic, major Cretaceous–Palaeogene orogenic belt, the occurrence of Cenozoic metamorphic core complexes (Doi Inthanon, Rayong, Khanom, Morley et al., 2011) and other rift basins on the site of Late Cretaceous tectonic activity, and Palaeogene inversion is reminiscent of the conditions required for core complex formation (i.e. hot, orogenically thickened crust, Buck, 1991). However, the key difference with most collisional orogenic belts is that the crust appears to be hot enough to be weak, and capable of flow even if it is of normal thickness (Morley and Westaway, 2006). The timing of the initiation of collapse is poorly constrained, but there are at least hints that local collapse during the late Eocene from Khanom (Section 3.1) and Doi Inthanon. U–Pb dates from monazite and zircon grains from top to the east mylonites at Doi Inthanon yielded ages around 40 ± 0.5 Ma (Macdonald et al., 2010). While apatite fission track data from the Rayong low angle normal fault suggests cooling/exhumation began around 38 Ma (S. Meffre pers. comm., 2010).

The diachronous timing of rift events, and the anomalously high magnitude of post-rift basin subsidence (6-10+ km during the Miocene-Recent in the Pattani and Malay Basins, Morley and Westaway, 2006), are key features of the Gulf of Thailand. Rifting commenced in the western part of the gulf, and later spread to the west. It is very likely that this early rifting was focused on the area of strongest inversion during the Palaeogene (i.e. the area of greatest gravity potential). This has yet to be fully demonstrated by systematic mapping of structures, but it is the author's experience from the Gulf of Thailand that all the early formed basins show evidence for folded and thrust Khorat Group underlying the syn-rift basins (also see Fyhn et al., 2010). The very deep post-rift basins have also been explained as a consequence of lower crustal flow driven by sediment loading, which is also focused on the eastern Gulf of Thailand (Morley and Westaway, 2006).

9. Conclusions

The Late Cretaceous–Early Palaeogene evolution of SE Asia remains poorly understood, and a diverse range of tectonic models have been proposed (and are still potentially viable) for the region during this time period. Publication of exisiting radiometric ages, and new collections for radiometric dating are needed to better understand metamorphic, magmatic and structural events in the region. There also exists a wealth of information in seismic reflection data and outcrop data that has yet to be fully exploited. In most of the

provinces discussed here, a definitive tectonic model cannot be concluded, the main purpose has been to explore weaknesses and strengths of existing models, and suggest preferred solutions, and to indicate what type of data is needed to further our understanding. SE Asia represents continental growth by accretion over an unusually prolonged period of time, and determining how this growth has progressed during the different stages from the Triassic to the Present is important for a variety of reasons (e.g. the impact on Cenozoic deformation, lithospheric strength profiles, sedimentary basin development, mineralization, models for granite genesis, conditions for exhumation of middle/lower crustal rocks). The key conclusions from this review are as follows:

1) Collision termination of Palaeo-Pacific subduction along southern Asian margin invoking an inferred Luconia fragment either in the Late Cretaceous or Palaeogene is discounted. Instead relocation of the subducting margin further west, possibly in response of back-arc extension (which created the Proto-South China Sea) is preferred. A small Luconia fragment is regarded as having collided with NW Borneo during the Palaeogene and was only directly responsible for deformation confined to NW Borneo.
2) Collision of the Mawgyi and Woyla Terranes was possibly diachronous in the mid Cretaceous on the Myanmar–Sumatra margin. However, the unconformable relationship between the Paung Chaung Limestone and underlying Triassic units requires further investigation, and the occurrence of serpentitite sheets within the limestone needs explanation, since they suggest the limestone maybe part of the accretionary complex, and not overlying it. Formation of the Andaman ophiolite in a supra-subduction zone setting around 94 Ma is inferred to mark the time of subduction initiation. The onset of subduction is proposed to have caused the Late Cretaceous granite pluton emplacement and associated thermal event in western and central Thailand.
3) Significant Late Cretaceous–Early Palaeogene orogenic events (emplacement of granite plutons, migmatites, amphibolite grade metamorphism, ductile strike–slip shear zones, basin forming faulting, thrust and strike–slip related uplift and erosion of up to 4 km thickness of Khorat Group sediments) affected the core of SE Asia. These events seem to have largely occurred in a subduction zone or extensional setting on the Pacific and Tethyan margins, and are not related to collisions with continental fragments on either margin. The deformation represents the final period when the crust of SE Asia was 'conditioned' prior to India–Eurasian collision. From about 60 to 55 Ma onwards some deformation in Indochina, particularly in Myanmar, may be related to interaction with the NE margin of Greater India as it moved northwards (Figs. 8, 23).
4) When the effects of India–Eurasian collision entered the area a number of segments of strike–slip faults attributed to escape tectonics were already present, many were active during the Late Cretaceous–Palaeogene (e.g. Ton Le Sap region of Mae Ping Fault, Ranong and Khlong Marui Faults, western margin of Khorat Plateau; Fig. 26).
5) The Khorat Basin is a very large basin with a simple sag-like geometry. Although the basin lies between two Andean type margins there are not the relationships to major thrust fronts to indicate it is a classic foredeep basin. Perhaps its origins lie in the dynamic topography created above two converging subduction zones. The basin is an important response to an unusual tectonic setting and does not fit easily with standard basin classification.
6) South and west of the Khorat Basin strike–slip and compressional deformation of Jurassic–Cretaceous sedimentary rocks in Central, Eastern and Peninsular Thailand and Malaysia is extensive. The full extent and timing of this deformation has yet to be determined, although apatite fission track data indicated the Palaeogene was an important time of exhumation. In particular it is likely that regional analysis of seismic reflection data from the Gulf of Thailand will show a complex structural history exists for the Late Cretaceous–Palaeogene.
7) The Cenozoic rift basins were largely developed on the crust that was pre-conditioned by Indosinian and Cretaceous tectonic events, to be the weakest parts of the Indochina region, in regionally hot, weak crust. The result was the development of metamorphic core complexes, other low-angle normal faults, collapse of the weakest areas of the crust first (i.e. the eastern Gulf of Thailand) and lateral (east to west) and south to north propagation of the extension away from this core area. Subsequent anomalously large post-rift subsidence was also focused on the eastern Gulf of Thailand crust. Extension has been a feature of the South China Sea area since the Cretaceous, with Oligocene extension in the South China Sea representing the latest stage. Cenozoic strike–slip fault activity (e.g. ASRR fault zone) is viewed here as being superimposed on this extending margin, rather than being a (late-stage) driver of extension.

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